Co-sufficient weighting: leveraging sparsity for more powerful variable selection in the linear model

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Abstract

We show that for the problem of controlled variable selection in the Gaussian linear model, informative and p-value-independent weights (for weighted multiple testing) can be derived entirely from sufficient statistics and a belief in sparsity using only the data itself and no external quantitative side information. Using these weights, which we call *co-sufficient weights* (CSW), in weighted versions of Holm's procedure and the Benjamini–Hochberg (BH) procedure results in substantially higher power than their unweighted analogues when the true coefficient vector is sparse, while retaining Holm's exact validity and BH's highly robust empirical validity regardless of sparsity. We demonstrate the superior performance of our proposed weighted methods compared to their unweighted analogues and other competing methods in a wide range of simulations and an application to HIV drug resistance.

1 Introduction

1.1 Motivation

Under the setting of the homoskedastic Gaussian linear model

$$\boldsymbol{y} \sim \mathcal{N}\left(\boldsymbol{X}\boldsymbol{\beta}, \sigma^2 \boldsymbol{I}_n\right), \tag{1.1}$$

where $\boldsymbol{X} \in \mathbb{R}^{n\times d}$ is a full column-rank matrix, $\boldsymbol{I}_n$ denotes the $n \times n$ identity matrix and $\boldsymbol{\beta} \in \mathbb{R}^d, \sigma^2 > 0$ are unknown, this paper considers the problem of controlled variable selection, i.e., recovering the set of signal indices $\mathcal{S} = \{j : \beta_j \neq 0\}$ while controlling an error rate such as the familywise error rate (FWER) or false discovery rate (FDR). Variable selection in the linear model is one of the most canonical inference problems in science, with applications across many domains such as biology [Yamashita et al., 2008, Abraham et al., 2014] and economics [Belloni et al., 2012, Kozak et al., 2020].

Often, analysts in these applications possess domain knowledge about properties of the coefficient vector $\boldsymbol{\beta}$, with a widely occurring example being knowledge of *sparsity*, meaning that the analyst believes most entries of $\boldsymbol{\beta}$ are zero. Such sparsity assumptions arise across

many scientific domains, such as genetics [Lucas et al., 2006], econometrics [Fan et al., 2011], and chemistry [Filzmoser et al., 2012]. Indeed, Section 5 of this paper studies such a setting, where we apply variable selection methods to identify genetic mutations associated with resistance to an HIV drug.

In this paper, we show that such a structural belief, particularly sparsity, can be leveraged to obtain a more powerful variable selection procedure. A key aspect of this question is the inherent immeasurability of the structural belief—sparsity here reflects only the imprecise belief that a small fraction of the covariates account for the majority of the signal present in $\boldsymbol{X}$, and does not come with an explicit quantification, such as a known bound on the proportion of signal covariates or a fully specified prior distribution on $\boldsymbol{\beta}$. Thus, we develop methods that leverage sparsity in $\boldsymbol{\beta}$ solely to improve power. In particular, we propose a novel way to use the linear model to construct weights for use in multiple testing. With such weights, a certain weighted version of Holm's procedure controls the FWER regardless of sparsity, while a weighted version of BH, like unweighted BH, exhibits robust empirical FDR control regardless of the sparsity level. Thus, an analyst can use our methods when they believe in sparsity (without having to explicitly quantify that belief) and will benefit when that belief is correct without having to worry about inflated errors if the belief is incorrect.

1.2 Related Works

Perhaps the simplest approach to variable selection in the linear model is to construct p-values for the individual hypotheses $H_j : \beta_j = 0$, for example using the t-test, and then plug these p-values into a multiple testing procedure. The literature on multiple testing is vast and we do not attempt to review it here (see Emmert-Streib and Dehmer [2019], He et al. [2026] for recent reviews), though in Section 2, we will review some methods that are particularly relevant to our work. Among the standard choices for multiple testing widely used for variable selection in the linear model are Holm's procedure [Holm, 1979] and the Benjamini–Hochberg procedure [Benjamini and Hochberg, 1995]. However, these multiple testing algorithms have no mechanism for leveraging sparsity in $\boldsymbol{\beta}$ to improve power.

Existing methods in the literature that leverage sparsity often do so at the cost of additional assumptions. For FWER control, one class of methods applies stability selection [Meinshausen and Bühlmann, 2010, Shah and Samworth, 2013] to the LASSO estimator [Tibshirani, 1996], a well-known sparse estimator of $\boldsymbol{\beta}$ that performs variable selection by shrinking some coefficients exactly to zero. In this framework, variables are selected according to how frequently they are included across repeated LASSO fits on subsamples of the data. Stability selection can then bound the FWER of the resulting procedure, but only under substantial assumptions on $\boldsymbol{\beta}$. Another class of methods [Wasserman and Roeder, 2009, Meinshausen et al., 2009, Schultheiss et al., 2021] relies on sample splitting—a set of variables is selected on one part of the data, typically with the LASSO, and only the selected variables are then tested, with a multiplicity correction, on the remaining part. These methods, however, require additional assumptions ensuring that all signal variables are selected in the first step, which rarely happens in practice. G'Sell et al. [2016] also discuss a LASSO-based method for FWER control, but at the cost of a similar assumption that the signals enter the LASSO path before any noise variable.

For FDR control, several works also exploit properties of the LASSO to obtain FDR-

controlling procedures; examples include [Bogdan et al., 2015, G'Sell et al., 2016]. However, these methods again rely on the same ordering assumption that signals enter the LASSO path before noise variables. Another example is Javanmard and Javadi [2019], who propose a multiple testing procedure for the linear model based on the debiased LASSO [Zhang and Zhang, 2014, Javanmard and Montanari, 2014, van de Geer et al., 2014]. Their primary objective, however, is to extend the BH procedure applied to t-test p-values to the $d > n$ setting; in particular, when $d \leq n$, their procedure reduces to the original BH procedure. To our knowledge, the only class of methods that achieves FDR control without such additional assumptions while explicitly leveraging sparsity is the fixed-X knockoffs framework [Barber and Candès, 2015], along with subsequent refinements that improve its performance [Spector and Janson, 2022, Luo et al., 2022, Ren and Barber, 2023, Lee and Ren, 2024]. Knockoff-based procedures introduce synthetic negative controls for each variable and construct importance measures by comparing each variable to its corresponding control. Sparsity can then be exploited through LASSO-based importance measures. However, this framework also has several limitations. It requires augmenting the design matrix with synthetic variables, effectively doubling its dimension; this can reduce power and, more importantly, imposes the constraint $n \geq 2d$ for fixed-X knockoffs to be applicable. The choice of knockoff variables also introduces ambiguity such that two analysts applying the same method to the same data could get different answers. More broadly, knockoffs belongs to a larger class of synthetic variable methods targeting FDR control [Wu et al., 2007, Dai et al., 2022, Xing et al., 2023, Guan et al., 2025, Molinari and Thoresen, 2025] that rely on discrete comparisons between real and synthetic variables to make selections. Aside from fixed-X knockoffs, the methods in this class require additional assumptions beyond the Gaussian linear model to control the FDR, which we see in our simulations can result in significant violations.

1.3 Summary of our contributions

We develop novel methods for exploiting sparsity to improve the power of controlled variable selection in the linear model when $n > d$ (unlike knockoff-based methods, we do not require $n \geq 2d$). Our methods are weighted multiple testing procedures for the hypotheses $\{H_j : \beta_j = 0\}_{j=1}^d$, where each hypothesis H_j is assigned a weight w_j. Our primary contribution is a data-dependent choice of these weights, which we call *co-sufficient weights* (CSW): we show that a belief in sparsity alone, without any additional side information, can be used to construct weights that are independent of their corresponding p-values and that tend to be larger for variables likely to be non-null. The construction leverages the sufficient statistic under each H_j and combines it with a LASSO-type estimator to exploit sparsity.

Our two proposed multiple testing procedures incorporate these weights into weighted versions of Holm's procedure and the BH procedure, respectively, and are substantially more powerful than their unweighted analogues when the true coefficient vector is sparse. The weighted version of Holm's procedure guarantees exact FWER control, regardless of sparsity. The weighted BH procedure, like the ordinary BH procedure in the general linear model setting, does not have strictly guarantee FDR control but empirically exhibits highly robust FDR control in practice. Extensive simulations and an application to HIV drug resistance demonstrate our methods' power gains in sparse settings over methods that do not exploit sparsity, as well as their outperformance of other sparsity-exploiting methods.

2 Background: Weighted multiple testing

Our proposed variable selection procedures are constructed by applying weighted multiple testing with a novel, carefully chosen set of weights. We therefore begin with a review of weighted multiple testing procedures. Suppose that we have d hypotheses $\{H_j\}_{j=1}^d$ to test, with corresponding p-values $\{p_j\}_{j=1}^d$, where p_j is super-uniform under H_j; define $\mathcal{H}_0 := \{j : H_j \text{ is true}\}$ as the set of true nulls. In addition, suppose that each hypothesis H_j has an associated weight w_j.

2.1 Methods for FWER control

To begin with, assume the w_j are non-random and $\sum_{j=1}^d w_j = 1$. A canonical example is the weighted analogue of the Bonferroni procedure, which rejects all hypotheses j such that $p_j \leq \alpha w_j$, for a pre-specified nominal FWER level $\alpha \in (0, 1)$. Larger weights for hypotheses that are not true translate to higher cutoffs, with weights larger than $1/d$ making such hypotheses easier to be discovered than under the unweighted Bonferroni procedure. The Bonferroni procedure can be uniformly improved by Holm's procedure [Holm, 1979], which preserves FWER control and which likewise can be improved by a weighted analog [Holm, 1979, Scheme 2]. In fact, a weighted version of Holm's procedure can be constructed as the closure [Marcus et al., 1976] of a weighted Bonferroni test. We describe this construction next as it motivates our FWER controlling procedure in Section 3.3.1.

First, for any $I \subseteq [1 : d]$, consider the intersection hypothesis $H_I := \cap_{j \in I} H_j$ and the weighted I-Bonferroni test that rejects H_I if

$$p_j \leq \alpha \cdot \frac{w_j}{\sum_{i \in I} w_i} \quad \text{for any } j \in I. \tag{2.1}$$

Then the closure of these weighted I-Bonferroni tests is a multiple testing procedure for the original $\{H_j\}_{j=1}^d$ that rejects H_j if and only if H_I is rejected whenever $j \in I$. This closure test is a weighted version of Holm's procedure and, because taking a closure uniformly improves the power [Romano et al., 2009], is at least as powerful than the original weighted Bonferroni procedure. Notably, weighted Holm's procedure admits a sequential representation that can be executed by examining at most d subsets of indices I.

Recent work has extended validity of the weighted Bonferroni procedure to the case when the weights w_j are random, under the following assumption (see [Ignatiadis et al., 2023, Theorem 4.3]).

Assumption 2.1.

(a) w_j *is independent of* p_j *for all* $j \in \mathcal{H}_0$.

(b) $\mathbb{E}\left[\sum_{j \in \mathcal{H}_0} w_j\right] \leq 1$.

Unfortunately, it is not clear that the random-weighted Bonferroni procedure can be improved via the closure principle above under Assumption 2.1, since Assumption 2.1(a)'s independence condition will in general fail for the weighted I-Bonferroni tests due to the weight renormalization in Equation (2.1). In Section 3.1, we will show how to overcome

this roadblock for the linear model, whose sufficient statistics can be leveraged to construct valid weights for these weighted I-Bonferroni tests. This leads to a weighted version of Holm's procedure with random weights[1], which, like its non-random counterpart, admits a sequential representation, is computationally inexpensive, and is at least as powerful as the random-weighted Bonferroni procedure.

2.2 Methods for FDR control

Among the earliest works in weighted multiple testing for FDR control is Genovese et al. [2006], which introduced a weighted version of the BH procedure by applying BH to the scaled p-values $p_j/(dw_j)$. The validity of this approach assumes non-random weights that sum to 1, but Ignatiadis et al. [2023] show that validity also holds for random weights satisfying Assumption 2.1 and the following additional condition.

Assumption 2.2.

(a) The set $\{(p_j, w_j)\}_{j\in\mathcal{H}_0}$ *is independent of the set* $\{(p_j, w_j)\}_{j\notin\mathcal{H}_0}$.

(b) The d *pairs* (p_j, w_j) *are mutually independent with each other across* $j \in \mathcal{H}_0$.

However, neither the unweighted BH procedure nor its weighted counterpart controls the FDR under the general dependence structure of linear regression p-values [Dobriban, 2026, Lei, 2026][2], which violate Assumption 2.2 no matter the choice of weights. Yet the BH procedure is widely known to exhibit robust empirical FDR control under dependence structures encountered in practice, to the extent that many authors have conjectured its strict validity for correlated two-sided z-tests [Reiner-Benaim, 2007, Benjamini, 2010, Sarkar and Zhang, 2025]; in fact, only recently have examples of FDR violation been produced Dobriban [2026], Lei [2026]. Although Assumption 2.2 does not seem to be necessary for practical FDR control, we emphasize that Assumption 2.1 very much is, including its item (a) (note that unweighted BH satisfies Assumption 2.1 with its weights $w_j = 1/d$). Without Assumption 2.1(a), weights can be chosen to yield severely inflated errors; see Appendix A for an example that satisfies Assumption 2.1(b) and Assumption 2.2, yet exhibits severe FDR violations.

Thus, since Assumption 2.2 is unsatisfiable in general for linear regression and apparently unneccesary for practical FDR control, while Assumption 2.1 is critical for practical FDR control (and sufficient for strict FWER control as described in Section 2.1), our focus in this paper will be on constructing weights that satisfy Assumption 2.1. Indeed, as expected, we find in Section 4 that the weighted BH procedure of Genovese et al. [2006], with linear model p-values and using the weights we propose in Section 3, controls the FDR empirically quite robustly.

[1] As discussed in Ignatiadis et al. [2023], one can also construct a different weighted version of Holm's procedure when the weights are e-values. However, this assumption on the weights is restrictive and, in particular, does not hold in our case.

[2] Note that the unweighted BH procedure controls the FDR with dependent p-values under a certain class of positive dependence structure called PRDS [Benjamini and Yekutieli, 2001], however this condition is not generally satisfied by linear regression p-values.

If strict FDR control in the linear model is desired, one can instead leverage conditional calibration [Fithian and Lei, 2022]. Unfortunately, conditional calibration in general can come with a large computational cost due to the need to evaluate conditional expectations many times, and we find it to be prohibitive for our procedure; see Appendix B for details.

2.3 Choice of weights

For weighted multiple testing to improve power in practice, one needs a principled way to choose weights that both satisfy the above assumptions and tend to be larger for non-null variables. Much of the existing literature achieves this by translating external or quantitative side information into weights. Such side information may take the form of a Bayesian prior [Dobriban et al., 2015], a group or hierarchical structure among the hypotheses [Hu et al., 2010, Li and Barber, 2018], an ordering of hypotheses by importance [Li and Barber, 2017], or external covariates [Ignatiadis and Huber, 2021]. Ramdas et al. [2019] provide a unified framework for incorporating many such types of side information into multiple testing. Related methods outside the weighted multiple-testing paradigm also leverage side information; for example, Lei and Fithian [2018], Chao and Fithian [2021] do not directly construct weights, but instead use an iterative procedure to calibrate hypothesis-specific rejection thresholds based on the side information. These approaches, however, are not directly compatible with our setting, where no external or quantitative side information is available. Our only additional information is a belief that $\boldsymbol{\beta}$ is sparse, which is not in the form of a concrete, well-defined structure or the explicit quantitative or covariate-based side information required by these procedures. In the next section, we show that weights for powerful weighted multiple testing can nevertheless be constructed for the linear model without any external side information, by leveraging sparsity alone.

3 Powerful weighted multiple testing in the linear model

3.1 Co-sufficient weighting

Given the weighted multiple testing procedures reviewed in Section 2, our goal now reduces to constructing suitable weights for the linear model, which can then be plugged into these procedures.

To construct these weights, we start with a non-negative variable-importance measure $\mathcal{A}_j(\boldsymbol{y})$ for $\boldsymbol{X}_j$, where we suppress its notational dependence on the design $\boldsymbol{X}$ which is fixed throughout our discussion. When not specifically talking about any column index j, we write $\mathcal{A}$ to refer generically to a variable importance measure. We allow $\mathcal{A}$ to be a randomized function, i.e., its value can depend on external randomness independent of the data $(\boldsymbol{y}, \boldsymbol{X})$ in addition to its inputs (e.g., it could use cross-validation with randomly chosen folds, or randomized algorithms like stochastic gradient descent or random forests). Let $\boldsymbol{S}^{(j)} = (\boldsymbol{X}_{-j}^T\boldsymbol{y}, \|(\boldsymbol{I}_n - \boldsymbol{P}_{-j})\boldsymbol{y}\|^2)$ denote the minimal sufficient statistic for the linear model under H_j, where $\boldsymbol{P}_{-j}$ denotes the projection matrix onto the column space of $\boldsymbol{X}_{-j}$. See, for example, [Luo et al., 2022, Appendix E.2] for a justification of sufficiency. Consider the following

weights,

$$w_j = \mathbb{E}_{H_j}\left[\frac{\mathcal{A}_j(\boldsymbol{y})}{\sum_{i=1}^d \mathcal{A}_i(\boldsymbol{y})} \,\middle|\, \boldsymbol{S}^{(j)}\right], \qquad \forall j \in [1:d],$$

where we assume the convention that $0/0 = 0$ for the ratio inside the expectation. Unfortunately it will not generally be tractable to compute the above conditional expectation exactly, so we can instead consider its Monte Carlo estimate,

$$\hat{w}_j = \frac{1}{B}\sum_{b=1}^{B} \frac{\mathcal{A}_j(\tilde{\boldsymbol{y}}_b^{(j)})}{\sum_{i=1}^d \mathcal{A}_i(\tilde{\boldsymbol{y}}_b^{(j)})}, \qquad \forall j \in [1:d], \tag{3.1}$$

where the $\tilde{\boldsymbol{y}}_b^{(j)}$ are sampled i.i.d. from the distribution of $\boldsymbol{y} \mid \boldsymbol{S}^{(j)}$ under H_j, conditionally independently of $\boldsymbol{y}$; this so called *co-sufficient* sampling [Stephens, 2012] is easily carried out for the linear model [Luo et al., 2022] and justifies calling the $\{\hat{w}_j\}_{j=1}^d$ *co-sufficient weights* (CSW). Note that $\hat{w}_j$ has the same expectation as w_j and is conditionally independent of $\boldsymbol{y}$ given $\boldsymbol{S}^{(j)}$.

The key to CSW is that $\hat{w}_j$ only depends on the data through $\boldsymbol{S}^{(j)}$, making it easy to establish Assumption 2.1 because linear model p-values typically satisfy the following assumption:

Assumption 3.1. *For any $j \in [1:d]$, the p-value p_j for testing H_j is independent of $\boldsymbol{S}^{(j)}$ under H_j.*

Assumption 3.1 is satisfied, e.g., by standard linear model t-test p-values, and makes the proof of the following theorem straightforward.

Theorem 3.1. *If the p-values $\{p_j\}_{j=1}^d$ satisfy Assumption 3.1, then the weights $\{\hat{w}_j\}_{j=1}^d$ defined in Equation (3.1) satisfy Assumption 2.1.*

Proof. Because $\hat{w}_j$ is conditionally independent of the data given $\boldsymbol{S}^{(j)}$, it is independent of any p-value p_j that is itself independent of $\boldsymbol{S}^{(j)}$ under H_j. Furthermore, we have that

$$\begin{aligned}\mathbb{E}\left[\sum_{j\in\mathcal{H}_0} \hat{w}_j\right] &= \mathbb{E}\left[\sum_{j\in\mathcal{H}_0} w_j\right] = \mathbb{E}\left[\sum_{j\in\mathcal{H}_0} \mathbb{E}_{H_j}\left[\frac{\mathcal{A}_j(\boldsymbol{y})}{\sum_{i=1}^d \mathcal{A}_i(\boldsymbol{y})} \,\middle|\, \boldsymbol{S}^{(j)}\right]\right] = \sum_{j\in\mathcal{H}_0} \mathbb{E}_{H_j}\left[\frac{\mathcal{A}_j(\boldsymbol{y})}{\sum_{i=1}^d \mathcal{A}_i(\boldsymbol{y})}\right] \\ &= \mathbb{E}\left[\sum_{j\in\mathcal{H}_0} \frac{\mathcal{A}_j(\boldsymbol{y})}{\sum_{i=1}^d \mathcal{A}_i(\boldsymbol{y})}\right] \leq 1,\end{aligned}$$

thereby showing that both the conditions of Assumption 2.1 are satisfied. □

Theorem 3.1 holds for any choice of $\mathcal{A}$, but we next suggest appealing choices for high power under sparsity and tractable computation.

3.2 Sparsity-aware weights

One design objective in choosing $\mathcal{A}_j$ is for $\hat{w}_j$ to have small conditional variance given $\boldsymbol{S}^{(j)}$, so that B can be as small as possible, saving computation (note that if this conditional variance were zero, $\hat{w}_j = w_j$ for any $B \geq 1$). While it is not clear how to fully achieve this, it turns out we can get close by choosing $\mathcal{A}_j$ to only depend on $\boldsymbol{y}$ through $\boldsymbol{S}^{(j)}$. This makes $\hat{w}_j$'s numerator deterministic given $\boldsymbol{S}^{(j)}$, and although each denominator term with $i \neq j$ will be a function of $\boldsymbol{S}^{(i)}$ (and hence not deterministic given $\boldsymbol{S}^{(j)}$), $\mathcal{A}$ functions that measure variable importance in the ith variable generally do not vary much when resampling $\boldsymbol{y}$ conditional on $\boldsymbol{S}^{(j)}$, as such resampling holds $\boldsymbol{X}_i^T\boldsymbol{y}$ fixed. In fact, for the LASSO-based $\mathcal{A}$ we recommend in Section 3.2.1, we found in Appendix C that the conditional variance of $\hat{w}_j$ is close to zero even for $B = 1$. Appendix C does show that the more scalable soft-thresholding-based $\mathcal{A}$ we propose in Section 3.2.2 can be marginally but nontrivially improved by $B > 1$, this $\mathcal{A}$'s main goal is computational efficiency and $B = 1$ still performs well and is of course the fastest choice of B. Thus, for the remainder of the paper, we will always consider $B = 1$, and in particular drop the subscript from $\tilde{\boldsymbol{y}}^{(j)}$ going forward.

To design a $\mathcal{A}$ that only depends on $\boldsymbol{S}^{(j)}$, we observe that the first $d-1$ components of $\boldsymbol{S}^{(j)}$, $\boldsymbol{X}_{-j}^T\boldsymbol{y}$, transformed by $(\boldsymbol{X}_{-j}^T\boldsymbol{X}_{-j})^{-1/2}$ to $\boldsymbol{z}^{(j)} := (\boldsymbol{X}_{-j}^T\boldsymbol{X}_{-j})^{-1/2}\boldsymbol{X}_{-j}^T\boldsymbol{y}$, follow a homoskedastic Gaussian linear model with coefficient vector $\boldsymbol{\beta}$ and a $(d-1) \times d$ design matrix given by $\boldsymbol{W}^{(j)} := (\boldsymbol{X}_{-j}^T\boldsymbol{X}_{-j})^{-1/2}\boldsymbol{X}_{-j}^T\boldsymbol{X}$:

$$\boldsymbol{z}^{(j)} \sim \mathcal{N}\left(\boldsymbol{W}^{(j)}\boldsymbol{\beta}, \sigma^2\boldsymbol{I}_{d-1}\right). \tag{3.2}$$

As $\boldsymbol{z}^{(j)}$ is a function of $\boldsymbol{S}^{(j)}$, a natural way to construct $\mathcal{A}$ to only depend on $\boldsymbol{S}^{(j)}$ is to have it estimate β_j from the above linear model (3.2). Note this neglects the last component of $\boldsymbol{S}^{(j)}$, $\|(\boldsymbol{I}_n - \boldsymbol{P}_{-j})\boldsymbol{y}\|$, from informing $\mathcal{A}$; we justify this decision in Appendix D.

Equation (3.2) reveals a failure mode for any method trying to extract signal about H_j from $\boldsymbol{X}_{-j}^T\boldsymbol{y}$: if $\boldsymbol{X}_j$ is orthogonal to the remaining columns of $\boldsymbol{X}$, then β_j is unidentfiable from $\boldsymbol{X}_{-j}^T\boldsymbol{y}$, and in particular, $\boldsymbol{W}_j^{(j)} = \boldsymbol{0}$. Thus, non-zero inner products between columns are critical for CSW to perform well, as these inner products are how useful information is transmitted to the weights. That said, columns having a zero inner product *in expectation* are not a problem for CSW, and indeed Section 4 shows excellent performance even when the columns of $\boldsymbol{X}$ are all uncorrelated with one another.

In the remainder of this subsection we propose concrete choices of $\mathcal{A}$ that estimate β_j in the linear model (3.2).

3.2.1 Our recommendation: a LASSO-based importance measure

To obtain our recommended importance measure, for $\lambda > 0$, define

$$\hat{\boldsymbol{\beta}}^{\lambda,j} = \arg\min_{\boldsymbol{\beta}} \left(\frac{1}{2(d-1)} \|\boldsymbol{z}^{(j)} - \boldsymbol{W}^{(j)}\boldsymbol{\beta}\|^2 + \lambda \sum_{i \neq j} |\beta_i| \right). \tag{3.3}$$

This is the LASSO estimator applied to the linear model (3.2), except β_j is excluded from the penalty term. We make this modification because regularizing β_j induces a positive probability of $\hat{w}_j$ being zero, in which case H_j cannot be rejected by weighted multiple

testing, no matter how small its p-value p_j is, and indeed we found penalizing β_j to hurt power empirically. λ can be chosen by cross-validation[3], and we denote this choice by $\hat{\lambda}_j$, leading to the natural variable importance measure proposal

$$\mathcal{A}_j^{\text{LASSO}}(\boldsymbol{y}) = \left|\hat{\beta}_j^{\hat{\lambda}_j, j}\right|.$$

Note that while $\hat{\boldsymbol{\beta}}^{\lambda,j}$ for a fixed λ is a function only of $\boldsymbol{S}^{(j)}$, selecting $\hat{\lambda}_j$ by cross-validation in general introduces randomness in the choice of the folds so that $\hat{\boldsymbol{\beta}}^{\hat{\lambda}_j,j}$, and hence $\mathcal{A}_j^{\text{LASSO}}$, does not strictly satisfy our design objective of being exclusively a function of $\boldsymbol{S}^{(j)}$, though we found this extra randomness to make little difference.

Computing $\hat{w}_j$ with $\mathcal{A}_j^{\text{LASSO}}$ requires d cross-validated LASSO fits, and computing $\{\hat{w}_j\}_{j=1}^d$ requires d^2 such fits, which can be computationally expensive when d is large. This cost can be mitigated, however, by the fact that once the samples $\left\{\tilde{\boldsymbol{y}}^{(j)} \overset{H_j}{\sim} \boldsymbol{y} \mid \boldsymbol{S}^{(j)} : 1 \le j \le d\right\}$ have been generated for each hypothesis H_j, the resulting d^2 optimization problems are independent and can be run in parallel. As another way to reduce computational cost, we next propose another importance measure that still leverages sparsity in $\boldsymbol{\beta}$ and can be computed much more efficiently due to shared computation across the d^2 fits.

3.2.2 Faster alternative: Importance measure based on distilled soft-thresholding

Closely related to the LASSO estimator with penalty $\lambda \ge 0$ is the soft-thresholding estimator, defined as applying the soft-thresholding function $\beta \mapsto \text{sign}(\beta)\,(|\beta| - \lambda)_+$ coordinate-wise to the ordinary least squares (OLS) estimate. In fact, under orthogonal design, soft-thresholding matches the LASSO exactly. Because $\boldsymbol{W}^{(j)}$ has more columns than rows, the OLS estimate for the linear model (3.2) is not well-defined[4], so we instead consider a distilled [Liu et al., 2021] version of soft-thresholding as follows. We first compute the soft-thresholded estimator $\hat{\boldsymbol{\beta}}_{\text{ST}}^{\lambda,-j}$ from $\boldsymbol{z}^{(j)}$ and $\boldsymbol{W}_{-j}^{(j)}$ (which, unlike $\boldsymbol{W}^{(j)}$, is invertible). The importance measure is then the magnitude of the univariate OLS coefficient of regressing the residual $\boldsymbol{r}_j = \boldsymbol{z}^{(j)} - \boldsymbol{W}_{-j}^{(j)}\hat{\boldsymbol{\beta}}_{\text{ST}}^{\lambda,-j}$ on $\boldsymbol{W}_j^{(j)}$, and is given by $\frac{\left|\boldsymbol{r}_j^T \boldsymbol{W}_j^{(j)}\right|}{\|\boldsymbol{W}_j^{(j)}\|^2}$. The intuition is as follows: if the jth variable is *important*, then $\boldsymbol{W}_{-j}^{(j)}$ alone cannot explain the signal in $\boldsymbol{z}^{(j)}$, so the residual $\boldsymbol{r}_j$ retains signal attributable to $\boldsymbol{W}_j^{(j)}$ and should be correlated with it.

It remains to choose the threshold λ. Since the procedure can ultimately be viewed as a prediction problem for the residual $\boldsymbol{r}_j$ from $\boldsymbol{W}_j^{(j)}$, we select λ by cross-validation[3], minimizing the prediction mean-squared-error of this problem; further details are given in Appendix E.2.

[3] We use a 10-fold cross-validation by default.

[4] Even if we chose a well-defined estimate like ridgeless least squares to soft-threshold, we would still want to avoid soft-thresholding the jth coordinate for the same reason we do not penalize it in Equation (3.3). But then the variable importance measure that only takes the jth coordinate of the estimate would be entirely unaffected by the soft-thresholding and thus in no way leverage its sparse bias.

Denoting the resulting choice by $\hat{\lambda}_j^{\mathrm{ST}}$, the distilled soft-thresholded importance measure is

$$\mathcal{A}_j^{\mathrm{dST}}(\boldsymbol{y}) = \frac{\left| \left(\boldsymbol{z}^{(j)} - \boldsymbol{W}_{-j}^{(j)} \hat{\boldsymbol{\beta}}_{\mathrm{ST}}^{\hat{\lambda}_j^{\mathrm{ST}},-j} \right)^T \boldsymbol{W}_j^{(j)} \right|}{\|\boldsymbol{W}_j^{(j)}\|^2}.$$

Note that evaluating $\mathcal{A}_j^{\mathrm{dST}}(\boldsymbol{y})$ for all j requires computing $\boldsymbol{W}^{(j)}$ together with the OLS fit of $\boldsymbol{z}^{(j)}$ on $\boldsymbol{W}_{-j}^{(j)}$ for every j. In Appendix E.1 we show that this can be accelerated substantially via faster matrix updates, avoiding a separate matrix inverse and square root for each j. Similar speed-ups apply to the cross-validation step, for the sub-matrices restricted to the training and testing folds. Indeed, we found $\mathcal{A}^{\mathrm{dST}}$ to be at least 20 times as fast as $\mathcal{A}^{\mathrm{LASSO}}$ even for $d = 100$; see Section 4.3.

3.2.3 Other importance measures

The set of possible importance measures one could consider is enormous, but we have focused on simple and natural choices based on sparse coefficient estimators in the linear model. Even just for these choices, one may wonder why it is best to take the magnitude of the coefficient estimate, as opposed to, say, its square, or square root, or any other monotone transformation, any of which generally results in a different CSW. We explore the power of $\left(\mathcal{A}_j^{\mathrm{LASSO}}\right)^m$ for various choices of exponentiation m in Appendix H.2, finding no single choice dominates the others. We found that higher exponents tend to have relatively higher power in sparser settings and relatively lower power in denser settings, and overall we felt that using $\mathcal{A}^{\mathrm{LASSO}}$ (i.e., $m = 1$) provides a good balance of power gain in sparse settings and limited power loss in dense settings, compared to unweighted multiple testing.

3.3 Putting it all together: New multiple testing methods

3.3.1 FWER control

Our recommended procedure to leverage our proposed weights is a weighted version of Holm's procedure and is summarized in Algorithm 1 below. We state its validity in Theorem 3.2 and prove it in Appendix F.1.

Theorem 3.2. *Suppose the p-values in the input of Algorithm 1 satisfy Assumption 3.1. Then for any $\alpha \in (0, 1)$, Algorithm 1 controls the FWER at level α.*

The proof in Appendix F.1 relies on the fact that Algorithm 1 is the sequential representation of the closure of the weighted I-Bonferroni test, discussed in Section 2.1, with weights

$$\hat{w}_j^{(I)} = \frac{\mathcal{A}_j(\tilde{\boldsymbol{y}}^{(j)})}{\sum_{i \in I} \mathcal{A}_i(\tilde{\boldsymbol{y}}^{(j)})}, \text{ where } \tilde{\boldsymbol{y}}^{(j)} \overset{H_j}{\sim} \boldsymbol{y} \mid \boldsymbol{S}^{(j)}, \qquad \forall j \in I, \tag{3.4}$$

that satisfy Assumption 2.1. Recall from the discussion in Section 2.1 that it is generally unclear how to construct I-specific weights satisfying Assumption 2.1 from global weights

Algorithm 1: CSW-Holm

Input: p-values: p_j for $H_j, \forall j \in [1:d]$, Importance measure: $\mathcal{A}$, Nominal FWER: α.
Output: A subset of $[1:d]$.
Initialize $\hat{R} \leftarrow \emptyset$
Initialize *counter*
Sample $\tilde{\boldsymbol{y}}^{(j)} \overset{H_j}{\sim} \boldsymbol{y} \mid \boldsymbol{S}^{(j)}$, for all $j \in [1:d]$.
for *counter* *in* 1 *to* d **do**
    Define $\hat{w}_j^{(\hat{R}^c)} = \frac{\mathcal{A}_j(\tilde{\boldsymbol{y}}^{(j)})}{\sum_{i \notin \hat{R}} \mathcal{A}_i(\tilde{\boldsymbol{y}}^{(j)})}, \forall j \notin \hat{R}$.
    $j^* \leftarrow \arg\min_{j \notin \hat{R}} \frac{p_j}{\hat{w}_j^{(\hat{R}^c)}}$.
    if $p_{j^*}/\hat{w}_{j^*}^{(\hat{R}^c)} \leq \alpha$ **then**
        $\hat{R} \leftarrow \hat{R} \cup \{j^*\}$
    end
    else
        Stop
    end
end
Return $\hat{R}$.

satisfying Assumption 2.1, but the CSW construction admits a natural way to do so. Indeed, Algorithm 1 is not simply a CSW version of an existing weighted Holm's procedure, since our weights are not e-values and $\hat{w}_j^{(I)}$ above is not simply an I-renormalization of the global weight $\hat{w}_j$.

3.3.2 FDR control

As described in Section 2, our recommended procedure for FDR control in the linear model, following Genovese et al. [2006], applies the Benjamini–Hochberg procedure [Benjamini and Hochberg, 1995] to the scaled p-values $\left\{\frac{p_j}{d\hat{w}_j}\right\}_{j=1}^{d}$, where the weights $\hat{w}_j$ are as given in Equation (3.1). We call this resulting procedure, the Co-sufficient Weighted Benjamini–Hochberg (CSW-BH) procedure.

As noted previously, like the BH procedure applied to linear model p-values, this procedure lacks strict FDR control but exhibits robust empirical FDR control; we present empirical evidence for this in Section 4.2.1 and, as discussed in Section 2.2, attribute it to CSW weights satisfying Assumption 2.1. One could also use conditional calibration (discussed earlier in Section 2.2) to obtain an exact FDR-controlling procedure that leverages these weights. However, we find this approach to be computationally prohibitive; see Appendix B for details.

3.3.3 The choice of p-values

Although this paper primarily focuses on obtaining informative weights for the linear model, the power of the resulting weighted multiple testing procedures also depends on the choice of p-values for testing $\{H_j\}_{j=1}^d$, the canonical choice for which is the usual t-test p-values. However, we recommend instead ℓ-test p-values [Sengupta and Janson, 2024], which leverage sparsity in β (in a conceptually similar but operationally very different way to co-sufficient weighting) to improve over the t-test while still satisfying Assumption 3.1. In particular, our implementation in Sections 4–5 of both the CSW procedures described above uses ℓ-test p-values.

4 Experiments

We consider an $n \times d$ design matrix $\boldsymbol{X}$ whose rows are drawn from $\mathcal{N}_d(\mathbf{0}, \boldsymbol{\Sigma}(\rho))$ and columns are normalized, where $\boldsymbol{\Sigma}(\rho)$ has the Toeplitz structure $\boldsymbol{\Sigma}(\rho)_{ij} = \rho^{|i-j|}$. The coefficient vector $\boldsymbol{\beta}$ is zero except at k randomly chosen signal indices, each assigned a non-zero value (the mechanisms for which are described below). The response is generated as $\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{X}\boldsymbol{\beta}, \boldsymbol{I}_n)$. We vary four aspects of this setting:

- **Inter-variable correlation:** we take $\rho \in \{0, 0.5, 0.9\}$, corresponding to covariate independence, medium correlation, and high correlation, respectively.
- **Aspect ratio:** we set n/d to either 1.5 or 3.
- **Signal generation:** we draw the signal entries of $\boldsymbol{\beta}$ either from Unif$\{+A, -A\}$ (discrete) or from $\mathcal{N}(0, A^2)$ (continuous); the amplitude A is specified below.
- **Simulation scale:** we consider small-scale and large-scale simulations, corresponding to $d = 100$ and $d = 1000$ respectively.

Each of our simulation setting involves varying one parameter while holding the others fixed, according to one of the following three regimes: (a) fixing the amplitude A and varying the sparsity k; (b) fixing $k/d = 5\%$ (high sparsity) and varying A; and (c) fixing $k/d = 50\%$ (low sparsity) and varying A. The only exception is an additional MCC setting considered for $d = 1000$, which we describe shortly. Unless otherwise specified, we use nominal FWER and FDR levels of 0.05 and 0.1, respectively.

For $d = 100$, we consider all combinations of the choices of ρ, n/d, signal generation, and simulation regimes ((a), (b), or (c)) described above. For $d = 1000$, however, we restrict attention to a subset of settings, primarily to reduce computational cost. Specifically, we fix $n/d = 3$ (since the knockoff-based methods are not applicable when $n/d = 1.5$), take $\rho \in \{0, 0.9\}$, and consider regimes (a) and (b) for each choice of ρ. In the $d = 100$ simulations, these four settings capture the broad performance trends exhibited by our methods across the remaining settings. We therefore focus on these settings for $d = 1000$ as well, with the expectation that they remain representative of the broader performance patterns. Additionally, we consider a setting in which the design matrix follows the MCC design of Luo et al. [2022]. This setting exhibits the whiteout phenomenon [Li and Fithian, 2021], under

which the knockoff+ procedure performs poorly, and serves as a key example demonstrating the advantage of the cknockoff procedure. This setting features $n = 3000$, $d = 1000$, and $k = 10$, and compares the methods across a range of nominal FDR levels; for further details, see [Luo et al., 2022, Section 5.1]. Altogether, we consider 40 simulation settings for the FWER-controlling methods and 41 settings for the FDR-controlling methods (each setting varies one data-generating aspect, so the total number of distinct data-generating distributions is around 200). Throughout, we report empirical error rates (FWER and FDR) and average power averaged over 500 independent replicates for $d = 100$ and 200 independent replicates for $d = 1000$.

Next, we list the methods that we compare. For FWER control, we consider four procedures: CSW-Holm using (1) $\mathcal{A}^{\text{LASSO}}$ ($\text{CSW}_{\text{LASSO}}$–Holm) and (2) $\mathcal{A}^{\text{dST}}$ (CSW_{dST}–Holm), both applied to the ℓ-test p-values; and Holm's procedure applied to the (3) ℓ-test (ℓ-Holm) and (4) t-test (t-Holm) p-values. For FDR control, we compare 11 procedures: CSW-BH using (1) $\mathcal{A}^{\text{LASSO}}$ ($\text{CSW}_{\text{LASSO}}$–BH) and (2) $\mathcal{A}^{\text{dST}}$ (CSW_{dST}–BH), both applied to the ℓ-test p-values; (3) knockoffs and (4) knockoffs+ [Barber and Candès, 2015]; (5) cknockoff and (6) cknockoff* [Luo et al., 2022]; the BH procedure applied to the (7) ℓ-test (ℓ-BH) and (8) t-test (t-BH) p-values; (9) Gaussian mirrors (GM) [Xing et al., 2023]; (10) multiple data-splitting (MDS) [Dai et al., 2022]; and (11) OATK [Guan et al., 2025]. For the $d = 1000$ simulations, we include only CSW_{dST}–Holm and CSW_{dST}–BH among the CSW procedures due to their greater computational efficiency.

The full set of results for all methods across all the settings described above is deferred to Appendix H.1; we present a selected subset in the main text. In particular, we fix $n/d = 3$ (and also additionally consider the $n/d = 1.5$ case for FDR simulations for reasons described in Section 4.2.2 below) and focus on regime (a) to understand the effect of varying the sparsity level k. Note that regimes (b) and (c) zoom in on two specific levels of sparsity and, as we found, exhibit trends already seen at the corresponding sparsity levels in regime (a). We also focus only on discrete signal generation, as the continuous signal regime exhibits similar trends.

4.1 FWER controlling simulations

We present the select set of power curves of the various FWER-controlling methods described above in Figure 1. Because all the methods provably control the FWER, we do not additionally present their empirical FWER here; these results are available in Appendix H.1. For the $d = 100$ simulations, we find that $\text{CSW}_{\text{LASSO}}$–Holm and CSW_{dST}–Holm outperform the unweighted procedures in the sparse regime. In particular, as shown in the top-rightmost panel of Figure 1, $\text{CSW}_{\text{LASSO}}$–Holm and CSW_{dST}–Holm achieve up to 54% and 94% higher power, respectively, than the best-performing non-CSW procedure. As k increases, the gap between the CSW power curves narrows, with $\text{CSW}_{\text{LASSO}}$–Holm performing almost indistinguishably from the unweighted procedures in the denser regimes, while CSW_{dST}–Holm slightly underperforms. This behavior of CSW_{dST}–Holm also carries over to the $d = 1000$ regime.

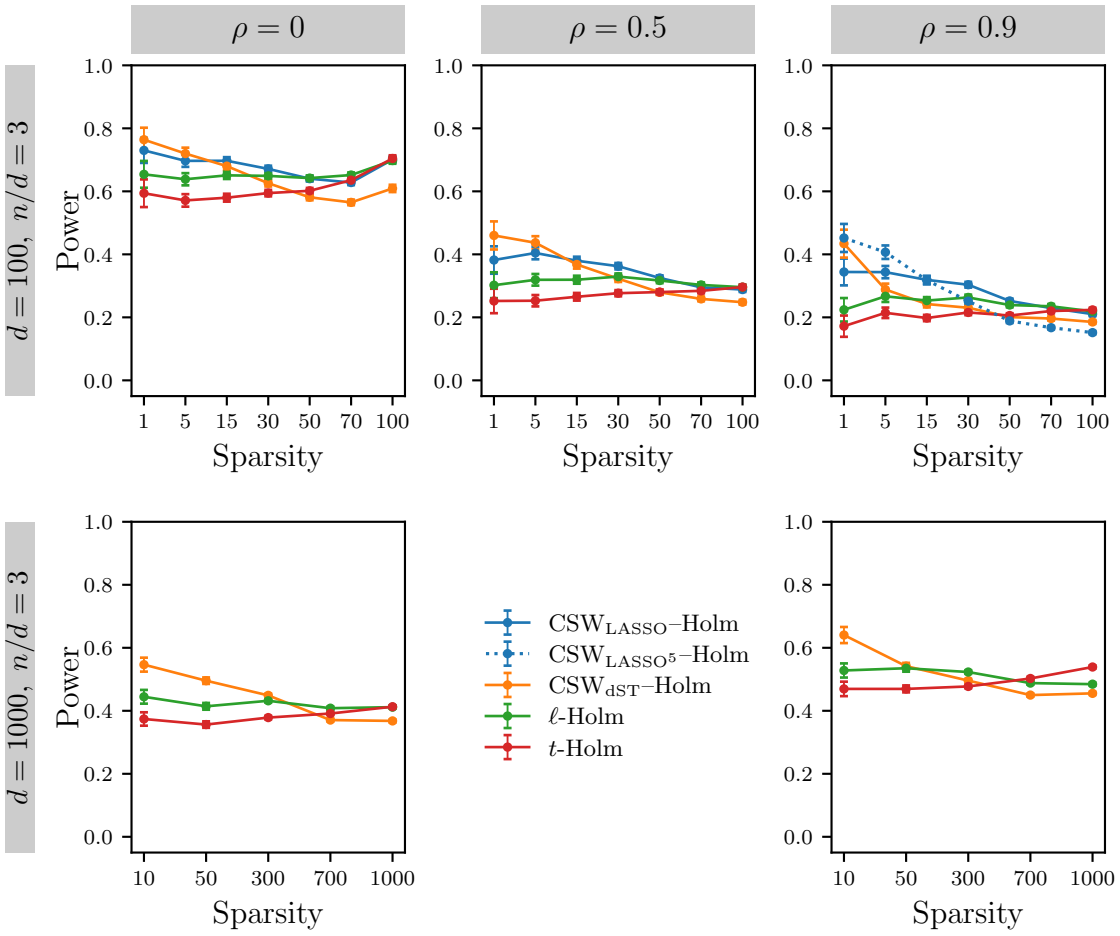


Figure 1: The power of various FWER controlling procedures under simulation regime (a), discrete signal generation and $n/d = 3$. The choice of the amplitude A is described in Appendix H.1. The error bars represent plus or minus two standard errors.

4.2 FDR controlling simulations

Among the FDR-controlling methods, we retain only $\text{CSW}_{\text{LASSO}}$–BH, CSW_{dST}–BH, knockoffs+, cknockoff* (which uniformly dominates cknockoff), ℓ-BH, and t-BH for our main text plots. We exclude GM, OATK, knockoffs, and MDS from the main text plots because we found them to exhibit poor empirical FDR control in a number of settings. For example, GM's empirical FDR reaches 0.67 in the top-middle panel of Figure 9, MDS's empirical FDR reaches 0.20 in the bottom-right panel of Figure 25, and OATK's and knockoffs' empirical powers in Figure 11 reach 0.40 and 0.37, respectively; recall the nominal FDR is 0.1, so all these violations are at least 100% above the nominal level. Notably, all of these methods exhibit their worst FDR violations in sparse settings, which is our primary motivating regime, and arguably, is widely encountered in scientific problems where variable selection is of interest. Additionally, in many of the sparse regimes where MDS does empirically control the FDR, its power is nearly zero even while other methods that control FDR achieve powers ≥ 0.5 in the same settings; see, e.g., Figure 10.

4.2.1 Summary of empirical FDRs

Figure 2 summarizes the empirical FDR of $\text{CSW}_{\text{LASSO}}$–BH and CSW_{dST}–BH across all the data-generating processes we simulated, and we find that the empirical FDRs of both methods are within Monte Carlo error of being bounded by the nominal FDR level. The figure excludes only the simulations from the MCC design corresponding to nominal FDR levels different from 0.1; these results are available in Appendix Figure 29, where we likewise observe empirical FDR control.

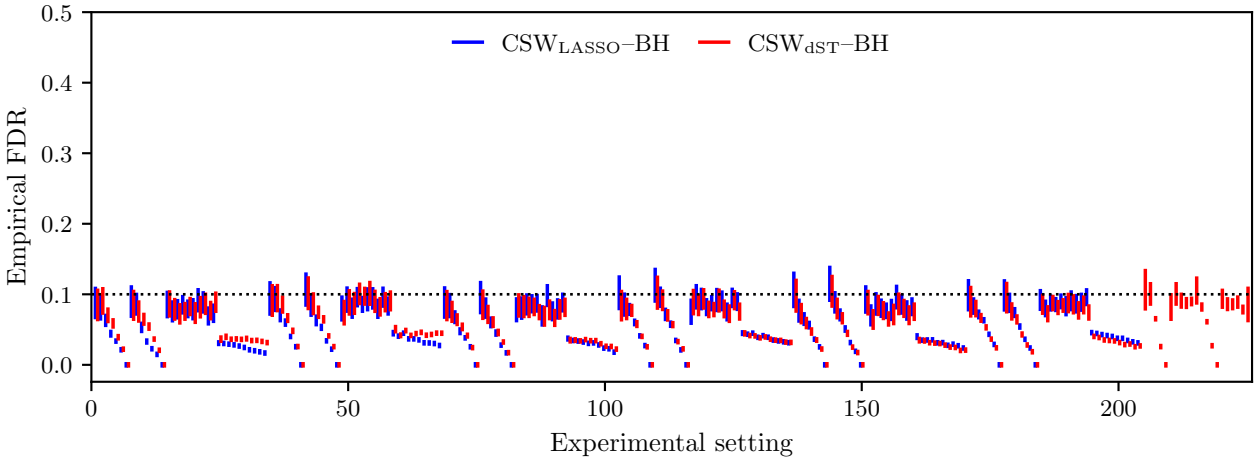


Figure 2: Empirical FDRs of $\text{CSW}_{\text{LASSO}}$–BH and CSW_{dST}–BH across all the simulations considered in this paper. The x-axis represents an enumeration of the different experimental settings with nominal FDR 0.1 and the settings corresponding to the large-scale regime with $d = 1000$ appear on the rightmost end. The error bars represent plus or minus two standard errors. The horizontal black dotted line represents the nominal FDR level of 0.1.

4.2.2 Powers

Turning to the power of FDR-controlling procedures, Figure 3 considers the same settings as Figure 1 plus the simulation settings with $n/d = 1.5$ for $d = 100$ (where the knockoff-based procedures are not applicable) and the MCC design in the $d = 1000$ regime. The first two rows of Figure 3 show that, for $d = 100$, $\text{CSW}_{\text{LASSO}}$–BH outperforms nearly all non-CSW procedures across the regimes considered, achieving up to 51% higher power than the best-performing non-CSW procedure, as highlighted in the rightmost panel of the middle row. There are, however, two exceptions. First, when the coefficient vector is dense ($\geq 50\%$), $\text{CSW}_{\text{LASSO}}$–BH is slightly outperformed by the unweighted BH procedures. Second, in the sparsest regime with $\rho = 0.9$ (upper-left panel), cknockoff* slightly outperforms $\text{CSW}_{\text{LASSO}}$–BH. However, this does not appear to represent a fundamental weakness of the CSW approach, as the same behavior is not observed for CSW_{dST}–BH. We also include a curve for $\text{CSW}_{\text{LASSO}^5}$–BH (formed by using $\mathcal{A}^5_{\text{LASSO}}$ as the importance measure, as discussed in Section 3.2.3), which shows that even a slight variation of $\text{CSW}_{\text{LASSO}}$–BH can close this gap. For comparison, we also include a curve for cknockoff* using a LASSO^5 variable importance measure, showing that this modification does not meaningfully change the performance

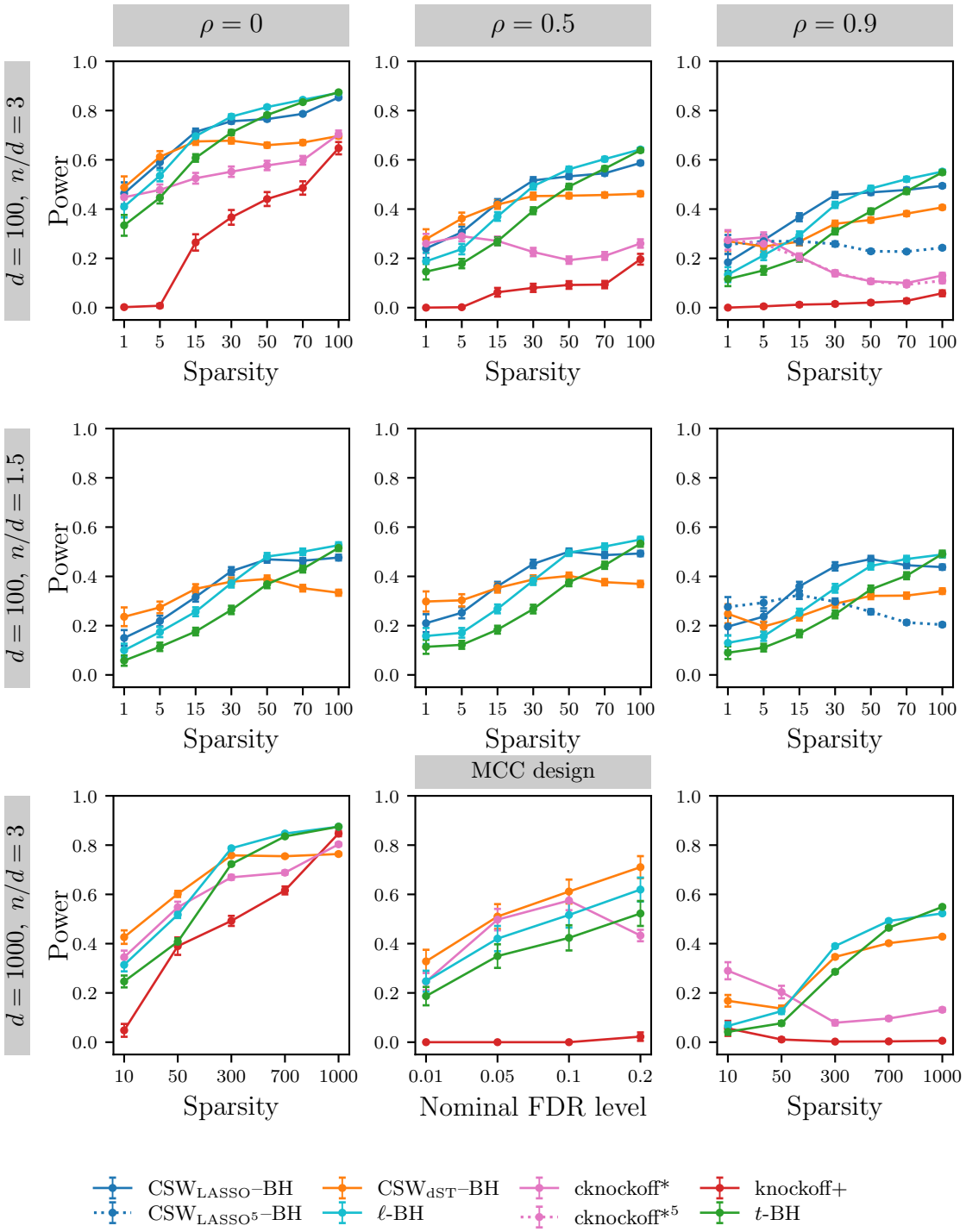


Figure 3: The power of various FDR controlling procedures under simulation regime (a) and discrete signal generation. The choice of the amplitude A is described in Appendix H.1. The error bars represent plus or minus two standard errors.

of cknockoff*.

The takeaway for CSW$_{\text{dST}}$–BH throughout Figure 3 is similar to that for CSW$_{\text{LASSO}}$–BH

except CSW_{dST}–BH seems to have relatively higher power than $\text{CSW}_{\text{LASSO}}$–BH in sparse regimes and relatively lower power in dense regimes. Indeed CSW_{dST}–BH can achieve more than twice the power of the best performing non-CSW method in the sparsest setting of the leftmost panel in the middle row. In the bottom-right panel of Figure 3, however, CSW_{dST}–BH is moderately outperformed in sparse settings by cknockoff*, though in the denser parts of that panel, CSW_{dST}–BH outperforms cknockoff* by a wide margin. In the MCC design of Luo et al. [2022], CSW_{dST}–BH outperforms all other methods including Luo et al. [2022]'s method cknockoff*.

4.3 Runtime comparison

To understand the computational efficiency of the procedures based on $\mathcal{A}^{\text{dST}}$ over $\mathcal{A}^{\text{LASSO}}$, we recorded the runtimes of CSW_{dST}–Holm and $\text{CSW}_{\text{LASSO}}$–Holm for the simulations in the top-left panel of Figure 1, averaging them across replicates for each sparsity level. We found that, across all sparsity levels considered, constructing weights by normalizing $\mathcal{A}^{\text{dST}}$ is at least 20 times faster than when using $\mathcal{A}^{\text{LASSO}}$. Although it was not computationally feasible to use $\mathcal{A}^{\text{LASSO}}$ in the large-scale simulations with $d = 1000$, we expect the computational advantage of $\mathcal{A}^{\text{dST}}$ to be even more pronounced in this setting.

5 Application to the HIV drug-resistance dataset

The HIV drug resistance dataset [Rhee et al., 2006] contains data corresponding to 16 regressions whose covariates are the presence of certain genetic mutations and whose outcomes are a measure of resistance to a particular drug. For each of the 16 regressions we follow the same pre-processing steps as Barber and Candès [2015] and compare our CSW methods with a selection of the methods considered in our simulations, informed by those simulations. In particular, in our FDR simulations, cknockoff* is the most powerful among the knockoffs-based procedures that exhibit consistent empirical FDR control, ℓ-BH is at least as powerful as t-BH throughout, and OATK, MDS, and GM did not exhibit consisten empirical FDR control, so for FDR control we compare $\text{CSW}_{\text{LASSO}}$–BH and CSW_{dST}–BH with only cknockoff* and ℓ-BH, at a nominal FDR level of 0.1. Similarly, for FWER control we compare $\text{CSW}_{\text{LASSO}}$–Holm and CSW_{dST}–Holm with only ℓ-Holm, at a nominal FWER level of 0.05.

Previous analyses [Barber and Candès, 2015, Fithian and Lei, 2022] have aggregated the discoveries to a gene-level and also compared the discovered set of genes against a set of mutations identified in the treatment-selected mutations (TSM) panel of Rhee et al. [2006], and we refer the reader to [Barber and Candès, 2015, Section 4] for further details. In Table 1 we tabulate the fraction of columns discovered by each of the methods across all 16 regressions, as well as the fraction of gene-level discoveries. In Figures 5 and 6 of the appendix, we present the number of genes present in the TSM list discovered by each of the methods for each of the drugs. For the FDR controlling procedures we generally find that cknockoff* produces far fewer discoveries than the other three methods, which perform similarly. For FWER control, all three procedures also perform similarly.

Thus, in this application where we expect sparsity to hold, our weighted procedures yield

Method	% of raw discoveries	% of gene-level discoveries
$\text{CSW}_{\text{LASSO}}$–BH (FDR)	11.62%	21.87%
CSW_{dST}–BH (FDR)	10.99%	20.94%
cknockoff*	4.83%	10.79%
ℓ-BH	12.01%	22.87%
$\text{CSW}_{\text{LASSO}}$–Holm (FWER)	6.90%	13.15%
CSW_{dST}–Holm (FWER)	6.87%	13.22%
ℓ-Holm	6.52%	12.51%

Table 1: Percentage of total columns (raw discoveries) and genes discovered by all the methods across all the 16 regressions.

a similar number of rejections as their unweighted counterparts. We argue that a primary reason for this behavior is that these regressions contain several strong signals but relatively few discoverable weaker ones. The standard nominal FDR and FWER levels that we use are sufficiently high that both weighted and unweighted methods discover the strong potential signals, leaving any differences in performance to be driven primarily by the weaker signals. Since the quality of our weights also depends on signal strength, the weighted procedures offer little advantage over their unweighted counterparts in identifying these weaker potential signals at the nominal FWER and FDR levels of 0.05 and 0.1, respectively.

To illustrate this phenomenon better and highlight the benefits of weighting, in Figure 4, we consider smaller nominal FWER and FDR levels and plot, for each of the 16 regressions, the ratio of the number of discoveries made by our corresponding weighted procedures to those made by their unweighted counterparts. We observe that all 16 curves in both panels are heavily biased above 1, often substantially so, particularly at smaller nominal error rates. At these more stringent levels, even the stronger potential signals become harder to discover, and weighting offers a clear advantage—our procedures often yield more discoveries, sometimes as many as twice those of their unweighted counterparts. We complement Figure 4 with Appendix G.2, where we present figures showing the exact numbers of discoveries made by these methods across varying nominal error rates. These figures show the benefits of weighting are real but can be more or less pronounced depending on the distribution of signal strengths relative to the nominal level of error control.

6 Discussion

Co-sufficient weighting uses, for each j, the sufficient statistic under H_j to leverage sparsity in $\boldsymbol{\beta}$ and construct a weight w_j. These weights can then be fed into downstream multiple testing procedures that outperform existing methods in their respective error-control regimes across a wide range of settings with sparse $\boldsymbol{\beta}$, while suffering at most only a slight sacrifice in power in dense regimes. We discussed two choices for obtaining these weights by normalizing either our recommended LASSO-based or a computationally faster distilled soft-thresholding variable importance measure.

We hope future work can address two limitations of the CSW approach: First, $\text{CSW}_{\text{LASSO}}$–BH and $\text{CSW}_{\text{LASSO}}$–Holm could benefit substantially from a faster implementation for obtaining

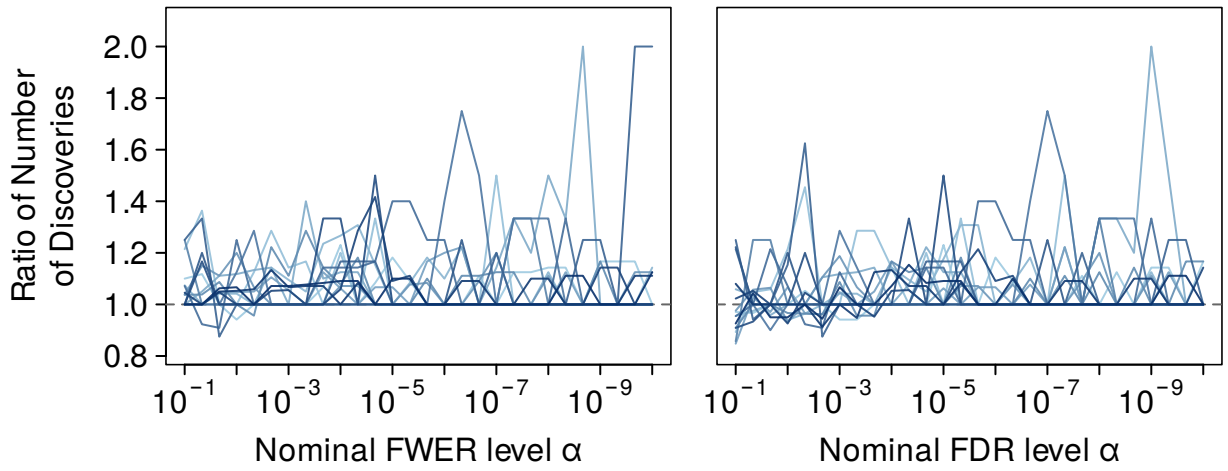


Figure 4: The ratio of number of discoveries of $\text{CSW}_{\text{LASSO}}$–Holm to that of ℓ-Holm (left) and that of $\text{CSW}_{\text{LASSO}}$–BH to that of ℓ-BH (right) across various levels of nominal FWER (left) and FDR (right) for all the 16 regressions from the HIV drug-resistance dataset.

all the weights. Part of this is implementational, and one could use customized warm starts and parallelization to perform the LASSO runs required to obtain the weights more efficiently. Second, BH (with or without weights) is known to be conservative in dense regimes, but we did not explore integrating CSW into a weighted analogue of Storey–BH [Storey et al., 2004] to address this. At a higher level, our work suggests the possibility of exploiting other forms of structural information, such as smoothness in $\boldsymbol{\beta}$, to improve multiple testing.

Stepping back further, this work highlights how sufficient statistics can be used to construct weights that leverage structural information, such as sparsity, that may otherwise be difficult to quantify. Our work thus belongs to a broader line of research exploiting sparsity to improve inference in the linear model [Sengupta and Janson, 2024, Paulson et al., 2025] without sacrificing validity. Despite addressing different problems, these methods share a fundamental principle: they exploit the independence of p_j and $\boldsymbol{S}^{(j)}$ under H_j to leverage information about the signal in $\boldsymbol{S}^{(j)}$, thereby improving p-value based inference using this information. In fact, our recommended p-values are the sparsity-leveraging p-values of [Sengupta and Janson, 2024], whose benefits compound with ours, together yielding even more substantial power gains over conventional multiple testing procedures than either alone.

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A Counter-example for FDR violation of the weighted BH procedure

Consider d independent uniform null p-values $\{p_j\}_{j=1}^d$, and choose weights $w_j = \frac{p_j}{\alpha}\mathbb{I}(p_j \leq \sqrt{2\alpha/d})$. These weights satisfy Assumption 2.1(b), since

$$\mathbb{E}[w_j] = \frac{1}{\alpha}\int_0^{\sqrt{2\alpha/d}} p\,dp = \frac{1}{\alpha} \times \frac{2\alpha}{2d} = \frac{1}{d},$$

and hence, $\mathbb{E}\left[\sum_{j=1}^d w_j\right] = 1$. Furthermore, $\{(w_j, p_j)\}_{j=1}^d$ satisfy Assumption 2.2(a) vacuously, since there are no non-null hypotheses. Assumption 2.2(b) is also satisfied, since each w_j depends only on p_j, and hence, the pairs $\{(w_j, p_j)\}_{j=1}^d$ are independent across j.

Now, for any j such that $p_j \leq \sqrt{2\alpha/d}$, we have $p_j/(dw_j) = \alpha/d$, which is the smallest possible cutoff of the BH procedure. Hence, any such hypothesis is rejected by the weighted BH procedure. Since all hypotheses are null, the FDR is therefore

$$\mathbb{P}\left(\exists j : p_j \leq \sqrt{\frac{2\alpha}{d}}\right) = 1 - \left(1 - \sqrt{\frac{2\alpha}{d}}\right)^d \approx 1 - \exp\left(-\sqrt{2\alpha d}\right) \to 1, \quad d \to \infty.$$

Thus, for sufficiently large d, the FDR of the weighted BH procedure can be made arbitrarily close to 1, despite the weights satisfying Assumption 2.1(b) and Assumption 2.2.

B Weighted dBH using co-sufficient weights

The dependence-adjusted Benjamini–Hochberg (dBH) procedure of Fithian and Lei [2022] modifies the usual BH procedure to guarantee theoretical FDR control under arbitrary p-value dependence, and in particular applies to the linear model. The vanilla BH procedure [Benjamini and Hochberg, 1995] at nominal FDR level α finds the largest index $\hat{i}$ such that $p_{(\hat{i})} \leq \frac{\alpha\hat{i}}{d}$, and rejects any hypothesis H_j with $p_j \leq \hat{\tau}$, where $\hat{\tau} = \frac{\alpha\hat{i}}{d}$. The dBH procedure instead assigns each H_j its own cutoff $\hat{\tau}_j$, chosen so that its *FDR-contribution* is controlled at α/d:

$$\mathbb{E}_{H_j}\left[\frac{\mathbb{I}(p_j \leq \hat{\tau}_j)}{\hat{R}_j}\,\middle|\,\boldsymbol{S}^{(j)}\right] \leq \frac{\alpha}{d},$$

where $\hat{R}_j$ is an estimate of $|\mathcal{R} \cup \{j\}|$ and $\mathcal{R}$ is the final rejection set. One plausible choice calibrates against a base BH procedure by setting $\hat{R}_j = |\hat{\mathcal{R}}^{\mathrm{BH}(\gamma\alpha)} \cup \{j\}|$, where $\hat{\mathcal{R}}^{\mathrm{BH}(\gamma\alpha)}$ is

the rejection set of BH applied to the p-values at level $\gamma\alpha$; the authors use $\gamma = 0.9$ under arbitrary dependence. A hypothesis H_j is then rejected if it passes two steps: (1) $p_j \le \hat{\tau}_j$, and (2) a randomized pruning step applied to the hypotheses rejected in step (1). The pruning step depends only on the $\hat{R}_j$ of the step-(1) rejections and on externally generated noise, and it can only shrink the step-(1) rejection set (and hence is detrimental to the power); it is invoked only when the $\hat{R}_j$ of a rejected index j fails to lower bound the eventual $|\mathcal{R} \cup \{j\}|$.

In [Fithian and Lei, 2022, Section 7.3], the authors note that a weighted variant of dBH is also valid: instead of controlling the above quantity at α/d, one controls it at $\alpha\tilde{w}_j$, where $\tilde{w}_j$ is a function of $\boldsymbol{S}^{(j)}$ satisfying $\mathbb{E}[\sum_{j\in\mathcal{H}_0} \tilde{w}_j] \le 1$ and $\mathcal{H}_0$ is the set of null indices. This makes our weights $\{\hat{w}_j\}_{j=1}^d$ an excellent candidate for use with dBH in the linear model, where an informative choice of weights should boost power just as it did for the weighted Bonferroni procedure. For completeness, we state the validity of the resulting procedure in the following theorem, whose proof follows from [Fithian and Lei, 2022, Theorem 2 and Remark 2.1]

Theorem B.1. *Suppose p-values $\{p_j\}_{j=1}^d$ satisfy Assumption 3.1. Then, for all $\alpha \in (0,1)$, the weighted dBH procedure described above, run at nominal FDR level α with weights $\{\hat{w}_j\}_{j=1}^d$, controls the FDR at level α.*

As with dBH, the weighted procedure must calibrate its underlying base weighted BH procedure; we can use $\hat{R}_j = |\hat{\mathcal{R}}^{\mathrm{BH}(\hat{\boldsymbol{w}};\gamma\alpha)} \cup \{j\}|$, where $\hat{\mathcal{R}}^{\mathrm{BH}(\hat{\boldsymbol{w}};\gamma\alpha)}$ is the rejection set of the weighted BH procedure at nominal FDR $\gamma\alpha$ with weight $\hat{w}_j$ for H_j. To leverage sparsity, a further natural choice is to take $\{p_j\}_{j=1}^d$ to be the ℓ-test p-values. Under these choices, the cutoff calibration takes the form

$$\mathbb{E}_{H_j}\left[\frac{\mathbb{I}(p_j/(dw_j) \le \hat{\tau}_j)}{\hat{R}_j}\,\middle|\,\boldsymbol{S}^{(j)}\right] \le \alpha\hat{w}_j. \tag{B.1}$$

In practice, the left-hand side has no closed form and must be evaluated by Monte Carlo sampling. Calibrating $\hat{\tau}_j$ to reasonable accuracy requires estimating this expectation quite precisely, hence many Monte Carlo draws, and recomputing the weights for each draw is computationally very expensive. The original paper Fithian and Lei [2022] proposes homotopy algorithms for evaluating this expectation and calibrating the base BH procedure with t-test p-values, but these do not extend to the weighted case.[5] We therefore leave a computationally feasible version of this weighted analog to future work.

C Empirical exploration of weights variability

To empirically assess the variability of our weights, we conduct the following experiment. For a given response $\boldsymbol{y}$, design matrix $\boldsymbol{X}$, and signal index j, we assess the conditional variability of $\hat{w}_j$ by generating M realizations of the weight defined in (3.1) with $B = 1$, using M conditionally independent draws of $\tilde{\boldsymbol{y}}^{(j)}$ from the conditional distribution of $\boldsymbol{y} \mid \boldsymbol{S}^{(j)}$ under H_j. Denote these realizations by $\{\hat{w}_j^{(1)}, \cdots, \hat{w}_j^{(M)}\}$. We then estimate the coefficient

[5] One can calibrate the base BH procedure to obtain $\hat{R}_j$, while retaining the weights only on the right-hand side of Equation B.1; we found, however, that this variant frequently triggers the randomized pruning step, significantly affecting its power.

of variation by taking the ratio of the sample standard deviation to the sample mean of these weights, denoting the resulting quantity by $\omega_{j,M}$, which can then be averaged across independently generated data sets.

We estimate this measure under four selected configurations from Section 4: Setting (a) with $n = 300$, $d = 100$, discrete signal generation, $k \in \{1, 5\}$, and $\rho \in \{0, 0.9\}$. Using $M = 50$, we summarize the estimated values of $\omega_{j,M}$ in Table 2. We find low estimated coefficient of variation for our recommended weights, based on $\mathcal{A}^{\text{LASSO}}$, supporting our claim of low variability in Section 3.2.

Although the coefficient of variation is higher for our computationally efficient alternative based on $\mathcal{A}^{\text{dST}}$, we do not investigate this phenomenon further or attempt to modify the procedure. This is because we found that the power benefits of increasing B for this importance measure is at most modest, and non-zero in only the dense regimes. Furthermore, the simulations in Section 4 and Appendix H.1 already demonstrate the power gain of the procedure corresponding to $B = 1$, especially under sparsity, and increasing B would only increase its computational expense.

ρ	k	Estimated $\omega_{j,M}$ using $\mathcal{A}^{\text{LASSO}}$	Estimated $\omega_{j,M}$ using $\mathcal{A}^{\text{dST}}$
0	1	0.06	0.22
0	5	0.05	0.26
0.9	1	0.04	0.1
0.9	5	0.03	0.55

Table 2: Estimates of $\omega_{j,M}$ based on weights formed using both the importance measures $\mathcal{A}^{\text{LASSO}}$ and $\mathcal{A}^{\text{dST}}$. We report the mean estimate from 200 independent replicates of each setting.

D Why $\mathcal{A}_j$ does not depend on the last component of $\boldsymbol{S}^{(j)}$

In Section 3.2, we restricted our variable importance measures to depend only on the first $d-1$ components of the sufficient statistic $\boldsymbol{S}^{(j)} = (\boldsymbol{X}_{-j}^T \boldsymbol{y}, \|(\boldsymbol{I}_n - \boldsymbol{P}_{-j})\boldsymbol{y}\|^2)$. In this section, we provide further justification for excluding the last component.

Recall that our weights are defined as

$$w_j = \mathbb{E}_{H_j}\left[\frac{\mathcal{A}_j(\boldsymbol{y})}{\sum_{i=1}^d \mathcal{A}_i(\boldsymbol{y})}\middle| \boldsymbol{S}^{(j)}\right] = \mathbb{E}_{H_j}\left[\frac{\mathcal{A}_j(\tilde{\boldsymbol{y}}^{(j)})}{\sum_{i=1}^d \mathcal{A}_i(\tilde{\boldsymbol{y}}^{(j)})}\middle| \boldsymbol{S}^{(j)}\right],$$

where $\tilde{\boldsymbol{y}}^{(j)} \sim \boldsymbol{y} \mid \boldsymbol{S}^{(j)}$, under H_j. [Luo et al., 2022, Proposition E.1] gives a closed-form representation of this draw: One can realize $\tilde{\boldsymbol{y}}^{(j)}$ as

$$\tilde{\boldsymbol{y}}^{(j)} = \boldsymbol{P}_{-j}\boldsymbol{y} + \hat{\sigma}_j \boldsymbol{V}^{(j)} \boldsymbol{u}, \tag{D.1}$$

where $\boldsymbol{V}^{(j)}$ is any orthogonal matrix for Col.sp.$(\boldsymbol{X}_{-j})^{\perp}$ and $\boldsymbol{u} \sim \text{Unif}(\mathbb{S}^{n-d})$ is drawn independently of the data, where Col.sp. returns the column space of a matrix. Now, let us

expand our notations and let $\boldsymbol{S}^{(j)}(\boldsymbol{y})$ denote the sufficient statistic under H_j computed from the response $\boldsymbol{y}$. By construction, for every j,

$$\boldsymbol{S}^{(j)}(\tilde{\boldsymbol{y}}^{(j)}) = \boldsymbol{S}^{(j)}(\boldsymbol{y}).$$

For $i \neq j$, however,

$$\boldsymbol{S}^{(i)}(\tilde{\boldsymbol{y}}^{(j)}) = \left(\boldsymbol{X}_{-i}^T \tilde{\boldsymbol{y}}^{(j)}, \|(\boldsymbol{I}_n - \boldsymbol{P}_{-i})\tilde{\boldsymbol{y}}^{(j)}\|^2\right).$$

The following lemma, proved in Appendix F.2, provides a useful representation of the last component.

Lemma D.1. *Define*

$$\boldsymbol{e}_i = \frac{(\boldsymbol{I}_n - \boldsymbol{P}_{-i})\boldsymbol{X}_i}{\|(\boldsymbol{I}_n - \boldsymbol{P}_{-i})\boldsymbol{X}_i\|}, \qquad \rho_{ij} = \boldsymbol{e}_i^T \boldsymbol{e}_j.$$

Suppose $\boldsymbol{V}^{(j)}$ is chosen such that its first column is $\boldsymbol{e}_j$ and its remaining columns form an orthonormal basis for the orthogonal complement of Col.sp.$(\boldsymbol{X})$. *Then,*

$$\|(\boldsymbol{I}_n - \boldsymbol{P}_{-i})\tilde{\boldsymbol{y}}^{(j)}\|^2 = (\boldsymbol{e}_i^T \boldsymbol{P}_{-j}\boldsymbol{y} + \rho_{ij}\hat{\sigma}_j u_1)^2 + (1 - u_1^2)\hat{\sigma}_j^2.$$

We next consider a heuristic approximation of this expression. Since u_1 is the first component of a vector $\boldsymbol{u}$ distributed uniformly on $\mathbb{S}^{n-d}$, it concentrates around zero as $n-d$ grows, so that for an n large enough compared to d, $1 - u_1^2 \approx 1$. Then Lemma D.1 suggests the approximation

$$\|(\boldsymbol{I}_n - \boldsymbol{P}_{-i})\tilde{\boldsymbol{y}}^{(j)}\|^2 \approx (\boldsymbol{e}_i^T \boldsymbol{P}_{-j}\boldsymbol{y} + \rho_{ij}\hat{\sigma}_j u_1)^2 + \hat{\sigma}_j^2.$$

This approximation suggests that the last component of the sufficient statistic for any variable $i \neq j$, evaluated at $\tilde{\boldsymbol{y}}^{(j)}$, approximately consists of the last component of $\boldsymbol{S}^{(j)}$, namely $\hat{\sigma}_j^2$, together with an additional non-negative term. Importantly, this latter term is also influenced by the signal strength β_j.

Thus, the signal associated with β_j can *bleed into* the last component of $\boldsymbol{S}^{(i)}(\tilde{\boldsymbol{y}}^{(j)})$ for other variables $i \neq j$, through a term that may grow with $|\beta_j|$ and is at least as large as the last component of $\boldsymbol{S}^{(j)}$. Consequently, if an importance measure $\mathcal{A}_j$ depends on the last component of the sufficient statistic in such a way that larger values of this component yield greater importance, then a strong signal at j may increase not only $\mathcal{A}_j(\tilde{\boldsymbol{y}}^{(j)})$, but also $\mathcal{A}_i(\tilde{\boldsymbol{y}}^{(j)})$ for $i \neq j$, potentially to an even greater extent because of the additional square term that can also depend on $|\beta_j|$. Since our weights are obtained by normalizing these importance measures, this effect can reduce the relative weight assigned to j, thereby weakening the ability of the weights to distinguish signal variables from null variables. This motivates our choice to exclude the last component of the sufficient statistic when constructing our importance measures.

E Tricks for faster computation

E.1 Faster computation of the co-sufficient weights

We first discuss high-level tricks that speed up the computation of the co-sufficient weights for both $\mathcal{A}^{\text{LASSO}}$ and $\mathcal{A}^{\text{dST}}$. Two things that can be sped up are the computation of $(\boldsymbol{X}_{-j}^T \boldsymbol{X}_{-j})^{-1/2}$ and $\tilde{\boldsymbol{y}}^{(j)}$ for all $j \in [1:d]$, as follows.

- **Computing $(\boldsymbol{X}_{-j}^T\boldsymbol{X}_{-j})^{-1/2}$ for all j.** We use the Cholesky square root rather than the principal square root, since the former admits the computational speed-ups described next.

  First compute $\boldsymbol{G} = \boldsymbol{X}^T\boldsymbol{X}$, its inverse $\boldsymbol{H} = \boldsymbol{G}^{-1}$, and the Cholesky square root $\boldsymbol{L}$ of $\boldsymbol{G}^{-1}$. Then, for any j, the Schur-complement formula gives $(\boldsymbol{X}_{-j}^T\boldsymbol{X}_{-j})^{-1} = \boldsymbol{H}_{-j,-j} - \boldsymbol{v}_j\boldsymbol{v}_j^T$, where

  $$\boldsymbol{v}_j = \frac{\boldsymbol{H}_{-j,j}}{\sqrt{H_{jj}}}.$$

  Efficient computations [Golub and Van Loan, 2013, Section 6.5.4] then yield the Cholesky square root of $(\boldsymbol{X}_{-j}^T\boldsymbol{X}_{-j})^{-1} = \boldsymbol{H}_{-j,-j} - \boldsymbol{v}_j\boldsymbol{v}_j^T$ from $\boldsymbol{L}$. Together, these steps make the Cholesky square root of $(\boldsymbol{X}_{-j}^T\boldsymbol{X}_{-j})^{-1}$ much faster to compute across j. We denote this Cholesky square root by $\boldsymbol{L}_j$, so that $\boldsymbol{L}_j\boldsymbol{L}_j^T = (\boldsymbol{X}_{-j}^T\boldsymbol{X}_{-j})^{-1}$, and set $(\boldsymbol{X}_{-j}^T\boldsymbol{X}_{-j})^{-1/2} := \boldsymbol{L}_j^T$ throughout. This choice gives $\boldsymbol{z}^{(j)} = \boldsymbol{L}_j^T\boldsymbol{X}_{-j}^T\boldsymbol{y}$ and $\boldsymbol{W}_{-j}^{(j)} = \boldsymbol{L}_j^{-1}$.

- **Computing $\tilde{\boldsymbol{y}}^{(j)}$ for all j.** By (D.1),

  $$\tilde{\boldsymbol{y}}^{(j)} = \boldsymbol{P}_{-j}\boldsymbol{y} + \|(\boldsymbol{I}_n - \boldsymbol{P}_{-j})\boldsymbol{y}\|\,\boldsymbol{V}^{(j)}\boldsymbol{u},$$

  where $\boldsymbol{P}_{-j} = \boldsymbol{X}_{-j}(\boldsymbol{X}_{-j}^T\boldsymbol{X}_{-j})^{-1}\boldsymbol{X}_{-j}^T$, the matrix $\boldsymbol{V}^{(j)}$ is an orthonormal basis for the orthogonal complement of the column space of $\boldsymbol{X}_{-j}$, and $\boldsymbol{u} \sim \mathrm{Unif}(\mathbb{S}^{n-d})$. Here the bottlenecks are $(\boldsymbol{X}_{-j}^T\boldsymbol{X}_{-j})^{-1}$ and $\boldsymbol{V}^{(j)}$. The former can again be computed from $\boldsymbol{G}^{-1}$ using the Schur-complement formula, exactly as in the previous item. For $\boldsymbol{V}^{(j)}$, one computes a single orthonormal basis $\boldsymbol{V}^{\perp}$ for the orthogonal complement of the column space of $\boldsymbol{X}$; then, for any j, $\boldsymbol{V}^{(j)}$ is obtained by prepending the column $\frac{(\boldsymbol{I}_n-\boldsymbol{P}_{-j})\boldsymbol{X}_j}{\|(\boldsymbol{I}_n-\boldsymbol{P}_{-j})\boldsymbol{X}_j\|}$ to $\boldsymbol{V}^{\perp}$. The vector $(\boldsymbol{I}_n - \boldsymbol{P}_{-j})\boldsymbol{X}_j = \boldsymbol{X}_j - \boldsymbol{P}_{-j}\boldsymbol{X}_j$ can then be evaluated from the quantities already computed above.

These tricks alone yield significant speed-ups in computing all d of the $\boldsymbol{z}^{(j)}$ and $\boldsymbol{W}^{(j)}$. Beyond them, evaluating $\mathcal{A}^{\mathrm{LASSO}}$ requires running cross-validated LASSO on

$$\left((\boldsymbol{X}_{-i}^T\boldsymbol{X}_{-i})^{-1/2}\boldsymbol{X}_{-i}^T\tilde{\boldsymbol{y}}^{(j)},\ (\boldsymbol{X}_{-i}^T\boldsymbol{X}_{-i})^{-1/2}\boldsymbol{X}_{-i}^T\boldsymbol{X}\right)$$

for all i and j. In principle one could use warm starts to avoid rerunning LASSO from scratch for each i and j, but we could not find a reliable way to do so with the standard `glmnet()` implementation in **R**, and we leave this to future work.

The computation of $\mathcal{A}^{\mathrm{dST}}$ across i and j, on the other hand, can be sped up substantially. For fixed λ, i, and j, computing $\mathcal{A}_i^{\mathrm{dST}}(\tilde{\boldsymbol{y}}^{(j)})$ requires the OLS fit of $(\boldsymbol{X}_{-i}^T\boldsymbol{X}_{-i})^{-1/2}\boldsymbol{X}_{-i}^T\tilde{\boldsymbol{y}}^{(j)}$ on $\boldsymbol{W}_{-i}^{(i)} = (\boldsymbol{X}_{-i}^T\boldsymbol{X}_{-i})^{-1/2}\boldsymbol{X}_{-i}^T\boldsymbol{X}_{-i}$, whose bottleneck is the inverse $(\boldsymbol{W}_{-i}^{(i)T}\boldsymbol{W}_{-i}^{(i)})^{-1} = (\boldsymbol{X}_{-i}^T\boldsymbol{X}_{-i})^{-1}$ which can again be computed from $\boldsymbol{G}^{-1}$ via the Schur-complement formula across i. The denominator of $\mathcal{A}_i^{\mathrm{dST}}(\tilde{\boldsymbol{y}}^{(j)})$, namely $\|\boldsymbol{W}_i^{(i)}\|^2 = \|\boldsymbol{P}_{-i}\boldsymbol{X}_i\|^2$, has the same bottleneck—the inversion of $\boldsymbol{X}_{-i}^T\boldsymbol{X}_{-i}$—and hence benefits similarly from the above computation. Thus, once $\boldsymbol{G}^{-1}$ is computed, it can be shared across all evaluations of $\mathcal{A}_i^{\mathrm{dST}}(\tilde{\boldsymbol{y}}^{(j)})$. Next, we describe the cross-validation step used to select λ for $\mathcal{A}^{\mathrm{dST}}$.

E.2 Cross-validation for computing the distilled soft-thresholding measure

To compute $\mathcal{A}_i^{\mathrm{dST}}(\boldsymbol{y})$ for a given i, we select $\hat{\lambda}_i^{\mathrm{ST}}$ by cross-validation. Specifically, we partition $[1:d-1]$ into $K=10$ folds, sequentially holding out one fold for testing and using the remaining folds for training. At each iteration, $(\boldsymbol{z}^{(i)}, \boldsymbol{W}^{(i)})$ is partitioned into training and testing sets, denoted by $(\boldsymbol{z}_{\mathrm{tr}}^{(i)}, \boldsymbol{W}_{\mathrm{tr}}^{(i)})$ and $(\boldsymbol{z}_{\mathrm{tst}}^{(i)}, \boldsymbol{W}_{\mathrm{tst}}^{(i)})$, respectively. Because $\boldsymbol{W}_{\mathrm{tr},-i}^{(i)}$ has fewer rows than columns, we cannot compute the usual OLS estimator to soft-threshold. Thus, we replace it with the ridgeless least-squares fit:

$$\arg\min_{\boldsymbol{\beta}} \ \|\boldsymbol{\beta}\|^2 \quad \text{subject to} \quad \boldsymbol{W}_{\mathrm{tr},-i}^{(i)}\boldsymbol{\beta} = \boldsymbol{z}_{\mathrm{tr}}^{(i)}. \tag{E.1}$$

We then obtain the λ-soft-thresholded version of this estimate, denoted by $\hat{\boldsymbol{\beta}}_{\mathrm{ST,tr}}^{\lambda,-i}$, and the OLS coefficient from regressing the residual $\boldsymbol{z}_{\mathrm{tr}}^{(i)} - \boldsymbol{W}_{\mathrm{tr},-i}^{(i)}\hat{\boldsymbol{\beta}}_{\mathrm{ST,tr}}^{\lambda,-i}$ on $\boldsymbol{W}_{\mathrm{tr},i}^{(i)}$, denoted by $\hat{\gamma}_{\mathrm{tr},i}^{\lambda}$. We evaluate the prediction error on the held-out fold as

$$\left\| \left(\boldsymbol{z}_{\mathrm{tst}}^{(i)} - \boldsymbol{W}_{\mathrm{tst},-i}^{(i)}\hat{\boldsymbol{\beta}}_{\mathrm{ST,tr}}^{\lambda,-i} \right) - \hat{\gamma}_{\mathrm{tr},i}^{\lambda}\boldsymbol{W}_{\mathrm{tst},i}^{(i)} \right\|^2. \tag{E.2}$$

Averaging this quantity across the K folds yields the cross-validated mean-squared error (MSE) for a given λ. We select the λ that minimizes this MSE.

Next, note that the standard solution to (E.1) is ${\boldsymbol{W}_{\mathrm{tr},-i}^{(i)}}^{+}\boldsymbol{z}_{\mathrm{tr}}^{(i)}$, where $\boldsymbol{A}^{+}$ denotes the Moore–Penrose pseudoinverse of a matrix $\boldsymbol{A}$. However, computing the pseudoinverse can be expensive. We instead propose the following computationally cheaper alternative.

Recall the Cholesky square root $\boldsymbol{L}_i$ of $(\boldsymbol{X}_{-i}^T\boldsymbol{X}_{-i})^{-1}$ from Appendix E.1, which is already available before cross-validation, and define $\boldsymbol{Q} = \boldsymbol{L}_i^T\boldsymbol{L}_i$. Since $\boldsymbol{W}_{-i}^{(i)} = \boldsymbol{L}_i^{-1}$, the reparameterization $\boldsymbol{\theta} = \boldsymbol{W}_{-i}^{(i)}\boldsymbol{\beta}$ gives $\boldsymbol{\beta} = \boldsymbol{L}_i\boldsymbol{\theta}$, so (E.1) is equivalent to

$$\arg\min_{\boldsymbol{\theta}} \ \|\boldsymbol{L}_i\boldsymbol{\theta}\|^2 \quad \text{subject to} \quad \boldsymbol{\theta}_{\mathrm{tr}} = \boldsymbol{z}_{\mathrm{tr}}^{(i)}. \tag{E.3}$$

To solve this problem, observe that

$$\begin{aligned}
\|\boldsymbol{L}_i\boldsymbol{\theta}\|^2 &= \boldsymbol{\theta}^T\boldsymbol{Q}\boldsymbol{\theta} \\
&= \boldsymbol{\theta}_{\mathrm{tr}}^T\boldsymbol{Q}_{\mathrm{tr,tr}}\boldsymbol{\theta}_{\mathrm{tr}} + \boldsymbol{\theta}_{\mathrm{tst}}^T\boldsymbol{Q}_{\mathrm{tst,tst}}\boldsymbol{\theta}_{\mathrm{tst}} + 2\boldsymbol{\theta}_{\mathrm{tr}}^T\boldsymbol{Q}_{\mathrm{tr,tst}}\boldsymbol{\theta}_{\mathrm{tst}} \\
&= {\boldsymbol{z}_{\mathrm{tr}}^{(i)}}^T\boldsymbol{Q}_{\mathrm{tr,tr}}\boldsymbol{z}_{\mathrm{tr}}^{(i)} + \boldsymbol{\theta}_{\mathrm{tst}}^T\boldsymbol{Q}_{\mathrm{tst,tst}}\boldsymbol{\theta}_{\mathrm{tst}} + 2{\boldsymbol{z}_{\mathrm{tr}}^{(i)}}^T\boldsymbol{Q}_{\mathrm{tr,tst}}\boldsymbol{\theta}_{\mathrm{tst}}.
\end{aligned}$$

The first term is constant under the constraint in (E.3). Hence, the solution is $\boldsymbol{\theta}^* = (\boldsymbol{z}_{\mathrm{tr}}^{(i)}, \boldsymbol{\theta}_{\mathrm{tst}}^*)$, where

$$\boldsymbol{\theta}_{\mathrm{tst}}^* = \arg\min_{\boldsymbol{\theta}_{\mathrm{tst}}} \left\{ \boldsymbol{\theta}_{\mathrm{tst}}^T\boldsymbol{Q}_{\mathrm{tst,tst}}\boldsymbol{\theta}_{\mathrm{tst}} + 2{\boldsymbol{z}_{\mathrm{tr}}^{(i)}}^T\boldsymbol{Q}_{\mathrm{tr,tst}}\boldsymbol{\theta}_{\mathrm{tst}} \right\} = -\boldsymbol{Q}_{\mathrm{tst,tst}}^{-1}\boldsymbol{Q}_{\mathrm{tst,tr}}\boldsymbol{z}_{\mathrm{tr}}^{(i)}.$$

Thus, the solution to (E.1) is $\boldsymbol{\beta}^* = \boldsymbol{L}_i\boldsymbol{\theta}^*$. Importantly, computing $\boldsymbol{\theta}_{\mathrm{tst}}^*$ only requires solving the linear system $\boldsymbol{Q}_{\mathrm{tst,tst}}\boldsymbol{\theta}_{\mathrm{tst}} = -\boldsymbol{Q}_{\mathrm{tst,tr}}\boldsymbol{z}_{\mathrm{tr}}^{(i)}$, which can be done efficiently using a Cholesky-based solver without explicitly computing $\boldsymbol{Q}_{\mathrm{tst,tst}}^{-1}$.

Our implementation exploits further computational savings. For each fixed i, we use the same training-test fold splits across all j when computing $\mathcal{A}_i^{\mathrm{dST}}(\tilde{\boldsymbol{y}}^{(j)})$. Consequently, $\boldsymbol{Q}_{\mathrm{tst,tr}}$ and $\boldsymbol{Q}_{\mathrm{tst,tst}}$, along with the Cholesky decomposition of $\boldsymbol{Q}_{\mathrm{tst,tst}}$ required to solve the linear system for $\boldsymbol{\theta}^*_{\mathrm{tst}}$, can be computed once for each i and fold split and reused across all j.

Moreover, once this ridgeless least-squares fit is available, evaluating the MSE in (E.2) for different values of λ requires only soft-thresholding this solution. Thus, evaluating the cross-validation criterion over a grid of λ values incurs no additional computationally expensive matrix operations.

Finally, for each i and j, we choose a grid of 50 equally spaced λ values on the logarithmic scale between $b_{\max}^{(i,j)}$ and $10^{-3}b_{\max}^{(i,j)}$, where

$$b_{\max}^{(i,j)} = \max_k \left|\hat{\beta}_k^{(i,j)}\right|, \qquad \hat{\beta}_k^{(i,j)} = (\boldsymbol{X}_{-i}^T\boldsymbol{X}_{-i})^{-1}\boldsymbol{X}_{-i}^T\tilde{\boldsymbol{y}}^{(j)}.$$

F Proofs

F.1 Proof of Theorem 3.2

Proof. Consider any subset $I \subseteq [1:d]$ and define the weights

$$\hat{w}_j^{(I)} = \frac{\mathcal{A}_j(\tilde{\boldsymbol{y}}^{(j)})}{\sum_{i\in I}\mathcal{A}_i(\tilde{\boldsymbol{y}}^{(j)})},$$

where $\tilde{\boldsymbol{y}}^{(j)}$ is sampled from the conditional distribution of $\boldsymbol{y} \mid \boldsymbol{S}^{(j)}$ under H_j. Note that the p-values $\{p_j\}_{j\in I}$ and the weights $\{\hat{w}_j^{(I)}\}_{j\in I}$ satisfy Assumption 2.1 by Theorem 3.1. Hence, it follows from [Ignatiadis et al., 2023, Theorem 4.3] that the weighted I-Bonferroni test using weights $\{\hat{w}_j^{(I)}\}_{j\in I}$ is valid. Thus, by the closure principle [Marcus et al., 1976] (discussed in Section 2.1), the closure of these Bonferroni tests is a valid FWER-controlling procedure. We will show that Algorithm 1 is precisely this closure.

First, suppose that Algorithm 1 rejects an index $i \in [1:d]$. We will show that this index is also rejected by the closure. Let $\emptyset = \hat{R}_0 \subset \hat{R}_1 \subset \cdots \subset \hat{R}_K$ denote the rejection sets of Algorithm 1, which makes $K \geq 1$ rejections, and let $i = j_m$ for some $1 \leq m \leq K$. Now consider any set I containing i and define $m^* = \min\{r : j_r \in I\}$. This minimum is well-defined since the set is nonempty, as it contains at least m. Because j_{m^*} was rejected by Algorithm 1, we have

$$p_{j_{m^*}} \leq \alpha\hat{w}_{j_{m^*}}^{(\hat{R}^c_{m^*-1})}.$$

By the definition of m^*, we also have $j_r \notin I$ for all $r < m^*$ (if any; otherwise, the statement holds vacuously). Hence, $\hat{R}_{m^*-1}$ is disjoint from I, implying that $I \subseteq \hat{R}^c_{m^*-1}$. By the definition of the weights, this yields

$$\hat{w}_{j_{m^*}}^{(\hat{R}^c_{m^*-1})} = \frac{\mathcal{A}_{j_{m^*}}(\tilde{\boldsymbol{y}}^{(j_{m^*})})}{\sum_{i\in\hat{R}^c_{m^*-1}}\mathcal{A}_i(\tilde{\boldsymbol{y}}^{(j_{m^*})})} \leq \frac{\mathcal{A}_{j_{m^*}}(\tilde{\boldsymbol{y}}^{(j_{m^*})})}{\sum_{i\in I}\mathcal{A}_i(\tilde{\boldsymbol{y}}^{(j_{m^*})})} = \hat{w}_{j_{m^*}}^{(I)}.$$

Consequently, $p_{j_{m^*}} \leq \alpha\hat{w}_{j_{m^*}}^{(I)}$. Since $j_{m^*} \in I$, the weighted I-Bonferroni test rejects H_I. As this holds for every I containing i, the closure also rejects i.

Conversely, suppose that Algorithm 1 does not reject some index i. We will show that the closure does not reject i either. For this, it suffices to identify a set I containing i such that the weighted I-Bonferroni test does not reject H_I. Since Algorithm 1 does not reject i, we must have $i \in \hat{R}_K^c$ and $K < d$. Define $I = \hat{R}_K^c$, so that none of the indices in I has been rejected by Algorithm 1. By the stopping rule of the algorithm, for every $j \in I$, we must have $p_j > \alpha \hat{w}_j^{(I)}$; otherwise, j would have been rejected by Algorithm 1. Thus, the weighted I-Bonferroni test does not reject H_I. Since $i \in I$, the closure does not reject i either. This completes the proof of equivalence. □

F.2 Proof of Lemma D.1

Proof. Let $\boldsymbol{V}^\perp$ be an orthonormal basis for the orthogonal complement of Col.sp.$(\boldsymbol{X})$. Since

$$\text{Col.sp.}(\boldsymbol{X}_{-i})^\perp = \text{span}(\boldsymbol{e}_i) \oplus \text{Col.sp.}(\boldsymbol{X})^\perp,$$

with $\boldsymbol{e}_i$ defined as in the statement of Lemma D.1, we have

$$\boldsymbol{I}_n - \boldsymbol{P}_{-i} = \boldsymbol{e}_i \boldsymbol{e}_i^\top + \boldsymbol{V}^\perp (\boldsymbol{V}^\perp)^\top.$$

Moreover, by the choice of $\boldsymbol{V}^{(j)}$,

$$\tilde{\boldsymbol{y}}^{(j)} = \boldsymbol{P}_{-j}\boldsymbol{y} + \hat{\sigma}_j \boldsymbol{e}_j u_1 + \hat{\sigma}_j \boldsymbol{V}^\perp \boldsymbol{u}_{-1},$$

where $\boldsymbol{u}_{-1} = (u_2, \ldots, u_{n-d+1})^\top$. Consequently, using the orthogonality of $\boldsymbol{e}_i$ and Col.sp.$(\boldsymbol{V}^\perp)$,

$$\begin{aligned}\|(\boldsymbol{I}_n - \boldsymbol{P}_{-i})\tilde{\boldsymbol{y}}^{(j)}\|^2 &= \left(\boldsymbol{e}_i^\top \boldsymbol{P}_{-j}\boldsymbol{y} + \rho_{ij}\hat{\sigma}_j u_1\right)^2 + \hat{\sigma}_j^2 \|\boldsymbol{u}_{-1}\|^2 \\ &= \left(\boldsymbol{e}_i^\top \boldsymbol{P}_{-j}\boldsymbol{y} + \rho_{ij}\hat{\sigma}_j u_1\right)^2 + (1 - u_1^2)\hat{\sigma}_j^2,\end{aligned}$$

where the last equality follows from $\|\boldsymbol{u}\| = 1$. □

G Further details on HIV drug resistance dataset analysis

G.1 Genetic mutations identified by the various procedures

Figures 5 and 6 summarize the genetic mutations identified by the FDR-controlling procedures (at a nominal level of 0.1) and the FWER-controlling procedures (at a nominal level of 0.05), respectively, considered in Section 5 for each of the 16 drug-specific regressions. These drugs are classified into three groups: protease inhibitors (PIs), nucleoside reverse transcriptase inhibitors (NRTIs), and nonnucleoside reverse transcriptase inhibitors (NNRTIs). The results in both figures are organized according to these groups.

As alluded to in Section 5, many prior works [Barber and Candès, 2015, Fithian and Lei, 2022, Luo et al., 2022] have also created such a summary, which is obtained by aggregating the raw discoveries of these methods to a gene-level summary. As also has been done in prior works, we compare these genetic discoveries against a pre-published list of influential genes from the TSM list. As we observed in Section 5, for each of these regressions the number of discoveries by our weighted and the vanilla unweighted procedures are very similar.

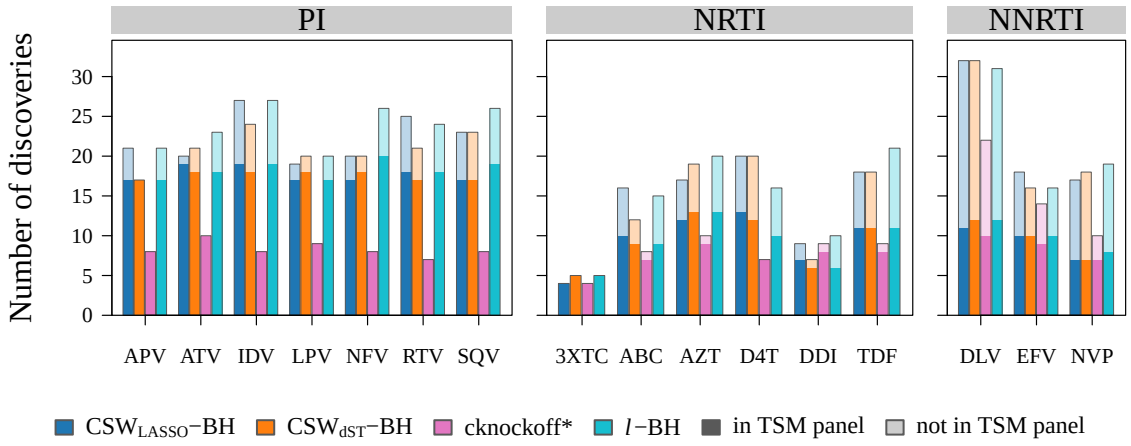


Figure 5: Number of genetic mutations identified by the various FDR controlling methods for various drug-specific regressions (marked on the x-axis) at nominal level 0.1 and their split based on presence/absence in the TSM list.

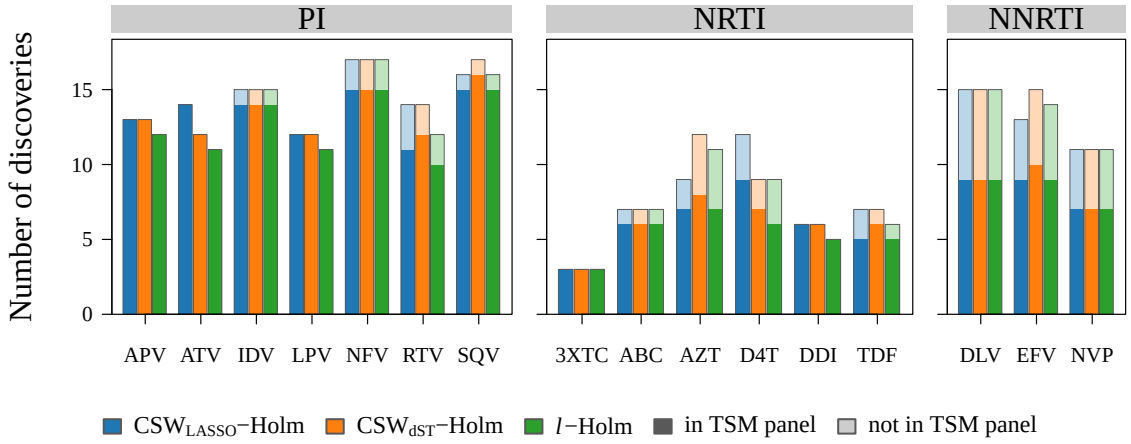


Figure 6: Number of genetic mutations identified by the various FWER controlling methods for various drug-specific regressions (marked on the x-axis) at nominal level 0.05 and their split based on presence/absence in the TSM list.

G.2 Further exploring the role of weighting

In Figures 7 and 8, we explore the number of discoveries for each of these procedures when the nominal FWER and FDR, respectively, are reduced. Particularly, for each of the 16 regressions, we track three quantities for these different nominal error levels: the common number of discoveries made by both the methods, the number of discoveries by the weighted procedure that is not discovered by the corresponding unweighted procedure, and the number of discoveries not discovered by the weighted procedure that are discovered by the corresponding unweighted method. As discussed in Section 5, these curves also demonstrate that our weighted procedures often make additional discoveries over their unweighted counterparts

for tightening levels of the error rates, thereby demonstrating the benefit of weighting.

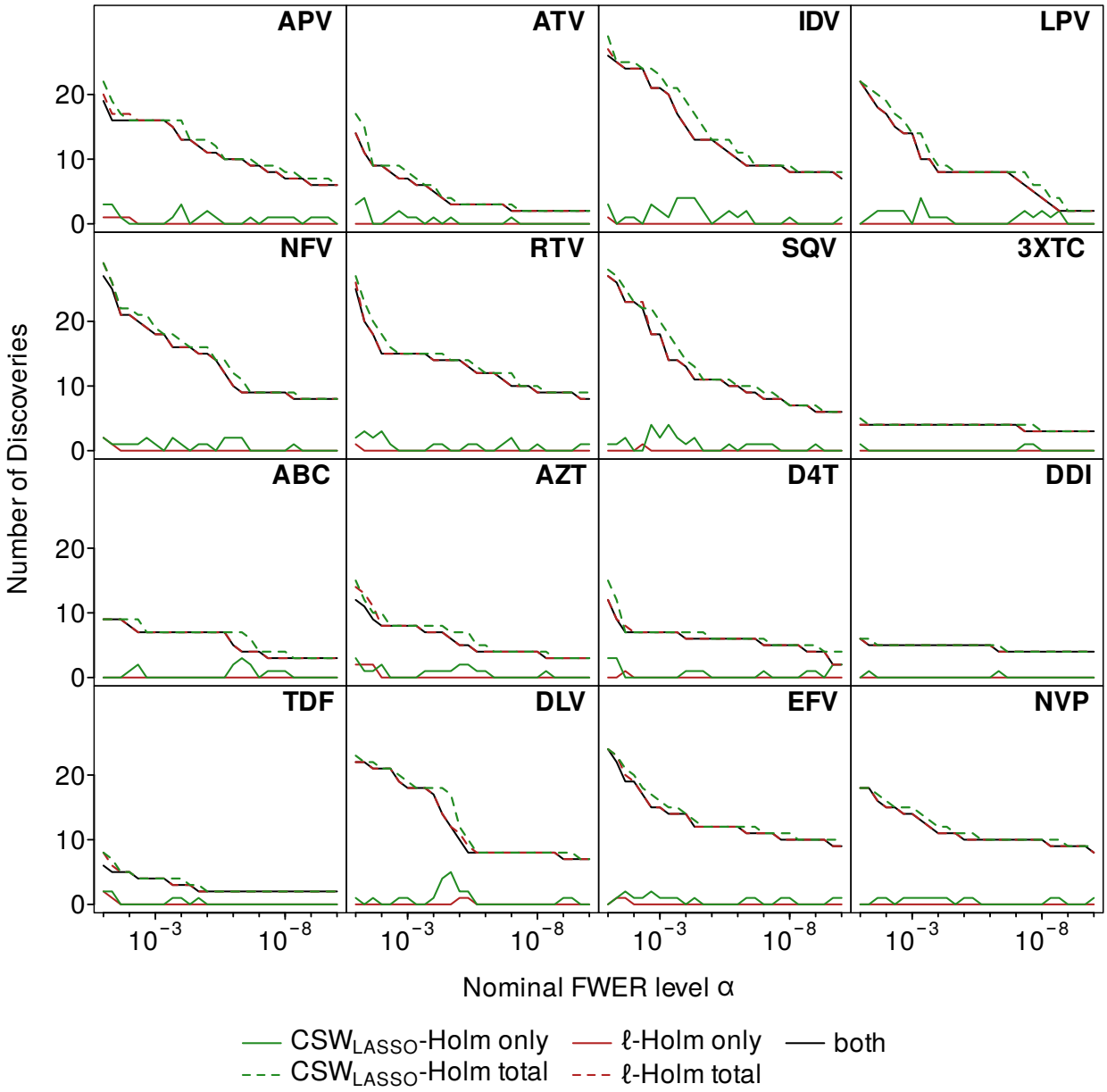


Figure 7: The number of discoveries by CSW_{LASSO}–Holm and ℓ-Holm for varying levels of nominal FWER for all the 16 regressions.

H Further details on simulations

H.1 Detailed simulation results

Here we present the complete set of empirical results for the simulation settings described in Section 4. Methods that do not exhibit consistent empirical FDR control across settings—

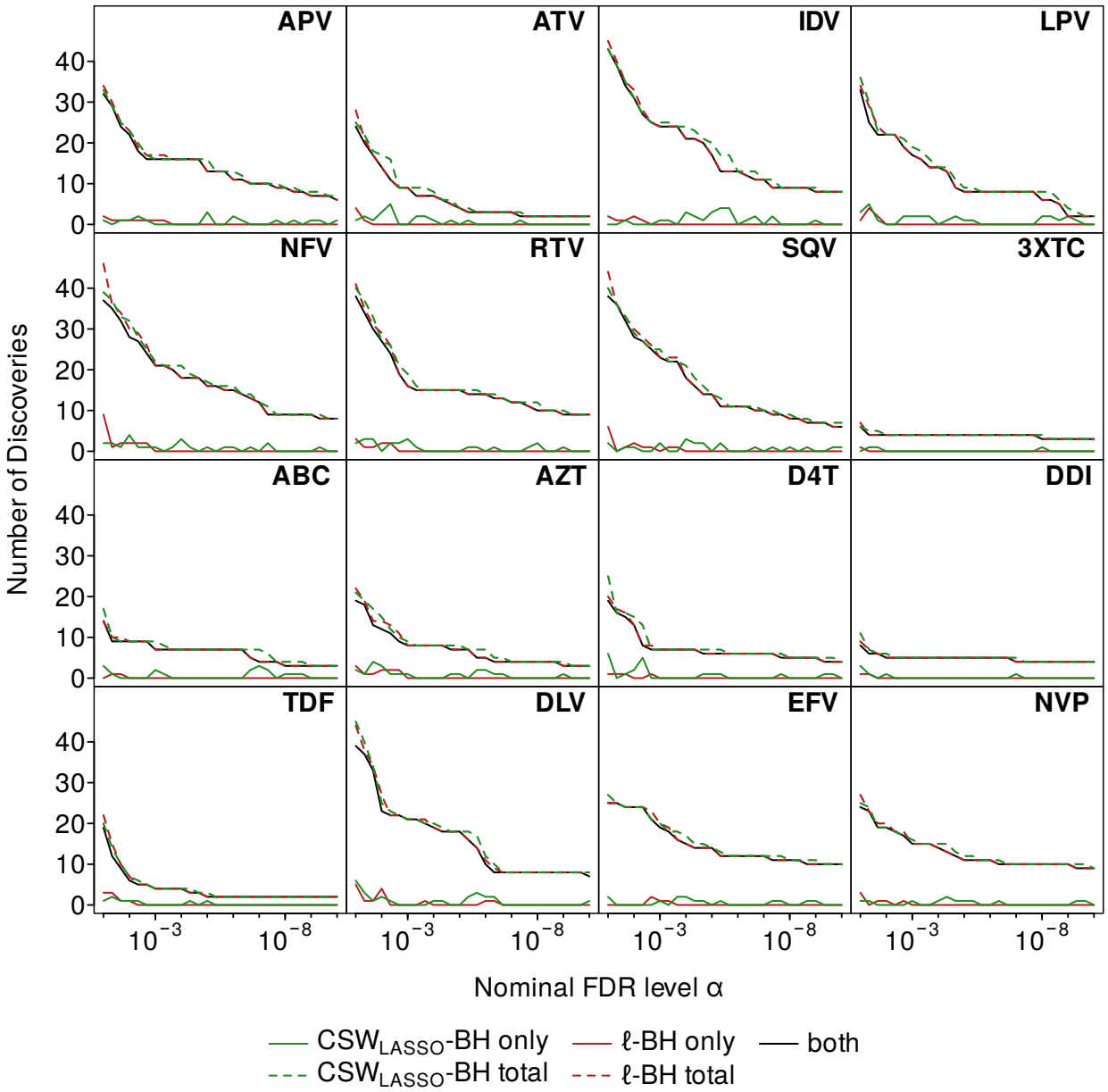


Figure 8: The number of discoveries by CSW$_{\text{LASSO}}$–BH and ℓ-BH for varying levels of nominal FDR for all the 16 regressions.

knockoffs, GM, MDS and OATK—are shown with dashed lines. In the $n/d = 1.5$ regime, the knockoff-based procedures (knockoffs, knockoffs+, cknockoff and cknockoff*) are omitted, as they require $n/d \geq 2$. In the $d = 1000$ regime, CSW$_{\text{LASSO}}$–Holm and CSW$_{\text{LASSO}}$–BH are omitted for computational reasons.

We next describe the choice of the amplitude A. These amplitudes were calibrated so that the power of the methods under comparison lies in an interesting range: not too close to 0 or 1. Accordingly, A is increased in settings that are intrinsically harder.

For regimes (b) and (c), the values of A are shown on the x-axes of the corresponding

figures. For regime (a), A is held fixed across sparsity levels and set to

$$A = 3.5 \times \kappa_{\text{design}} \times \kappa_{\text{signal}} \times \kappa_{\text{FWER}},$$

where

$$\kappa_{\text{design}} = \begin{cases} 1, & \rho = 0, \\ 1.3, & \rho = 0.5,\ n/d = 1.5, \\ 1, & \rho = 0.5,\ n/d = 3, \\ 3, & \rho = 0.9,\ n/d = 1.5, \\ 2.2, & \rho = 0.9,\ n/d = 3, \end{cases} \qquad \kappa_{\text{signal}} = \begin{cases} 1, & \text{discrete signal generation}, \\ 1.5, & \text{continuous signal generation}, \end{cases}$$

and $\kappa_{\text{FWER}} = 1.3$ for the FWER simulations and 1 for the FDR simulations. There are two exceptions to this rule. First, for the FWER simulations with $\rho = 0.9$ and $d = 1000$, we set $A = 15$ directly (recall that for $d = 1000$ we only consider discrete signal generation). Second, for the MCC design, every non-null coefficient is set to $+A$ with $A = 4.61$, for both the FDR and FWER simulations; [Luo et al., 2022, Section 5.1] chose this value so that the BH procedure at nominal FDR level 0.2 has power 0.5.

The full set of simulations are summarized in Figures 9 to 29. Under discrete signal generation, the full set of results supports the conclusions of Section 4, which focused on that case. Under continuous signal generation, similar trends hold, but the power curves lie much closer together. Because the signal strengths now vary, every method readily discovers the strongest signals and struggles with the weakest, so the methods differ mainly on the subset of the signals with intermediate strength.

H.2 Simulation results for exponentiated methods

Analogous to Figures 9 to 24, Figures 30 to 45 present the performance of $\text{CSW}_{\text{LASSO}^m}$–BH and $\text{CSW}_{\text{LASSO}^m}$–Holm for various values of m in the $d = 100$ simulations. As discussed in Section 3.2.3, these procedures use $\left(\mathcal{A}^{\text{LASSO}}\right)^m$ as the variable importance measure.

First, all exponentiated versions continue to exhibit empirical FDR control across every setting considered, while their FWER control is retained theoretically. Next, these figures reveal a general trend. In sparse settings, larger values of m tend to perform better, whereas in denser settings the trend reverses and smaller values of m take the lead. Moreover, in the sparser regimes, the gap between procedures with larger and smaller m appears to widen as the inter-variable correlation increases. Finally, as with the simulations in Section H.1, the gaps between the different curves narrow under continuous signal generation.

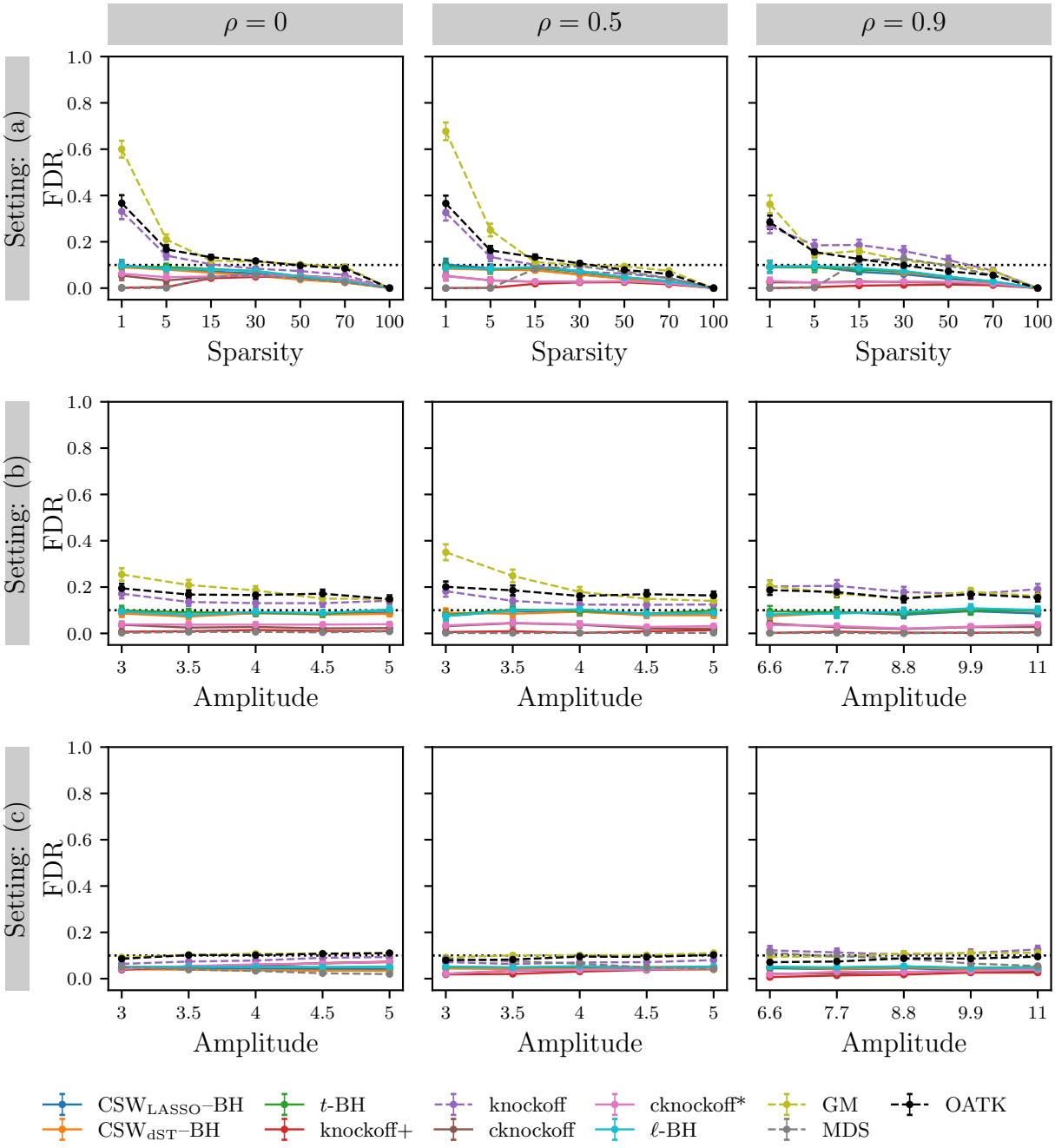


Figure 9: The empirical FDR of various FDR controlling procedures under discrete signal generation, $n/d = 3$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

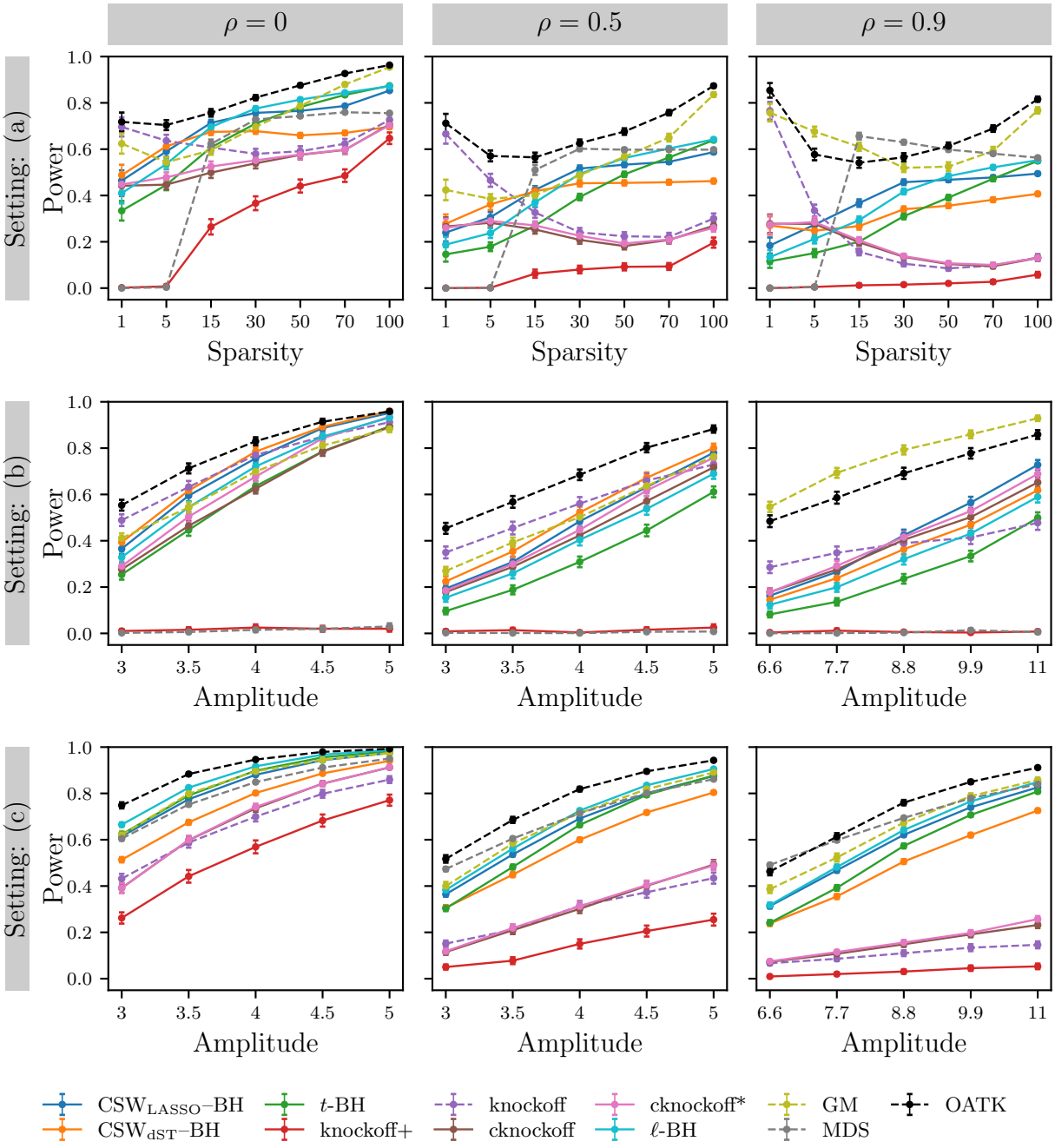


Figure 10: The empirical power of various FDR controlling procedures under discrete signal generation, $n/d = 3$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

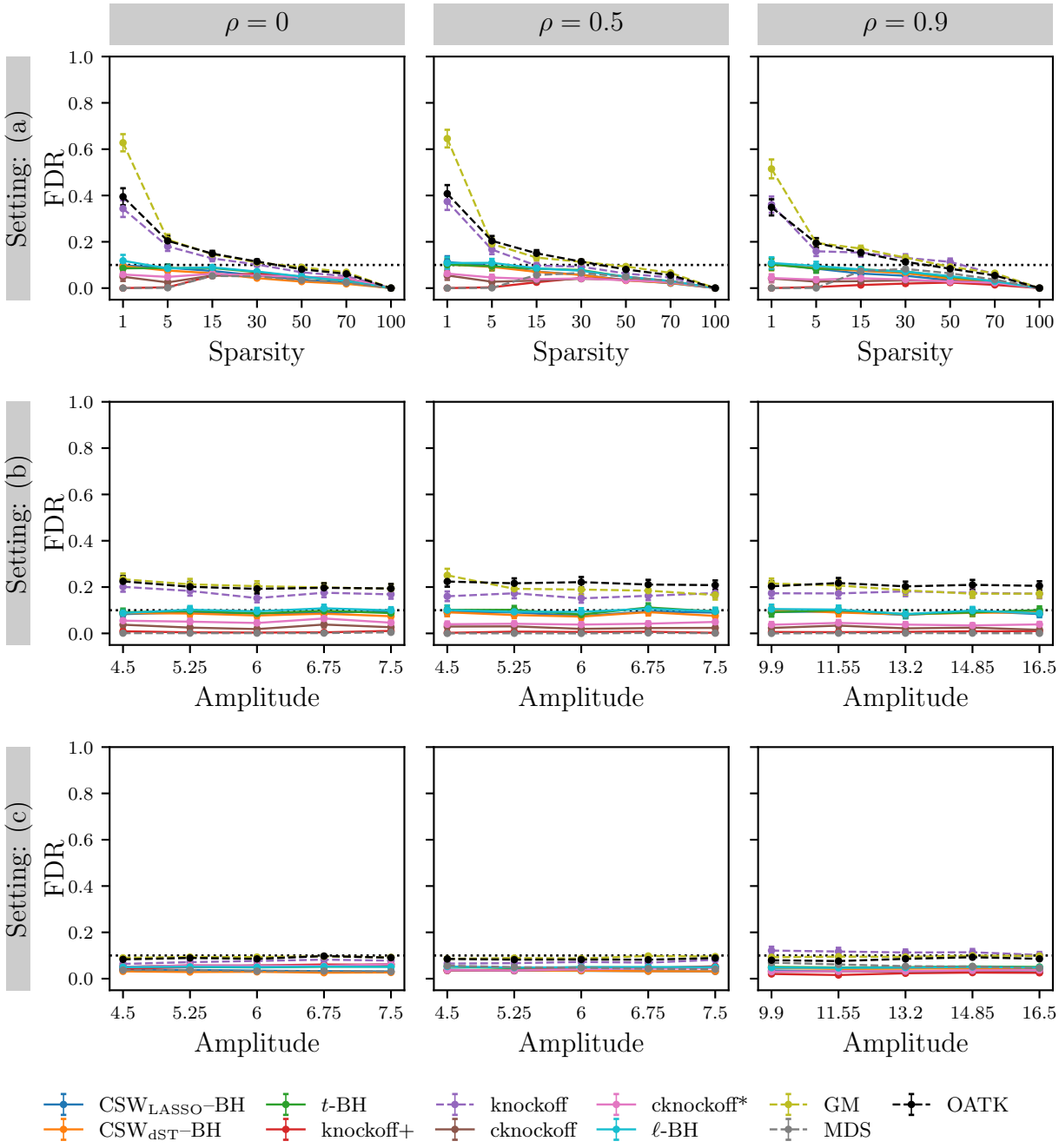


Figure 11: The empirical FDR of various FDR controlling procedures under continuous signal generation, $n/d = 3$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

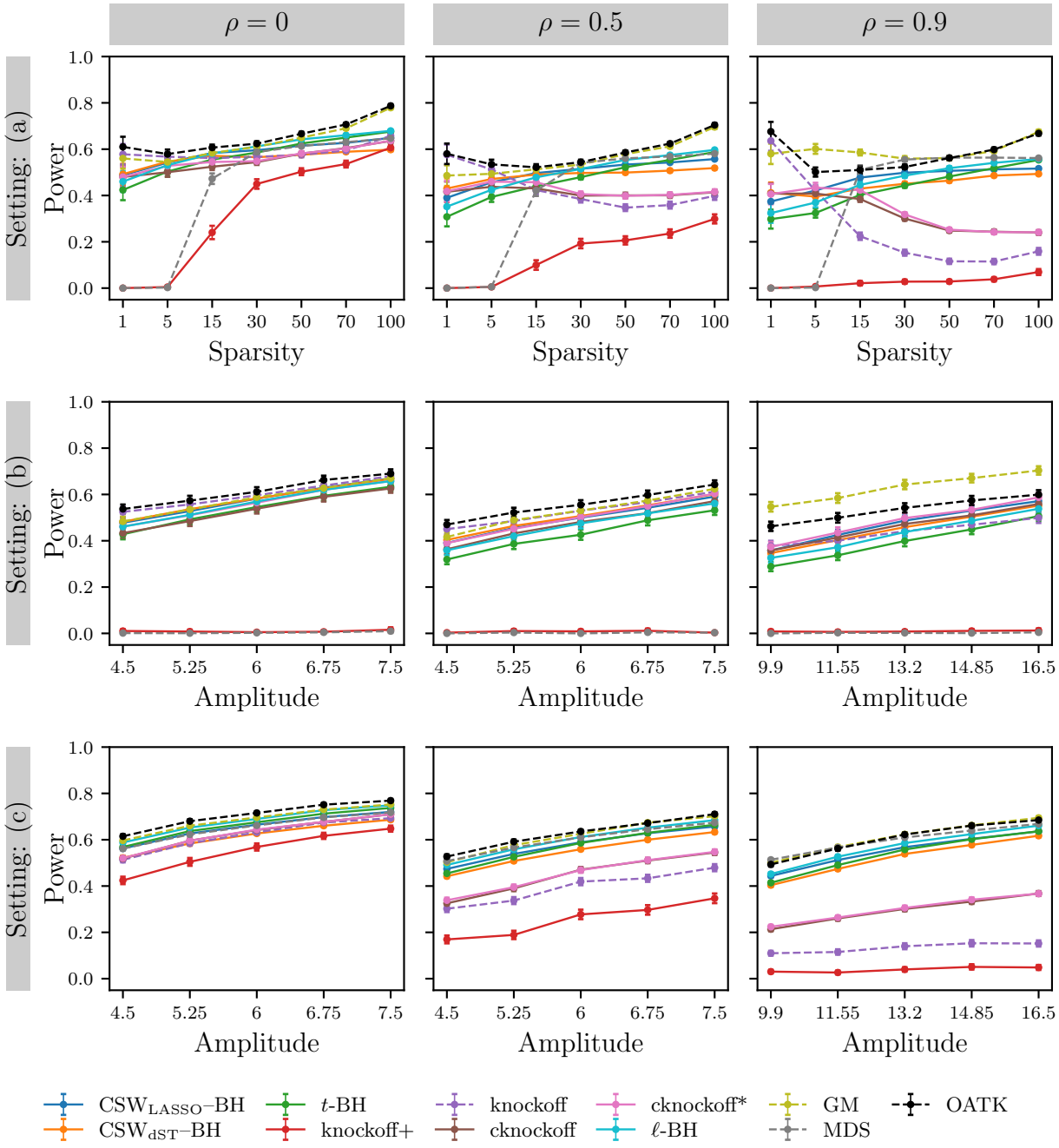


Figure 12: The empirical power of various FDR controlling procedures under continuous signal generation, $n/d = 3$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

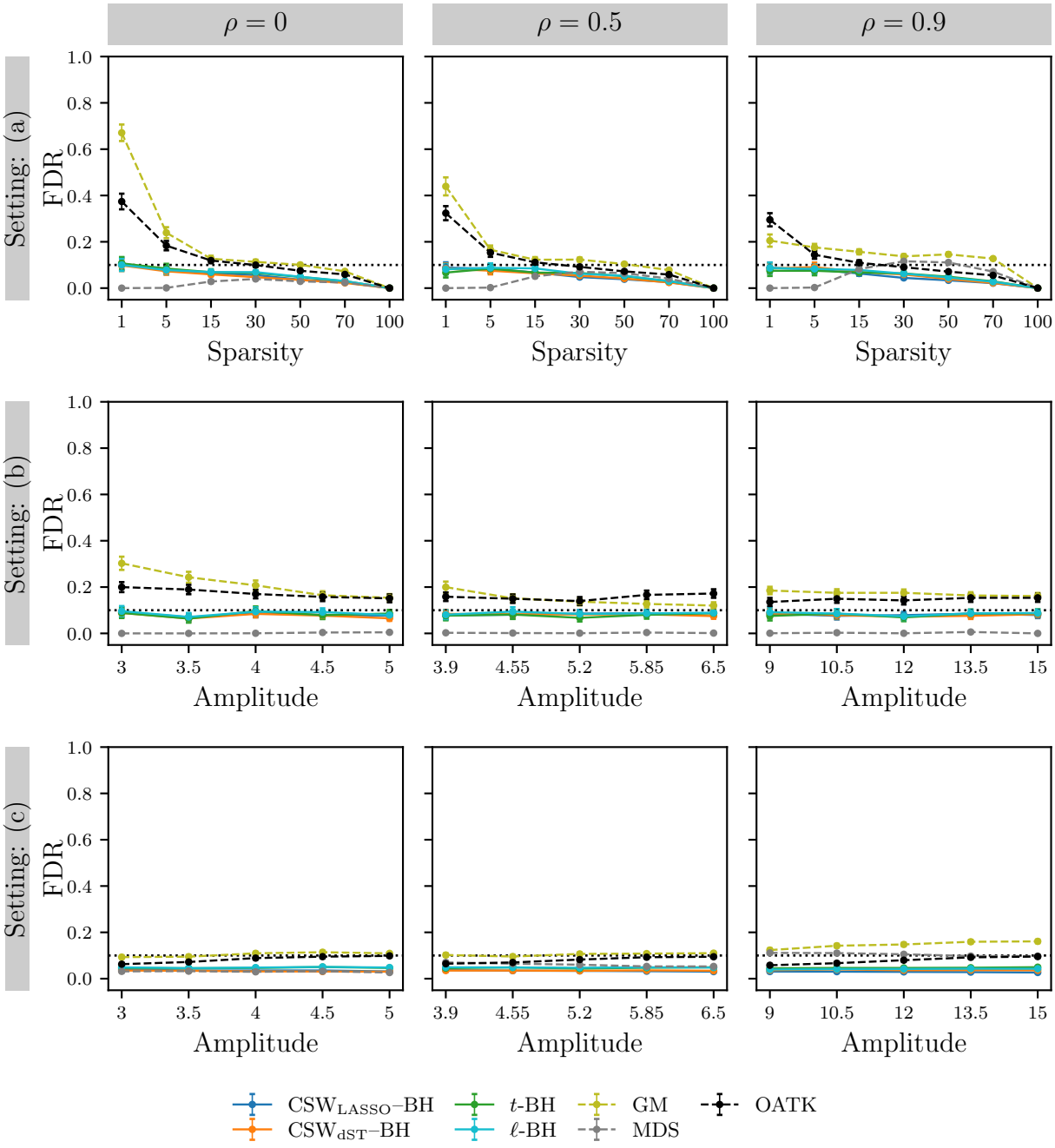


Figure 13: The empirical FDR of various FDR controlling procedures under discrete signal generation, $n/d = 1.5$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

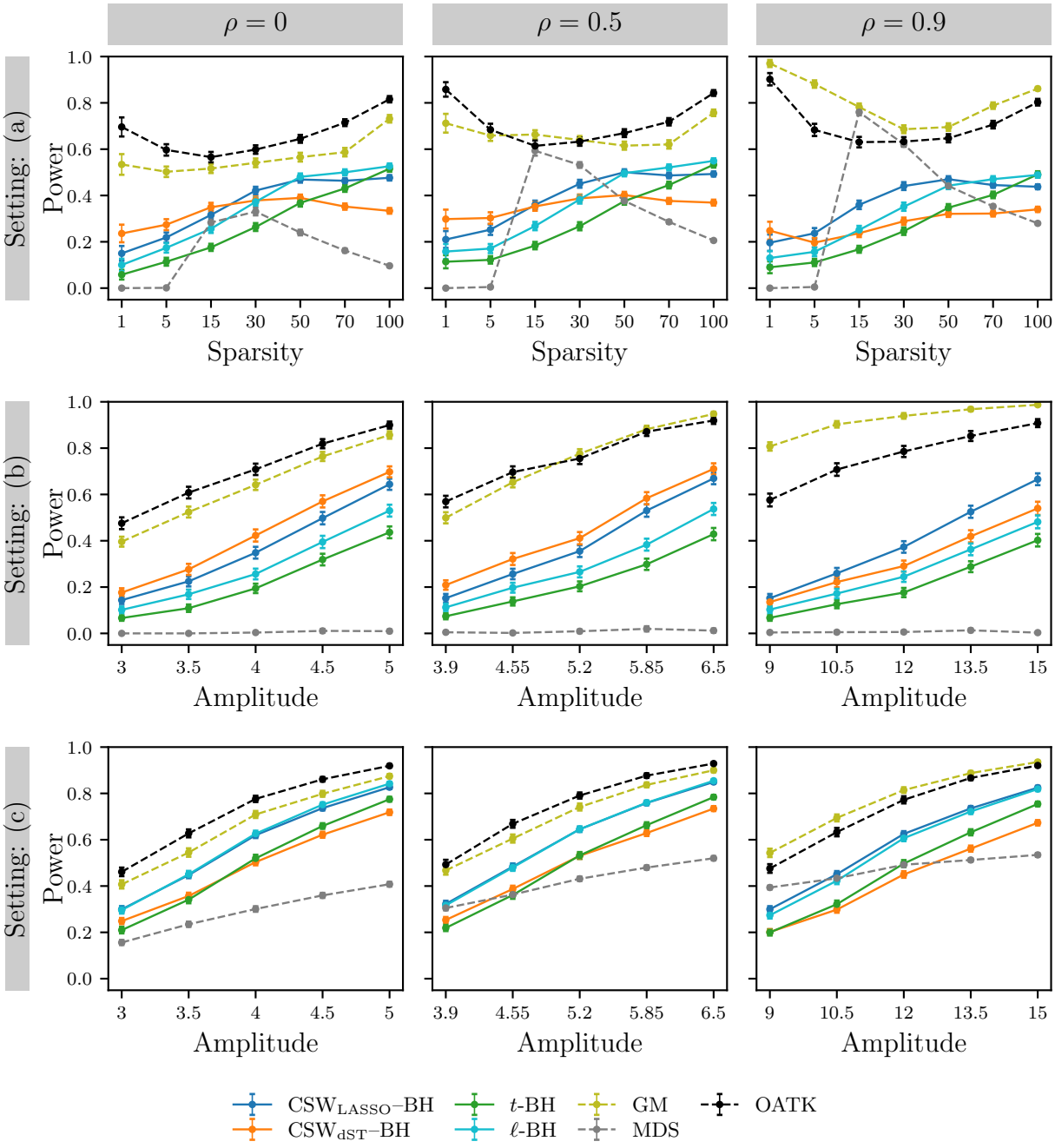


Figure 14: The empirical power of various FDR controlling procedures under discrete signal generation, $n/d = 1.5$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

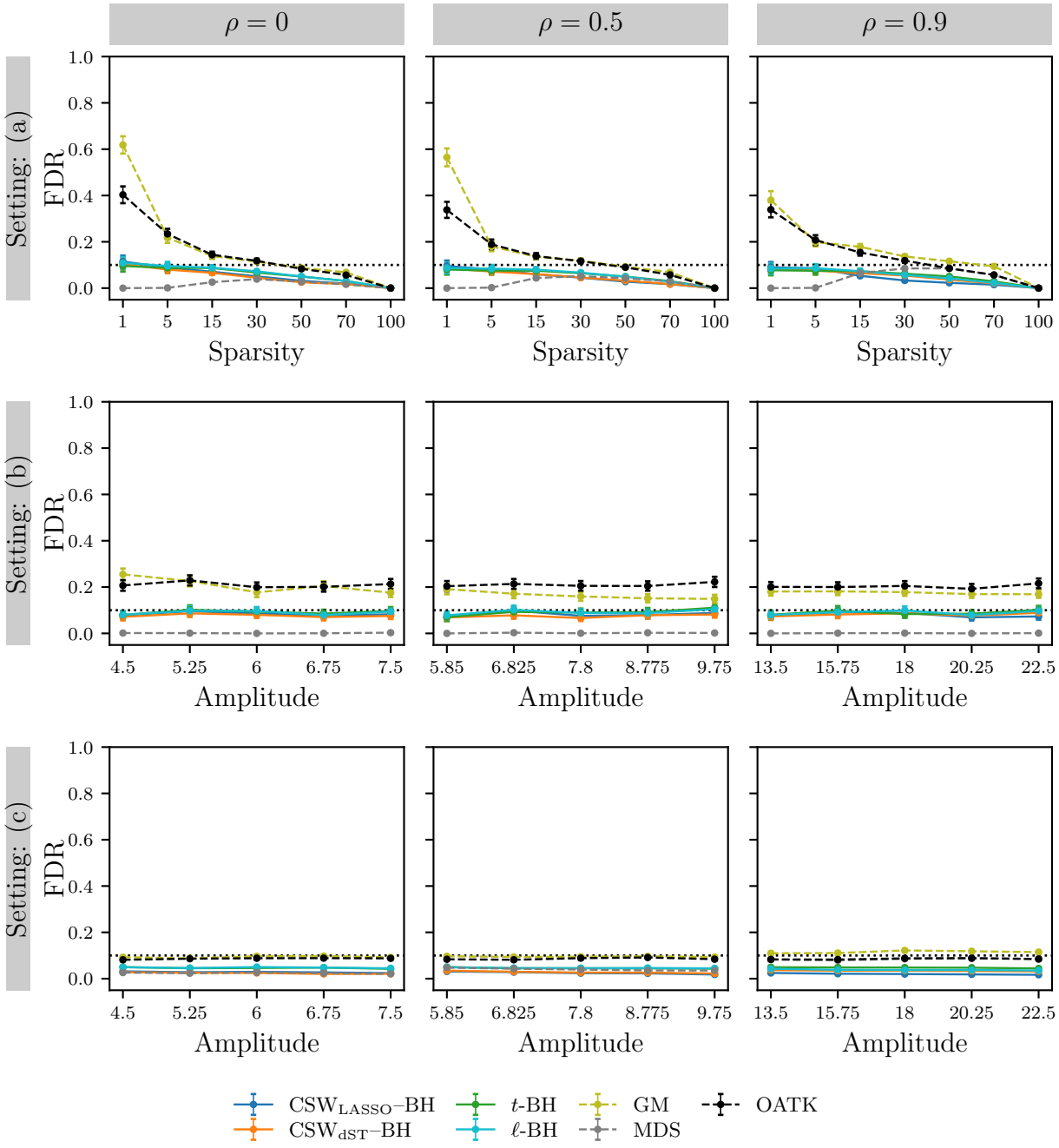


Figure 15: The empirical FDR of various FDR controlling procedures under continuous signal generation, $n/d = 1.5$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

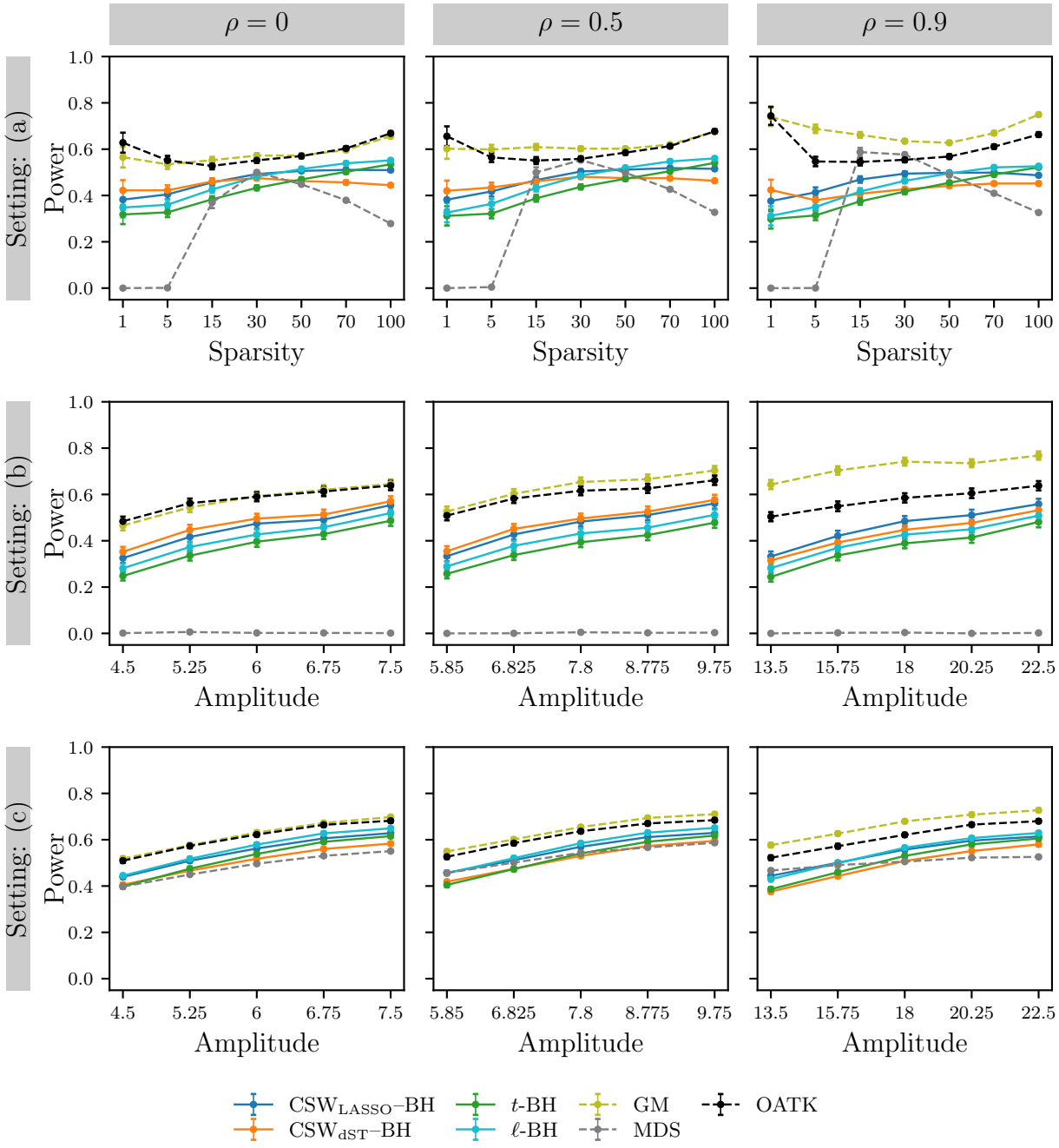


Figure 16: The empirical power of various FDR controlling procedures under continuous signal generation, $n/d = 1.5$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

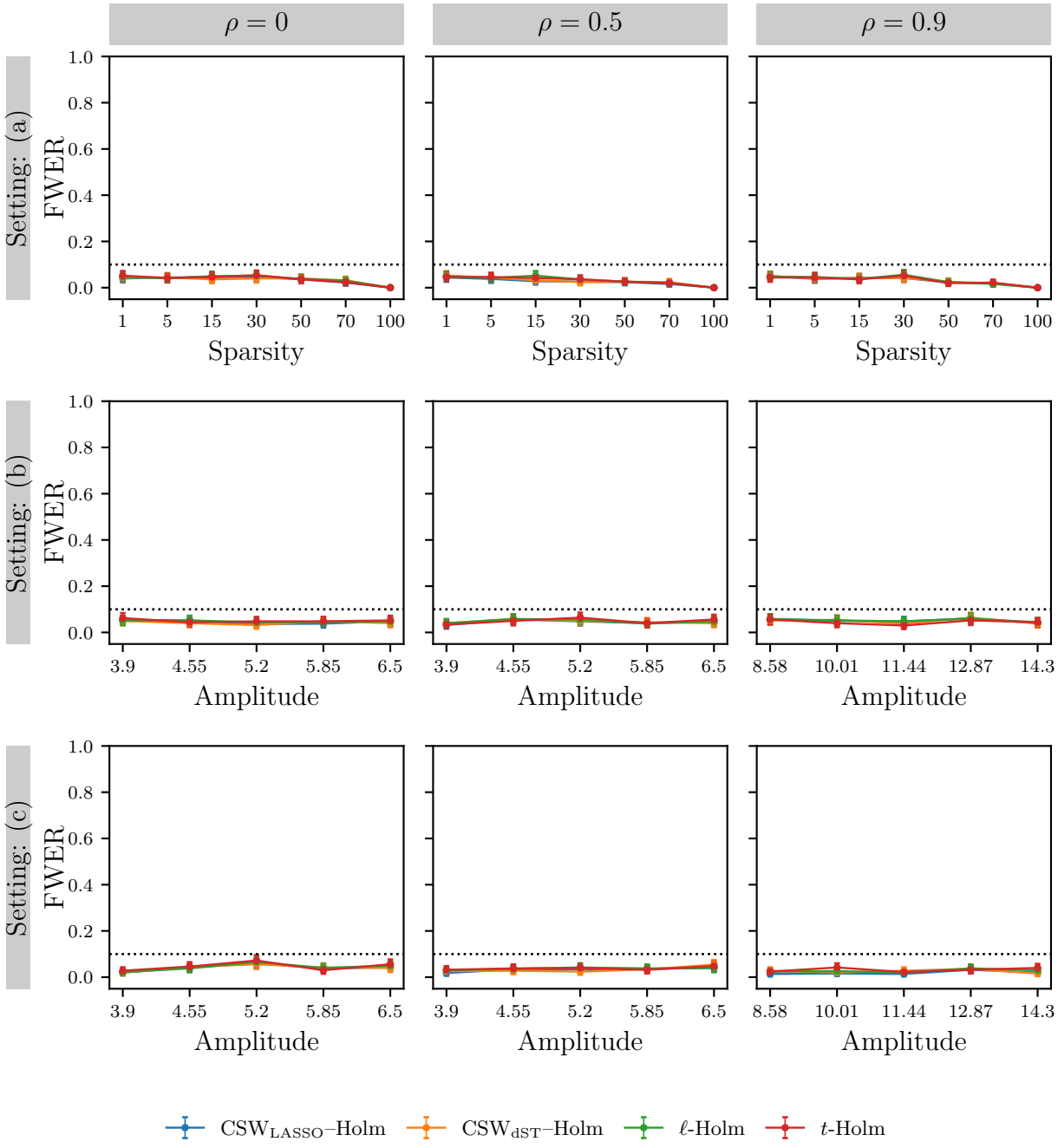


Figure 17: The empirical FWER of various FWER controlling procedures under discrete signal generation, $n/d = 3$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

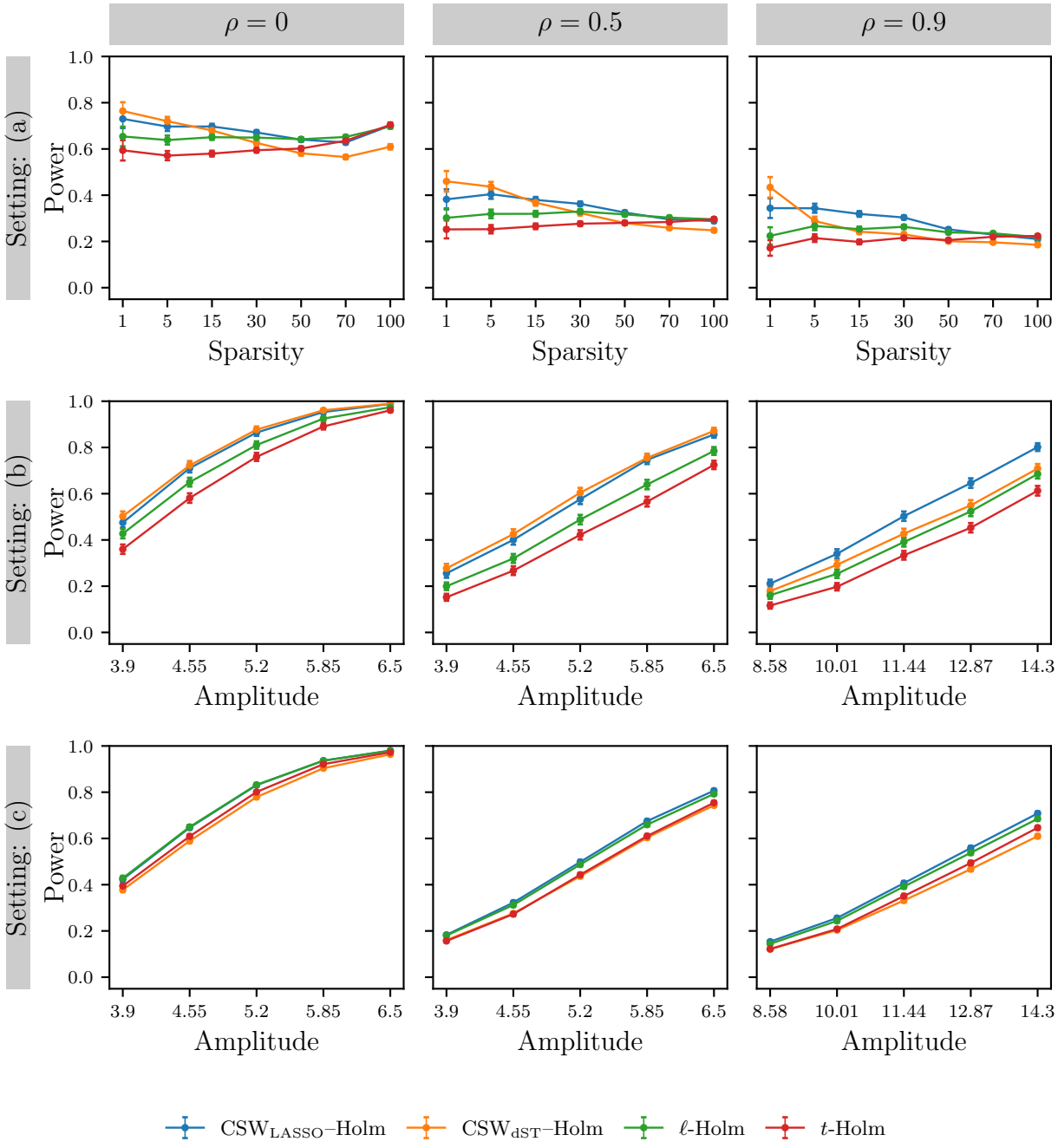


Figure 18: The empirical power of various FWER controlling procedures under discrete signal generation, $n/d = 3$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

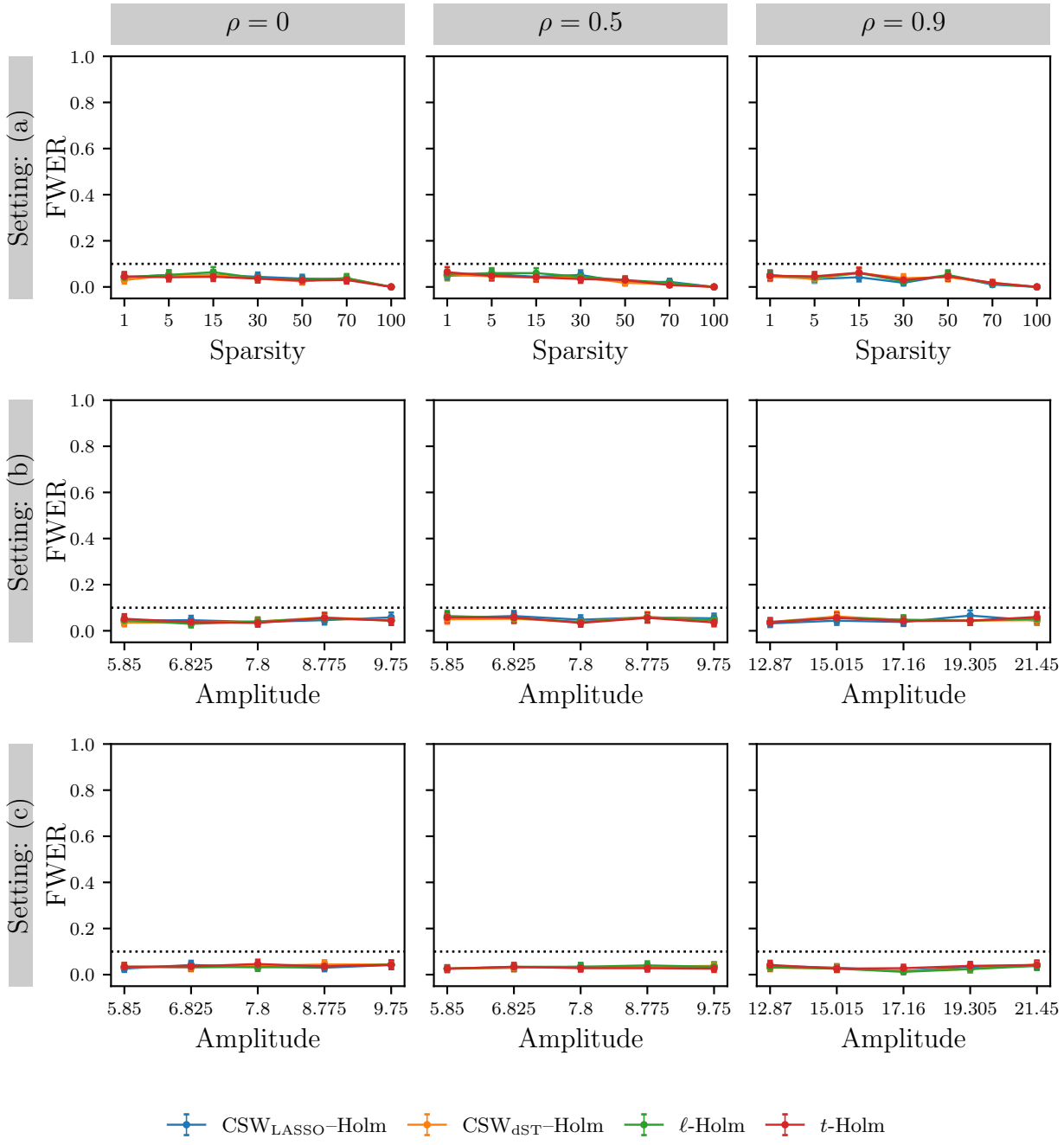


Figure 19: The empirical FWER of various FWER controlling procedures under continuous signal generation, $n/d = 3$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

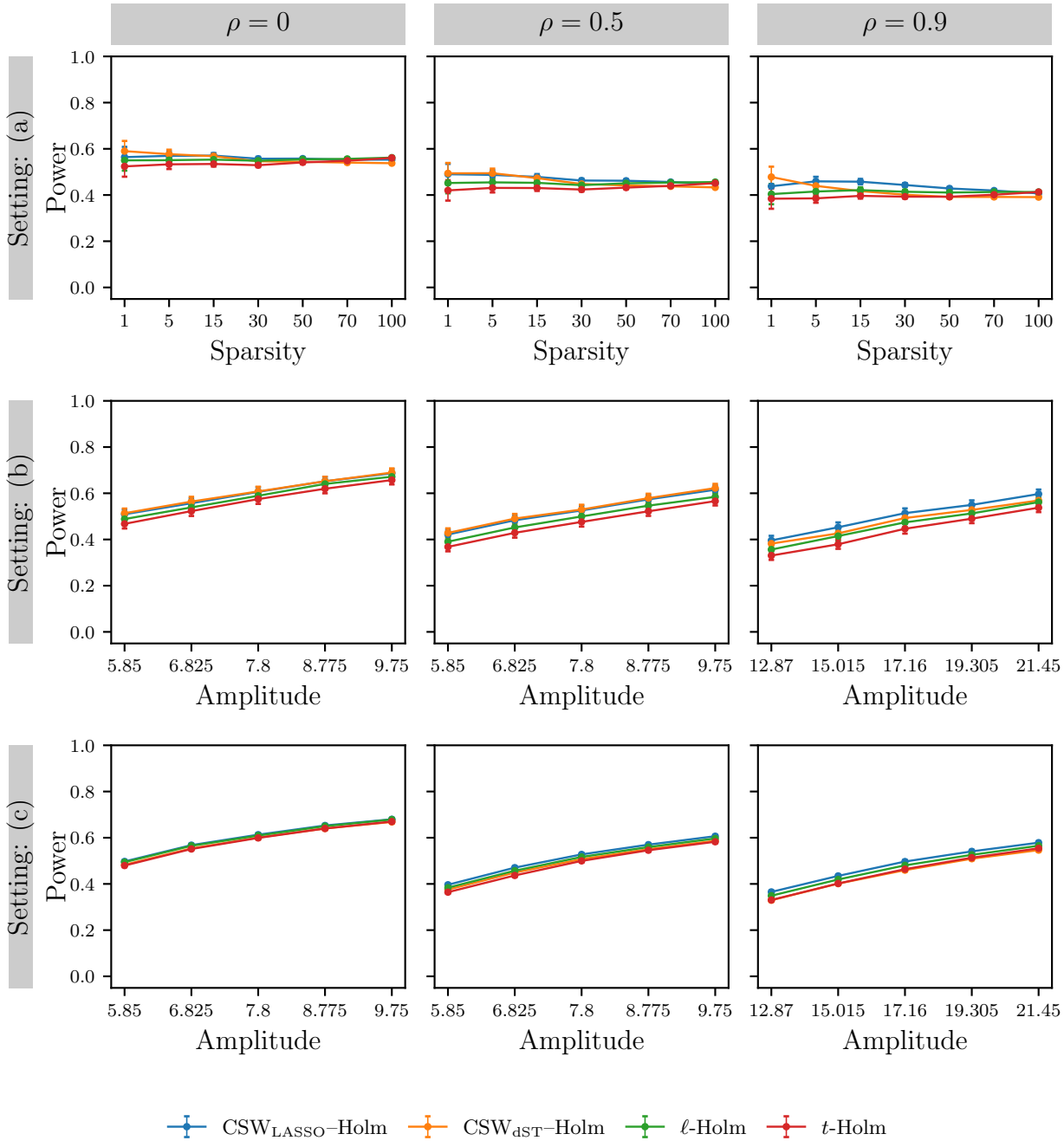


Figure 20: The empirical power of various FWER controlling procedures under continuous signal generation, $n/d = 3$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

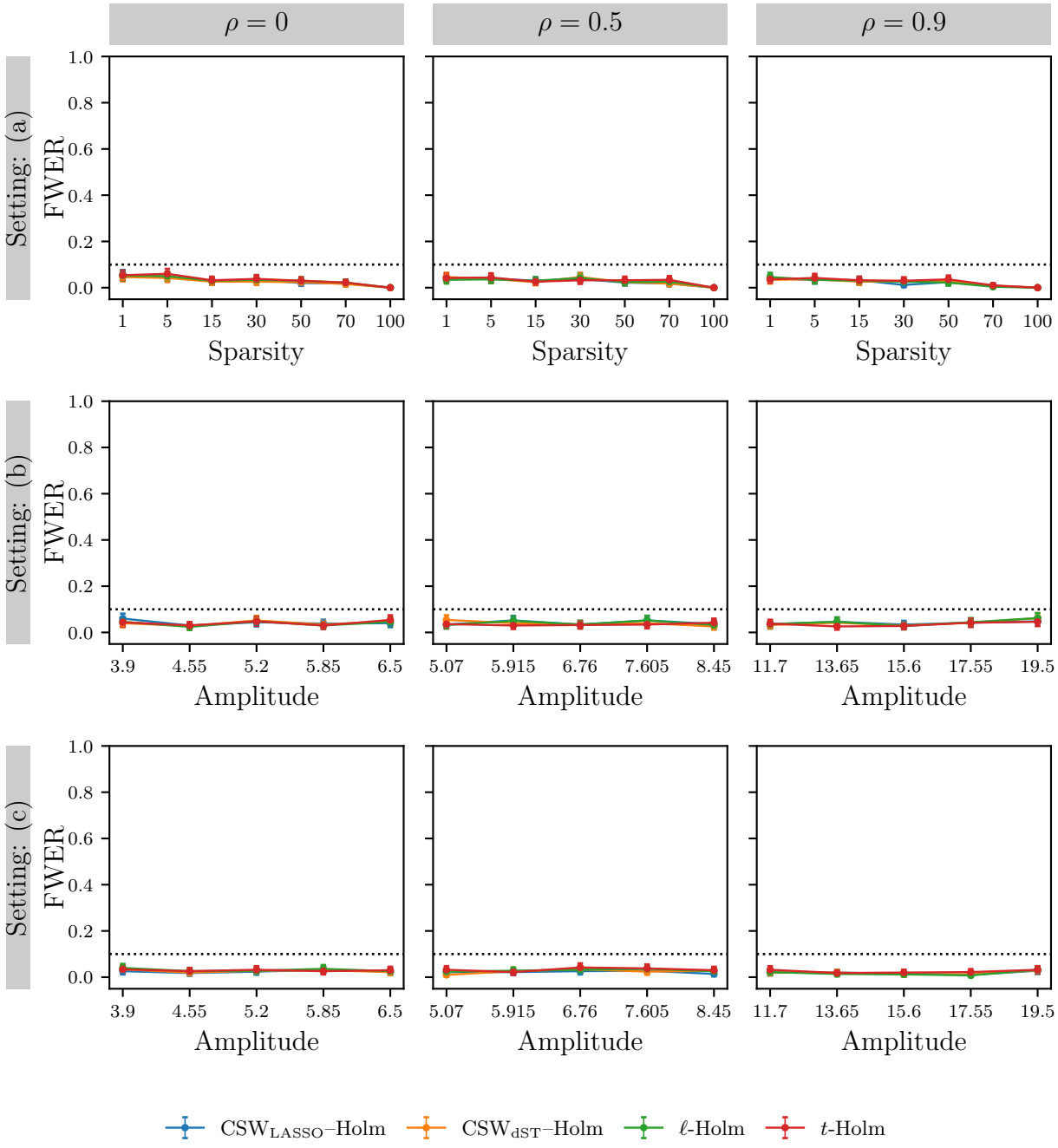


Figure 21: The empirical FWER of various FWER controlling procedures under discrete signal generation, $n/d = 1.5$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

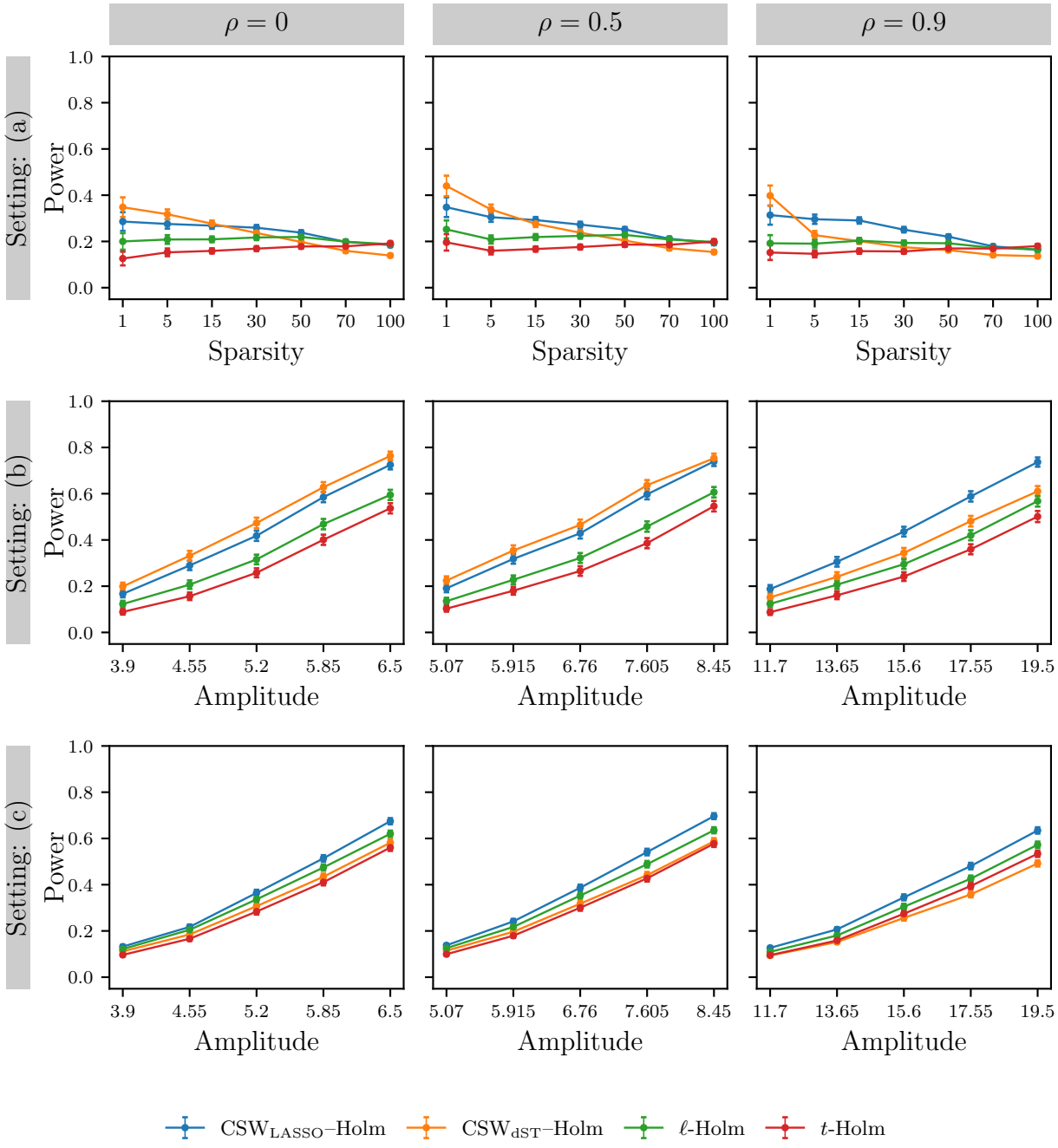


Figure 22: The empirical power of various FWER controlling procedures under discrete signal generation, $n/d = 1.5$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

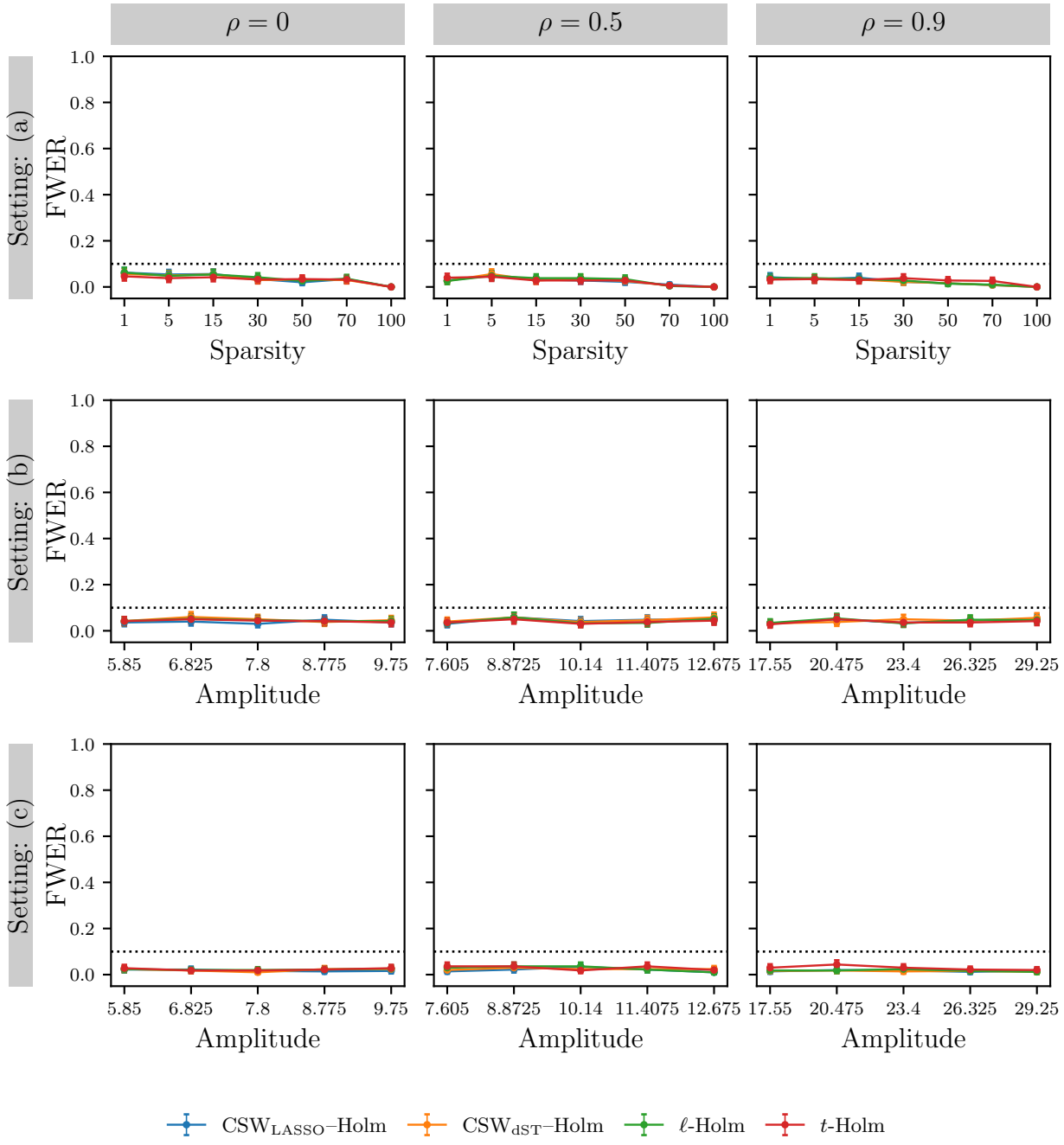


Figure 23: The empirical FWER of various FWER controlling procedures under continuous signal generation, $n/d = 1.5$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

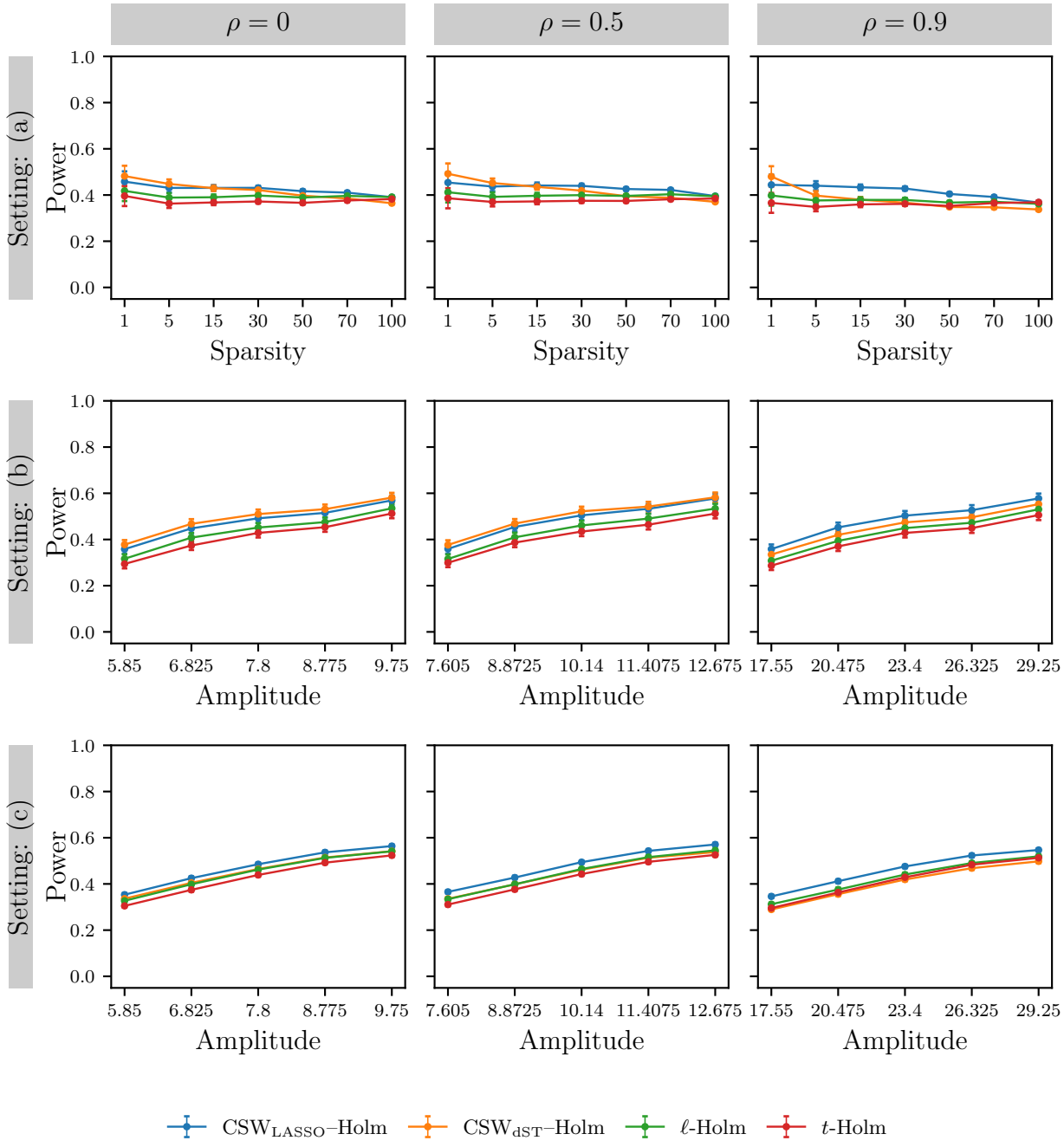


Figure 24: The empirical power of various FWER controlling procedures under continuous signal generation, $n/d = 1.5$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

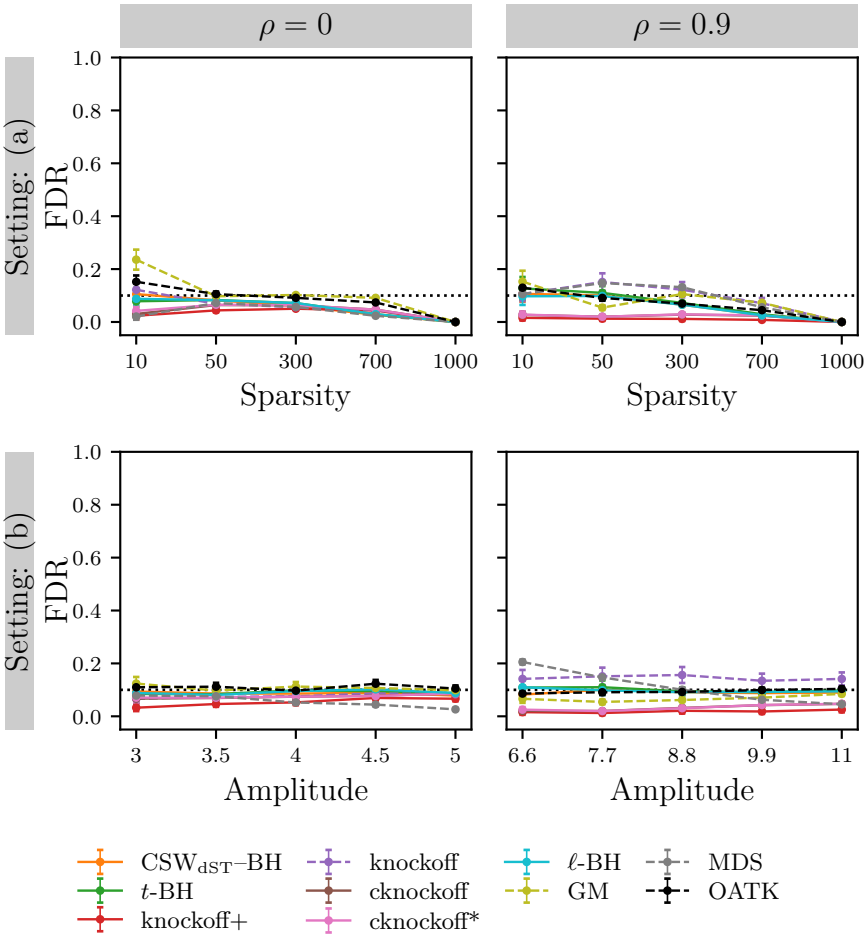


Figure 25: The empirical FDR of various FDR controlling procedures under discrete signal generation, $n/d = 3$, $d = 1000$, inter-variable correlations $\rho \in \{0, 0.9\}$ and simulation regimes (a) and (b). The error bars represent plus or minus two standard errors.

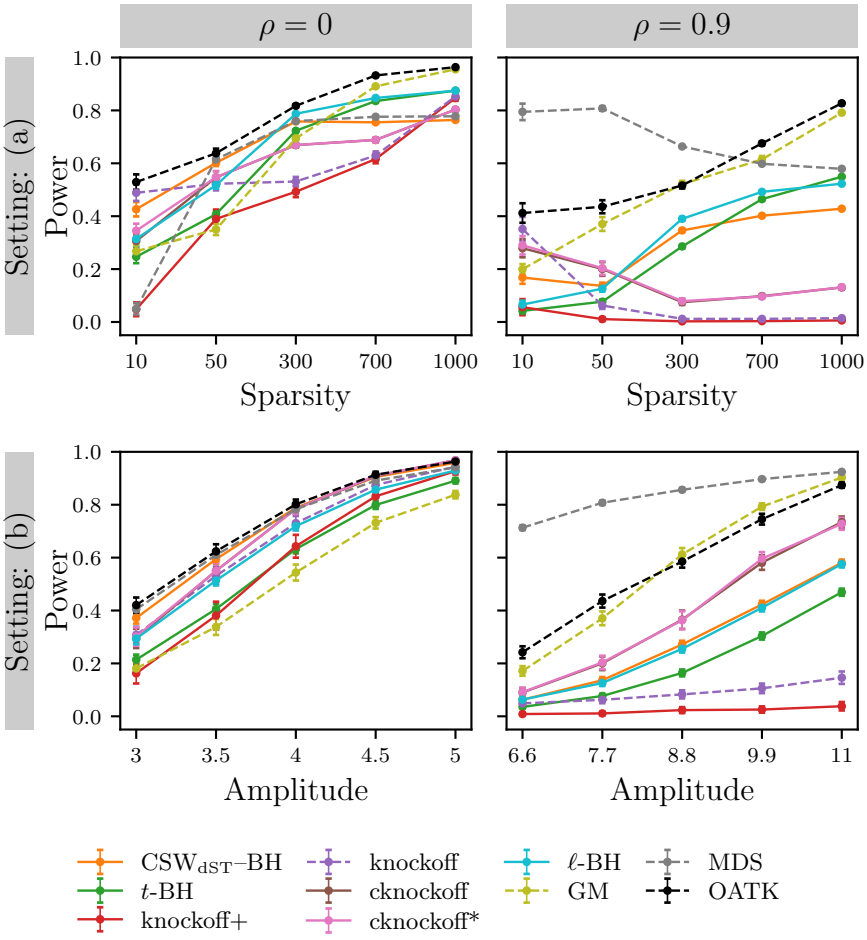


Figure 26: The empirical power of various FDR controlling procedures under discrete signal generation, $n/d = 3$, $d = 1000$, inter-variable correlations $\rho \in \{0, 0.9\}$ and simulation regimes (a) and (b). The error bars represent plus or minus two standard errors.

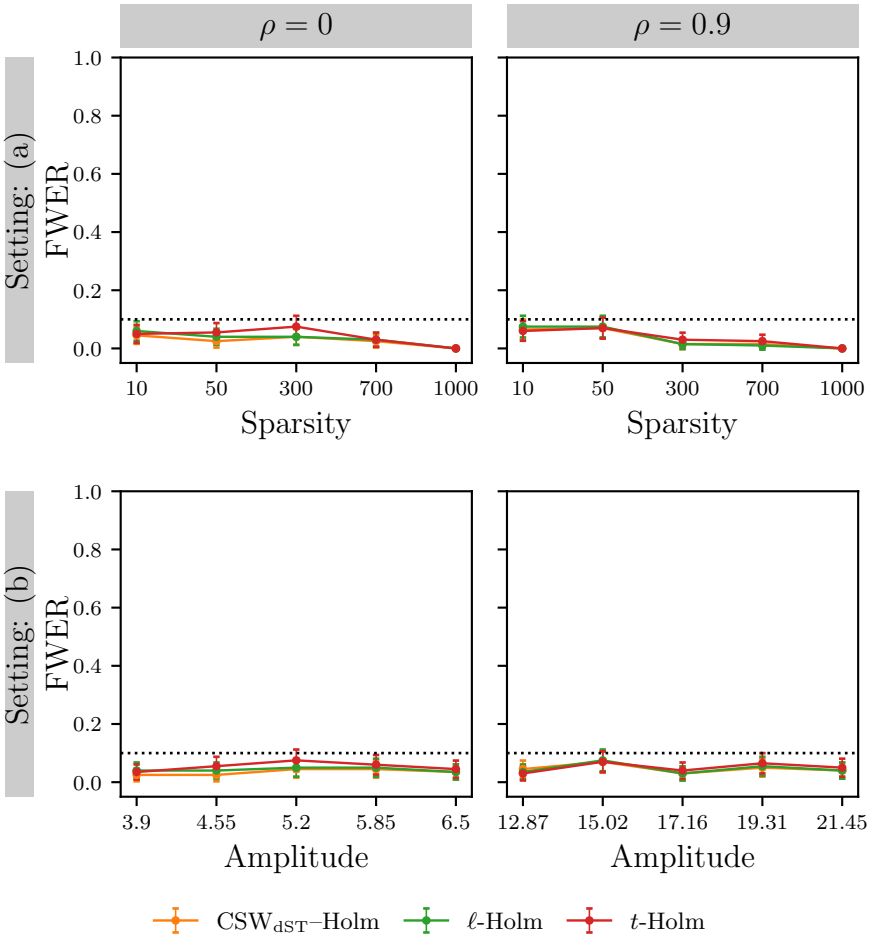


Figure 27: The empirical FWER of various FWER controlling procedures under discrete signal generation, $n/d = 3$, $d = 1000$, inter-variable correlations $\rho \in \{0, 0.9\}$ and simulation regimes (a) and (b). The error bars represent plus or minus two standard errors.

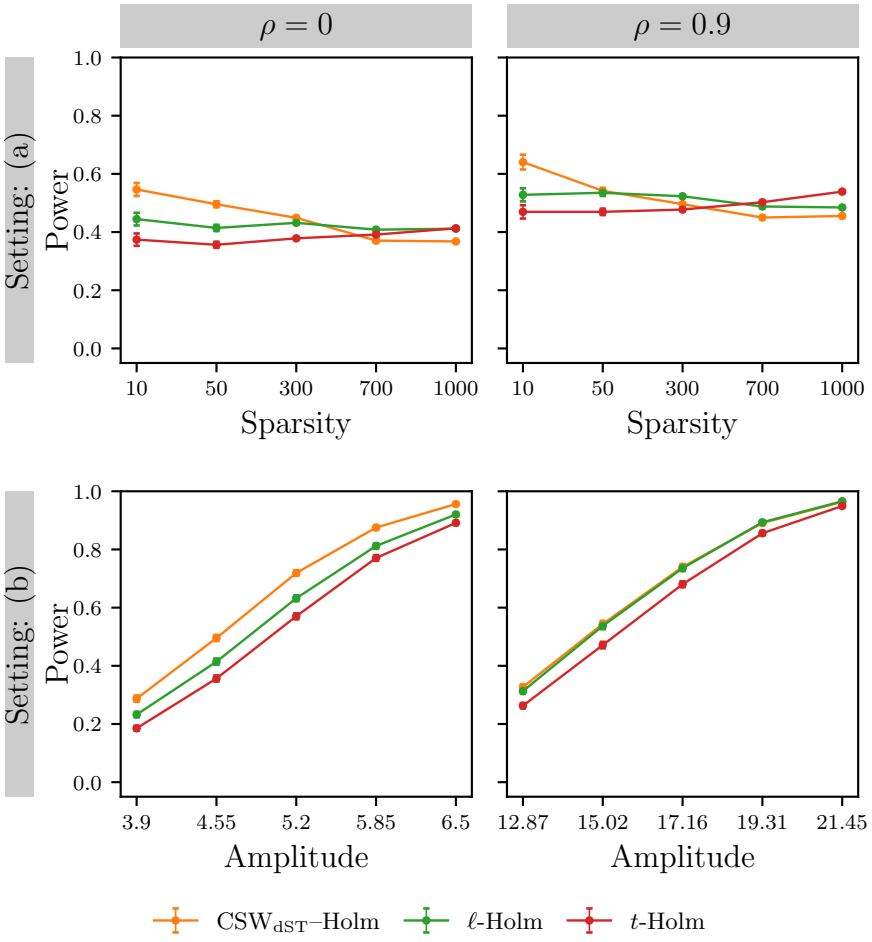


Figure 28: The empirical power of various FWER controlling procedures under discrete signal generation, $n/d = 3$, $d = 1000$, inter-variable correlations $\rho \in \{0, 0.9\}$ and simulation regimes (a) and (b). The error bars represent plus or minus two standard errors.

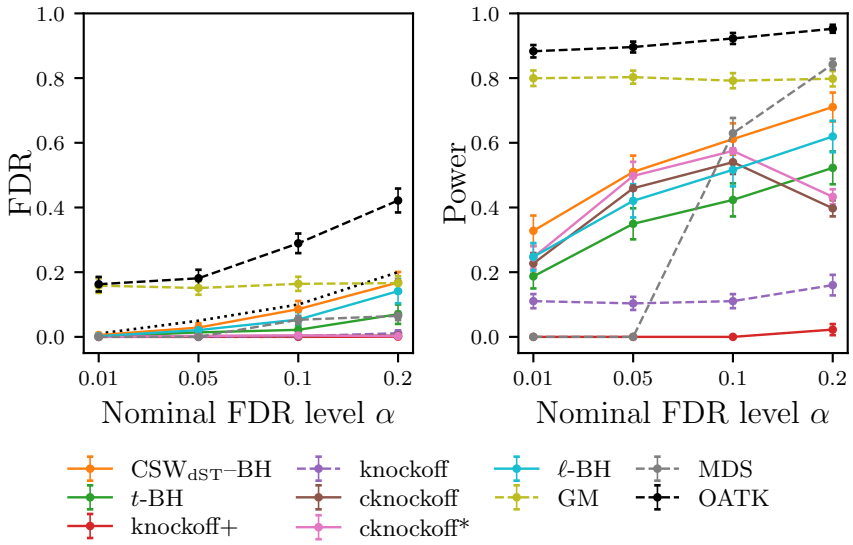


Figure 29: The empirical FDR (left) and power (right) of various FDR controlling procedures under the MCC design. The dotted line in the left panel marks FDR = nominal level. The error bars represent plus or minus two standard errors.

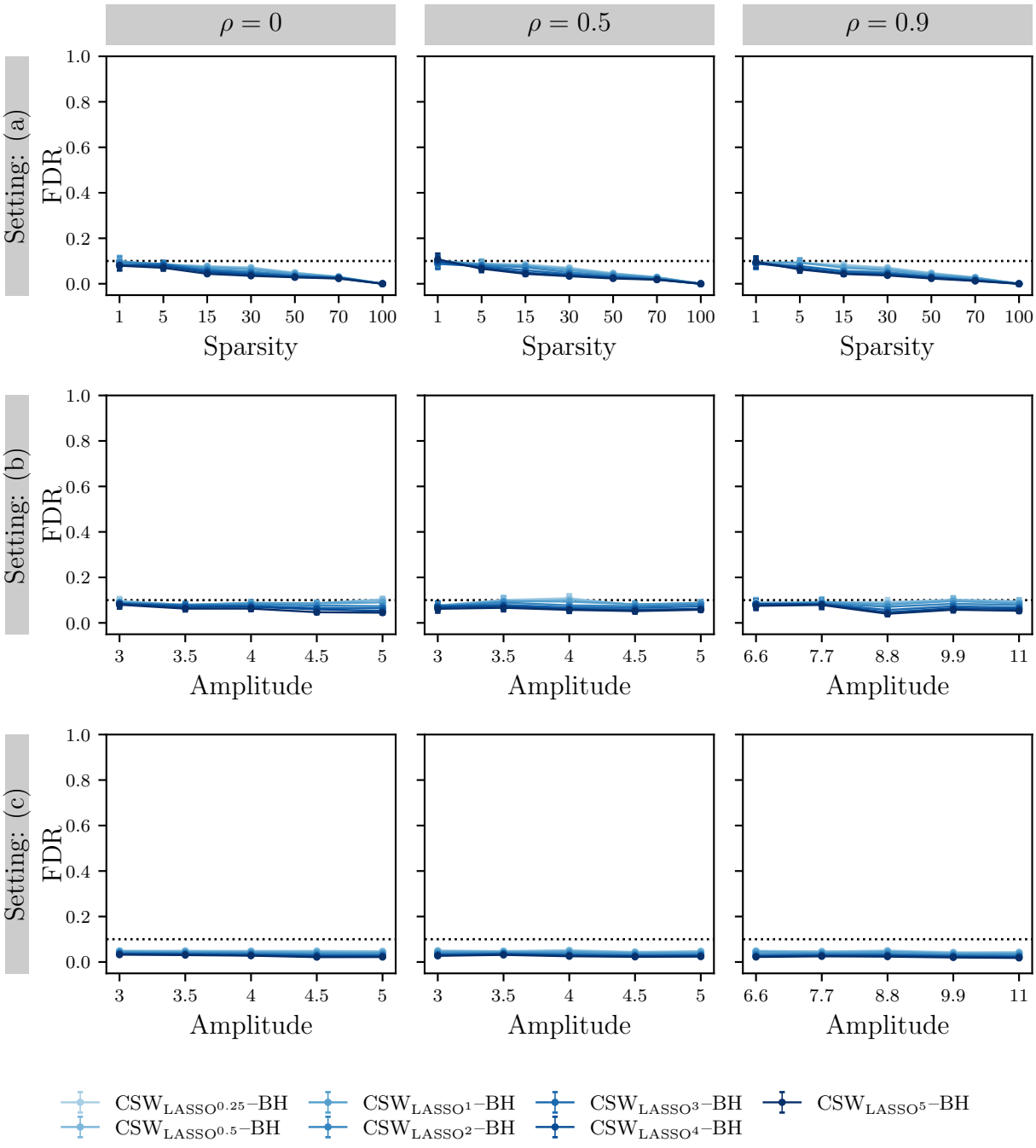


Figure 30: The empirical FDR of exponentiated versions of $\text{CSW}_{\text{LASSO}}$–BH under discrete signal generation, $n/d = 3$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

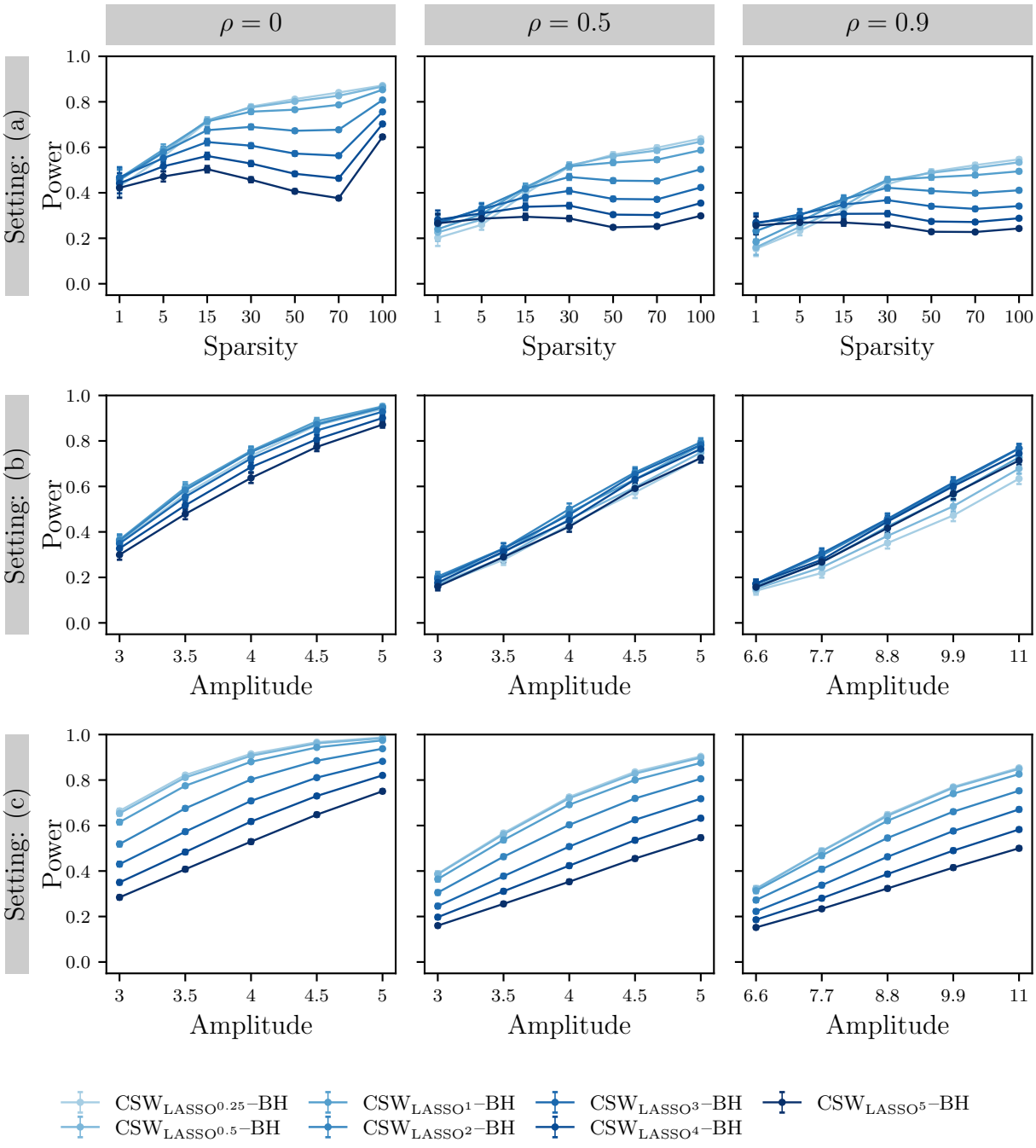


Figure 31: The empirical power of exponentiated versions of $\text{CSW}_{\text{LASSO}}$–BH under discrete signal generation, $n/d = 3$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

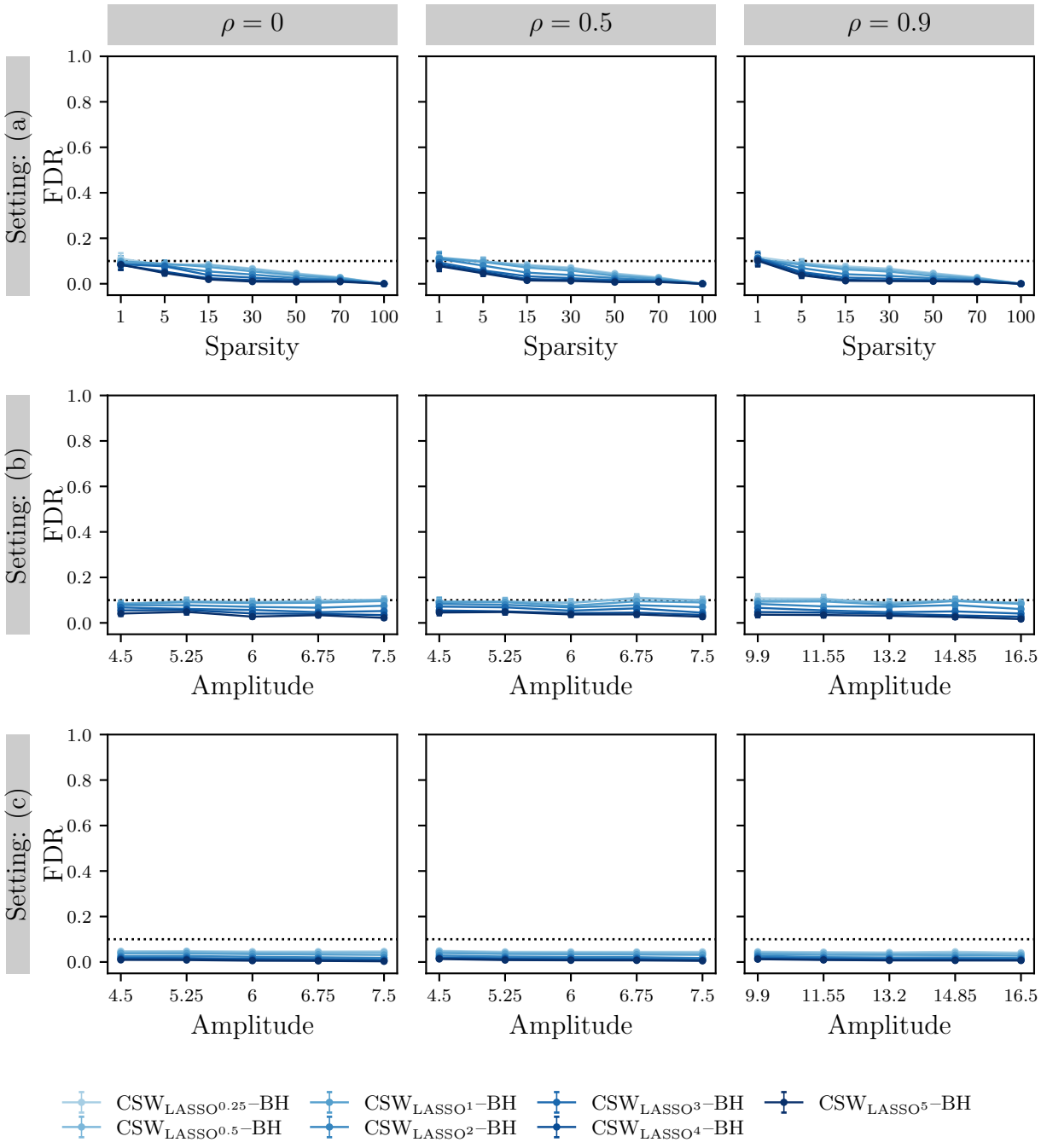


Figure 32: The empirical FDR of exponentiated versions of $\text{CSW}_{\text{LASSO}}$–BH under continuous signal generation, $n/d = 3$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

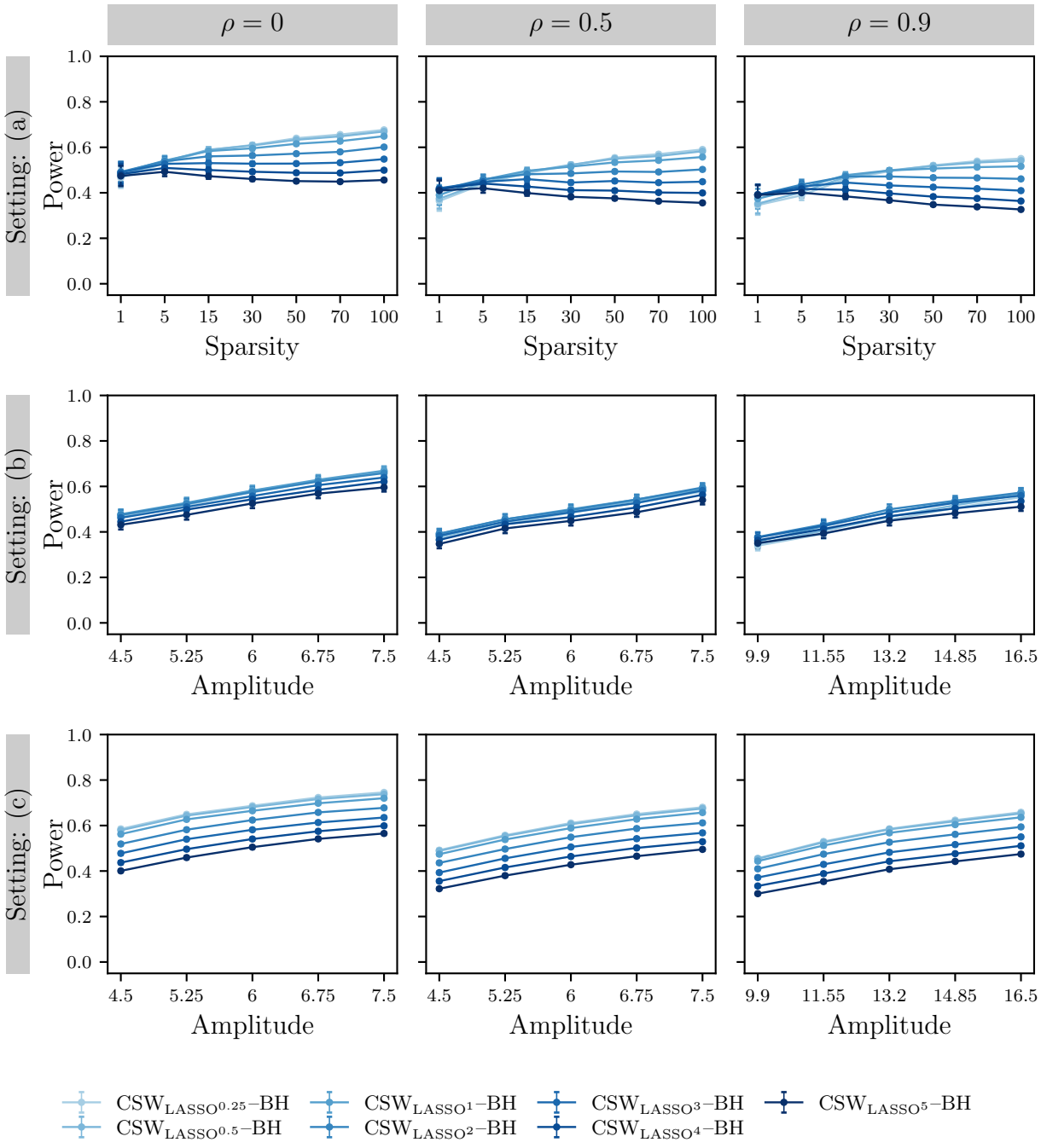


Figure 33: The empirical power of exponentiated versions of $\text{CSW}_{\text{LASSO}}$–BH under continuous signal generation, $n/d = 3$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

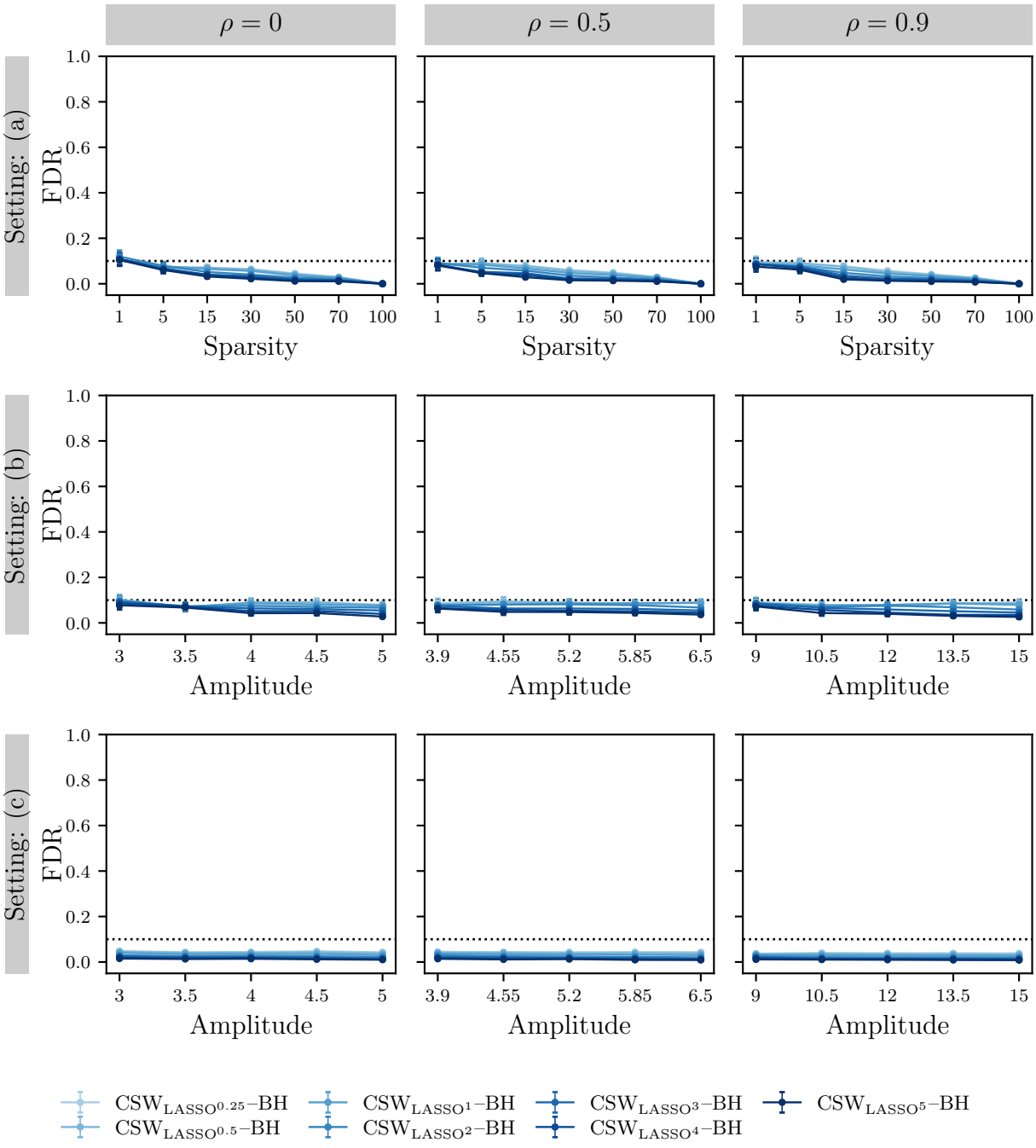


Figure 34: The empirical FDR of exponentiated versions of $\text{CSW}_{\text{LASSO}}$–BH under discrete signal generation, $n/d = 1.5$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

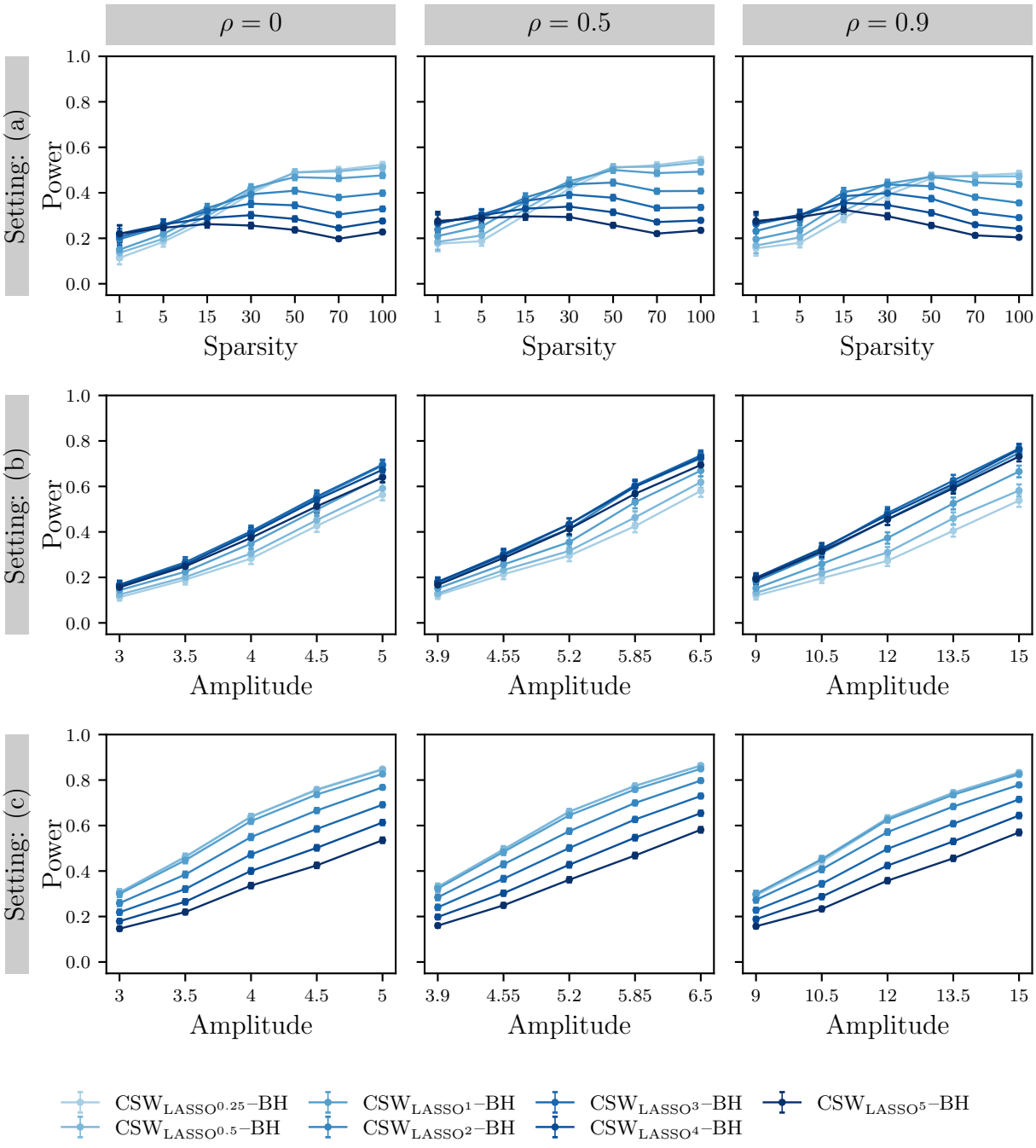


Figure 35: The empirical power of exponentiated versions of $\text{CSW}_{\text{LASSO}}$–BH under discrete signal generation, $n/d = 1.5$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

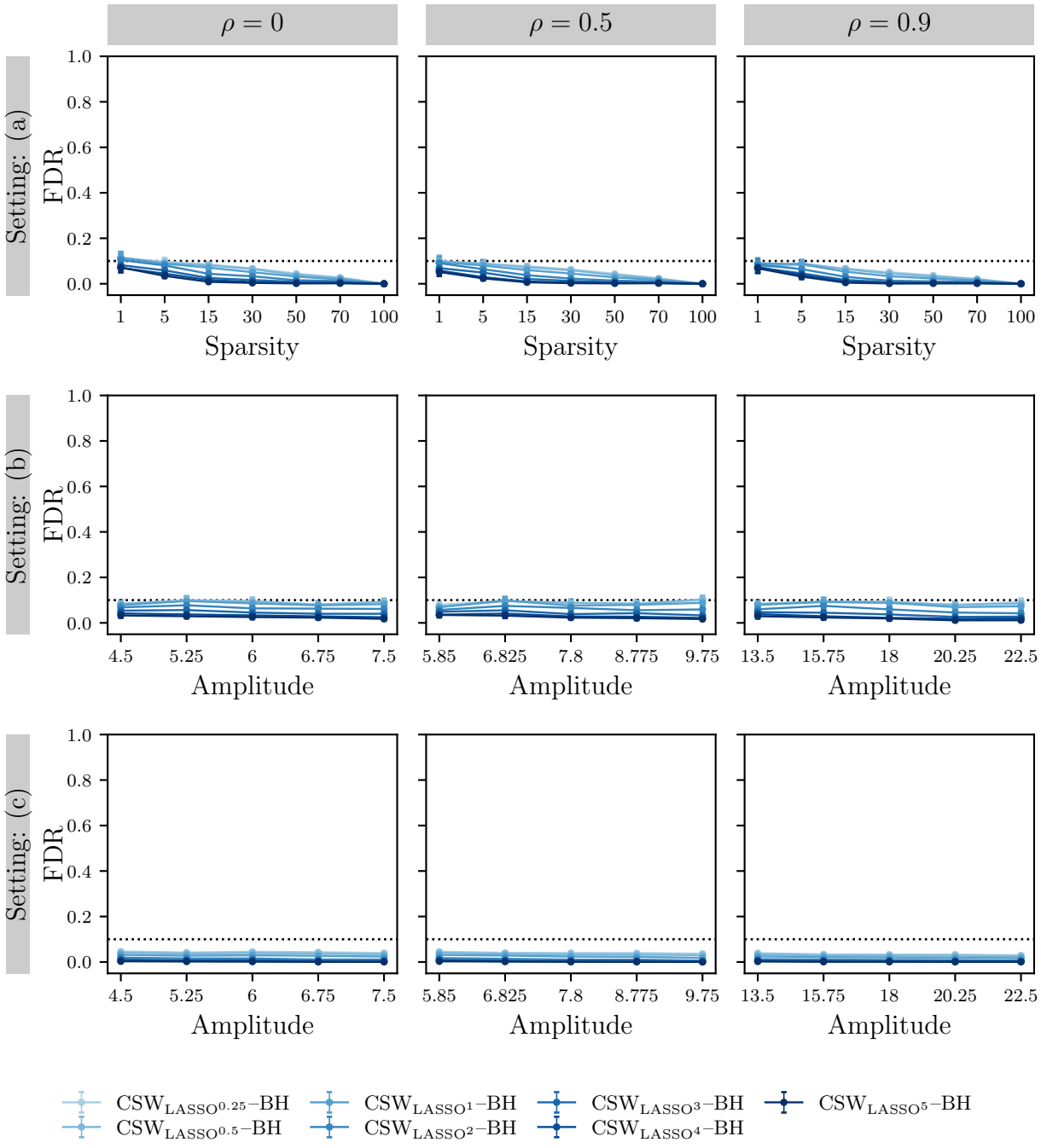


Figure 36: The empirical FDR of exponentiated versions of $\text{CSW}_{\text{LASSO}}$–BH under continuous signal generation, $n/d = 1.5$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

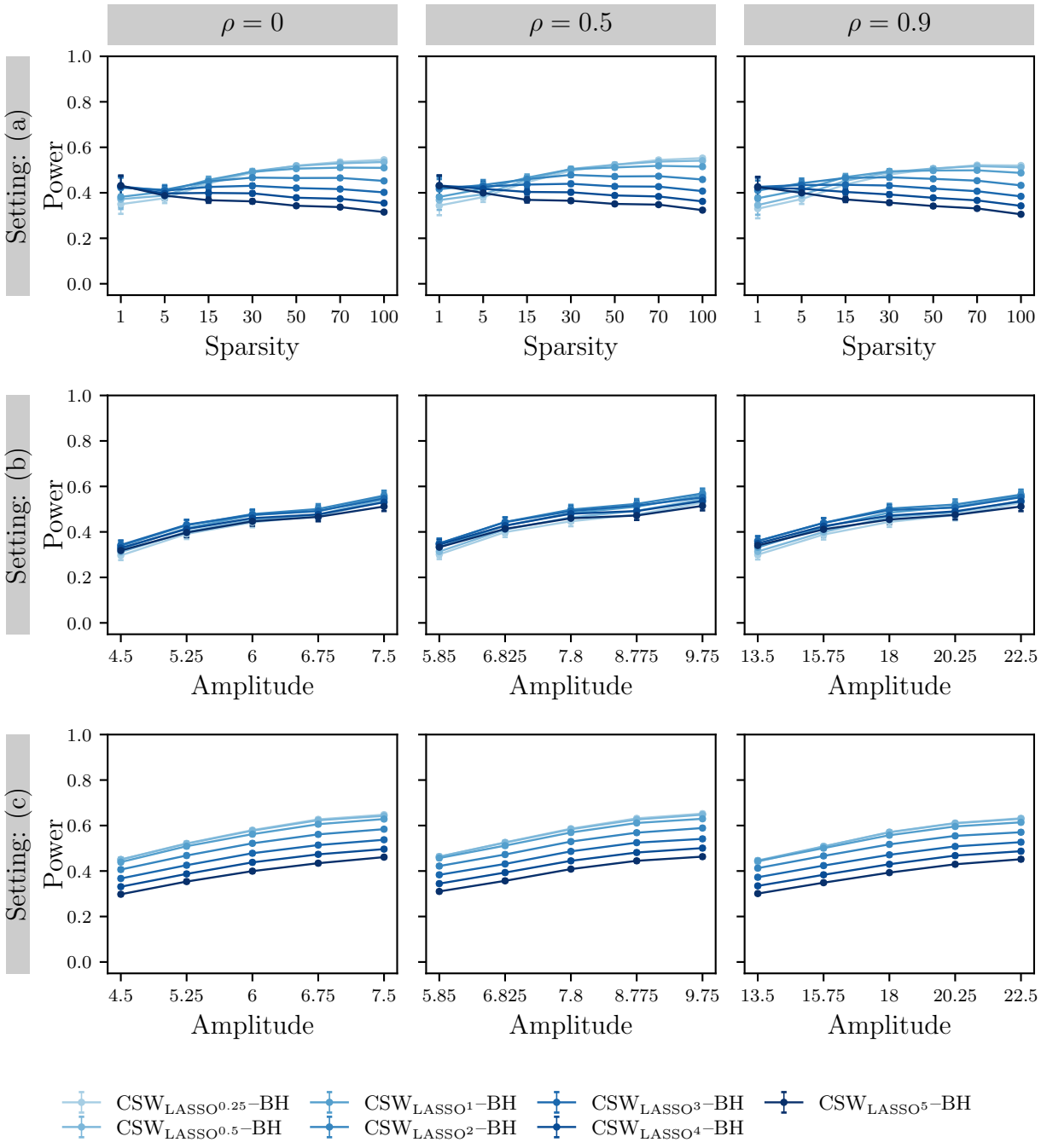


Figure 37: The empirical power of exponentiated versions of $\text{CSW}_{\text{LASSO}}$–BH under continuous signal generation, $n/d = 1.5$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

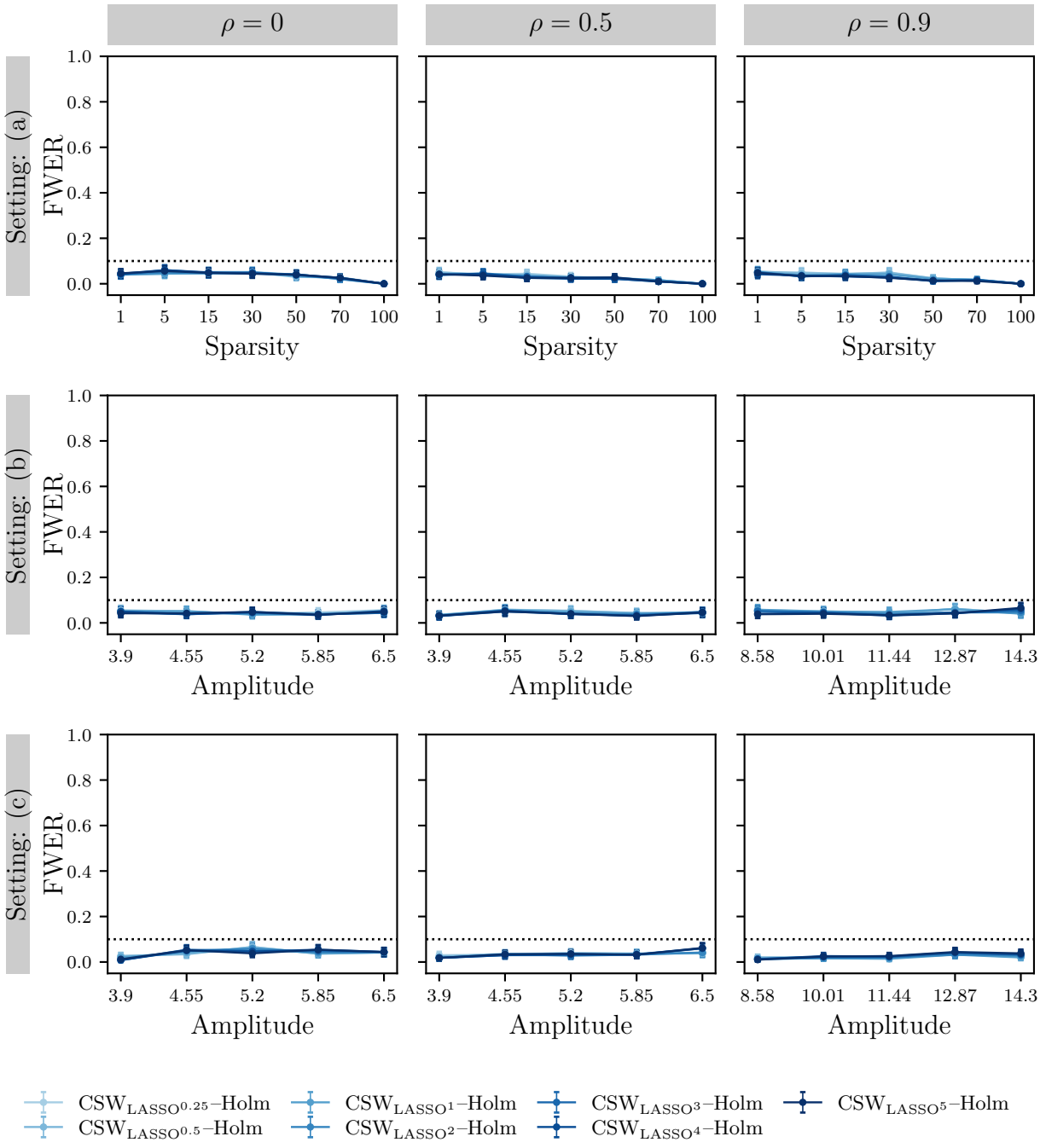


Figure 38: The empirical FWER of exponentiated versions of $\text{CSW}_{\text{LASSO}}$–Holm under discrete signal generation, $n/d = 3$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

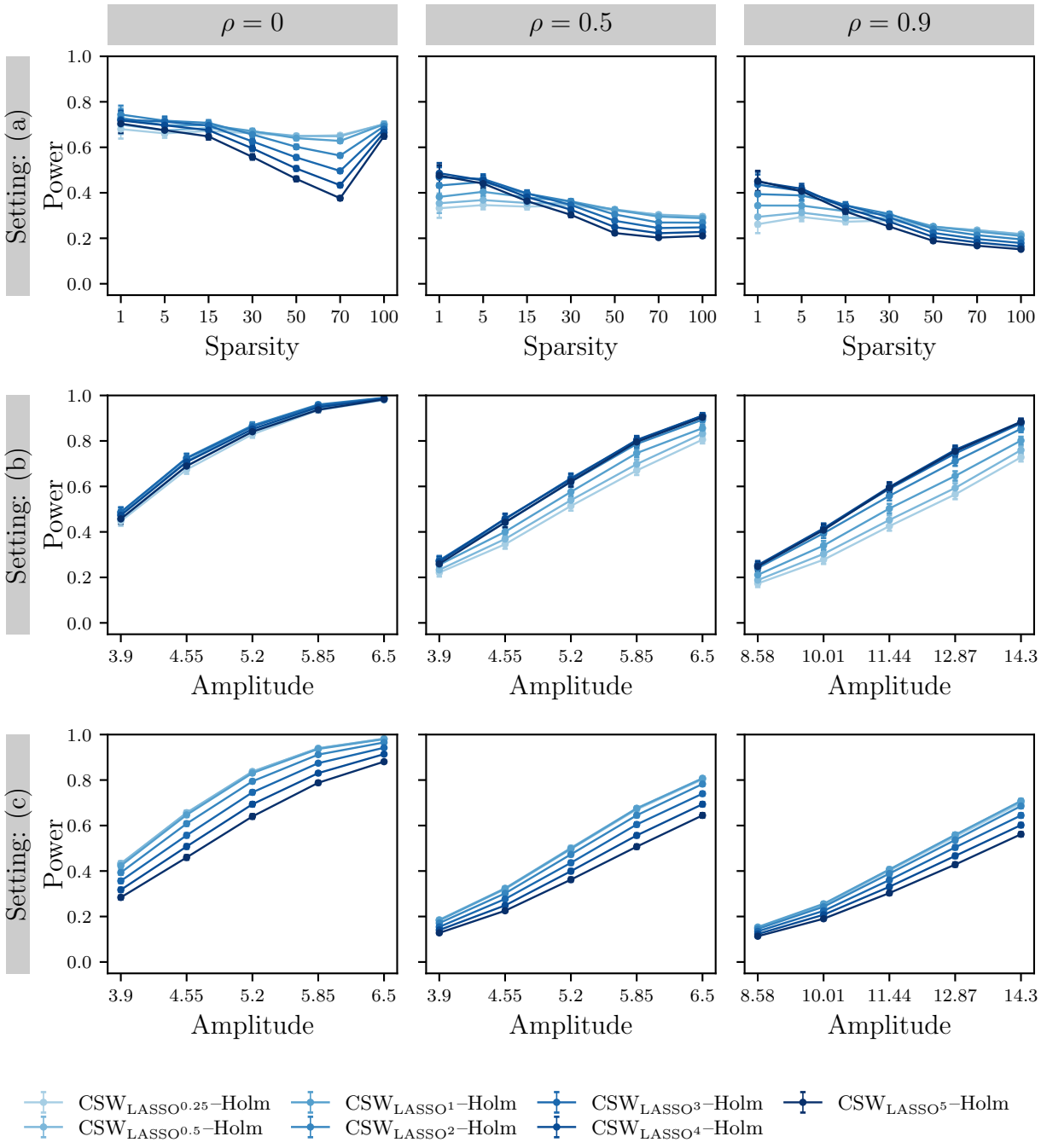


Figure 39: The empirical power of exponentiated versions of $\text{CSW}_{\text{LASSO}}$–Holm under discrete signal generation, $n/d = 3$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

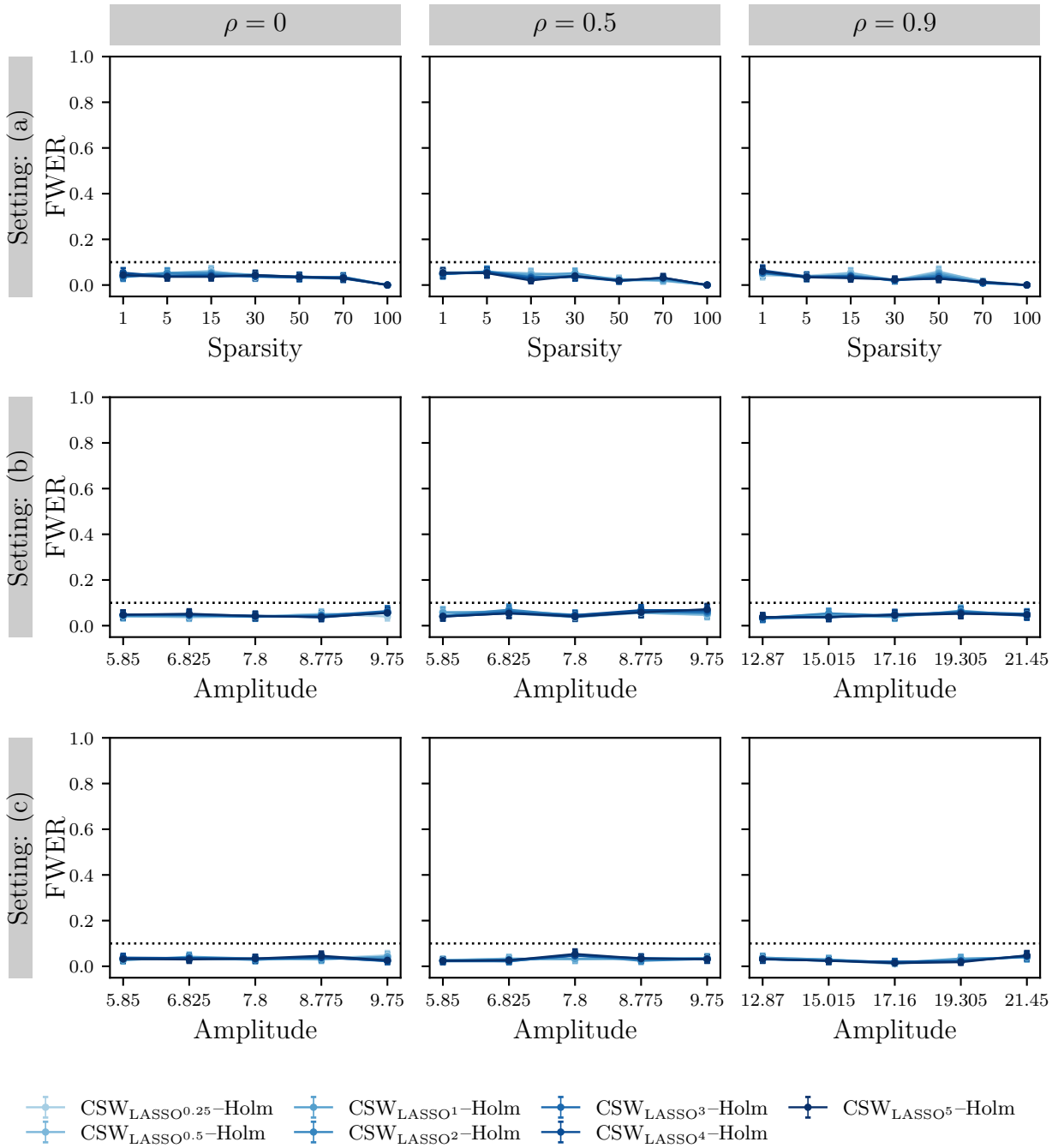


Figure 40: The empirical FWER of exponentiated versions of $\text{CSW}_{\text{LASSO}}$–Holm under continuous signal generation, $n/d = 3$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

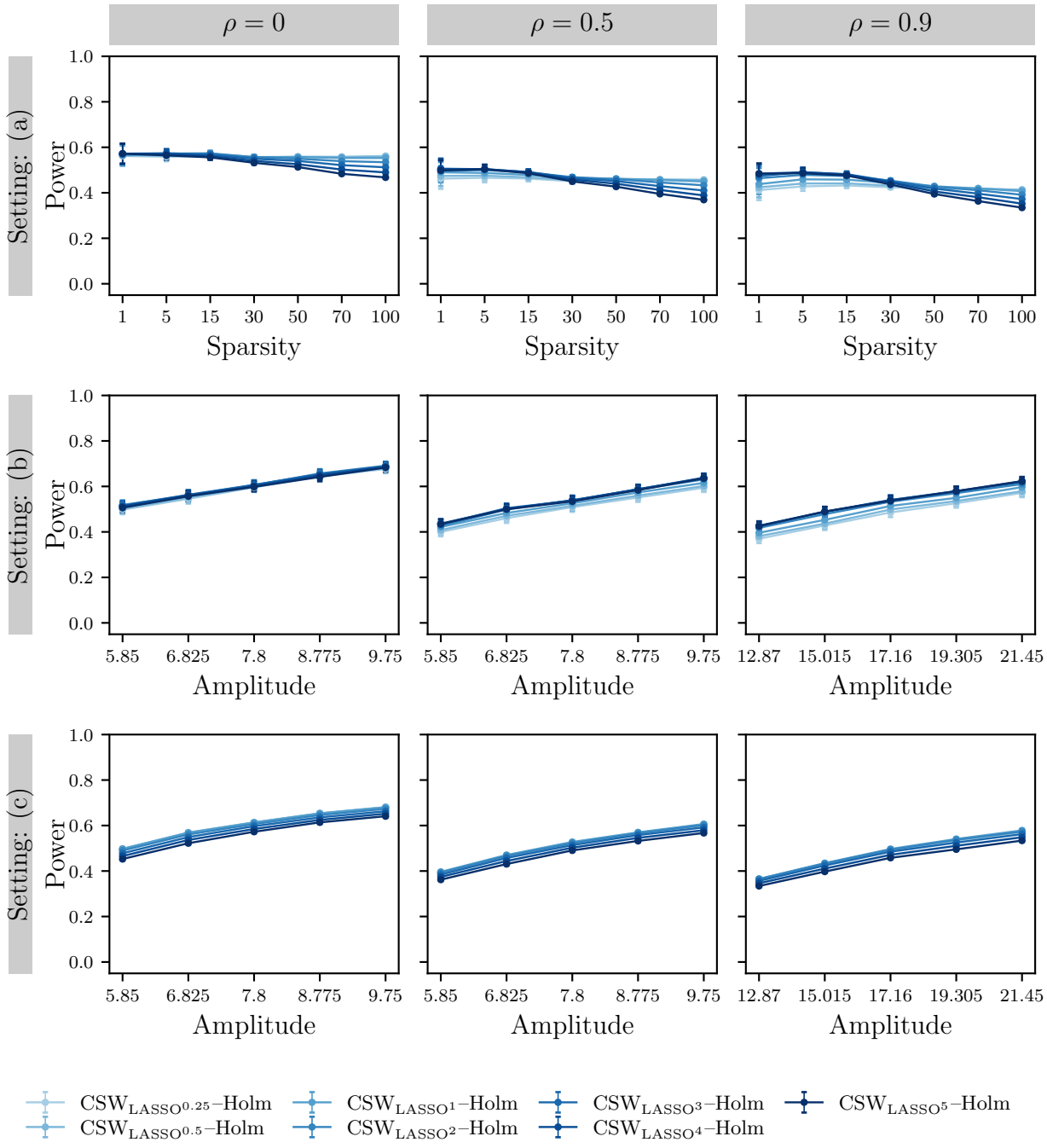


Figure 41: The empirical power of exponentiated versions of $\text{CSW}_{\text{LASSO}}$–Holm under continuous signal generation, $n/d = 3$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

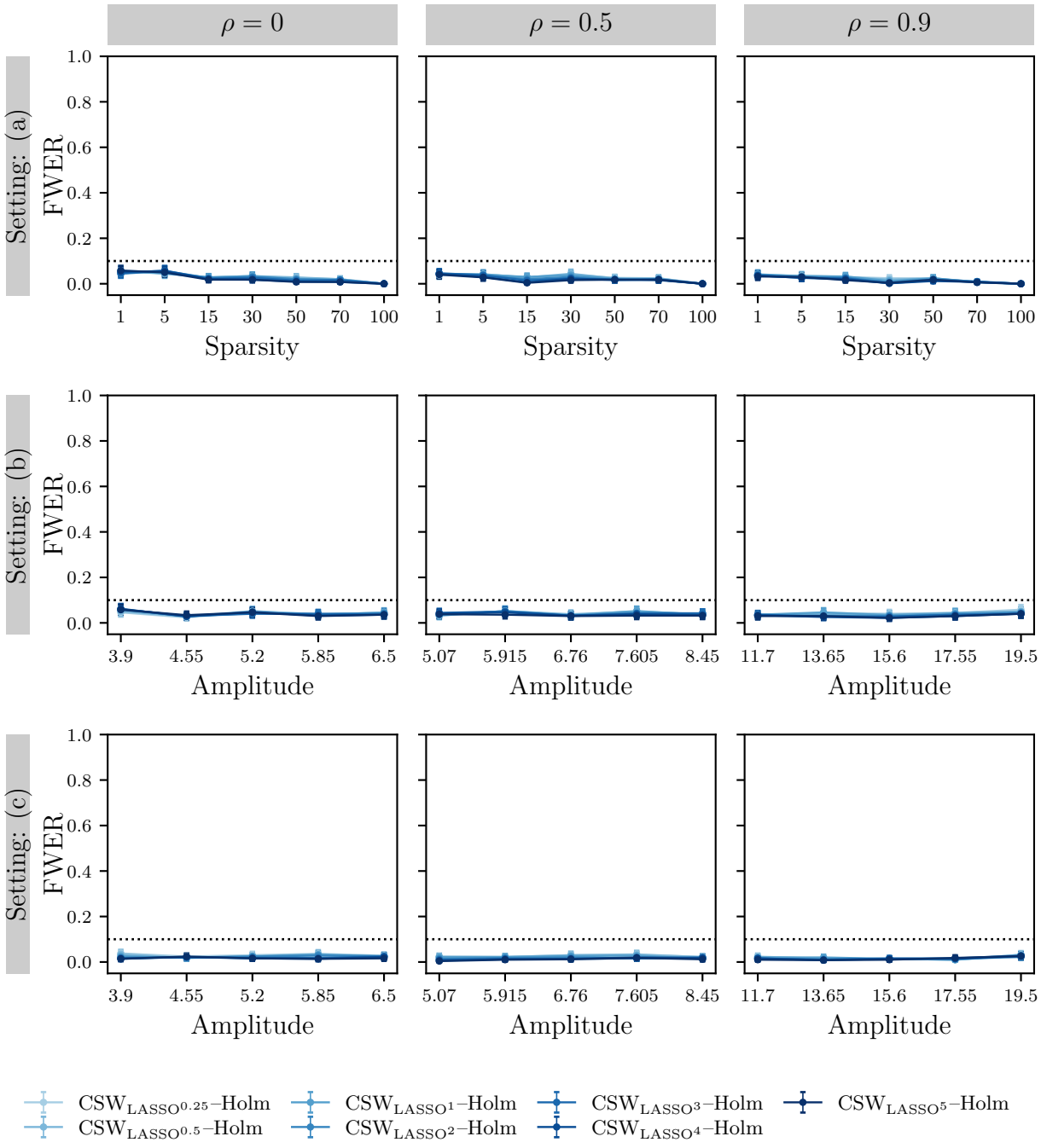


Figure 42: The empirical FWER of exponentiated versions of $\text{CSW}_{\text{LASSO}}$–Holm under discrete signal generation, $n/d = 1.5$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

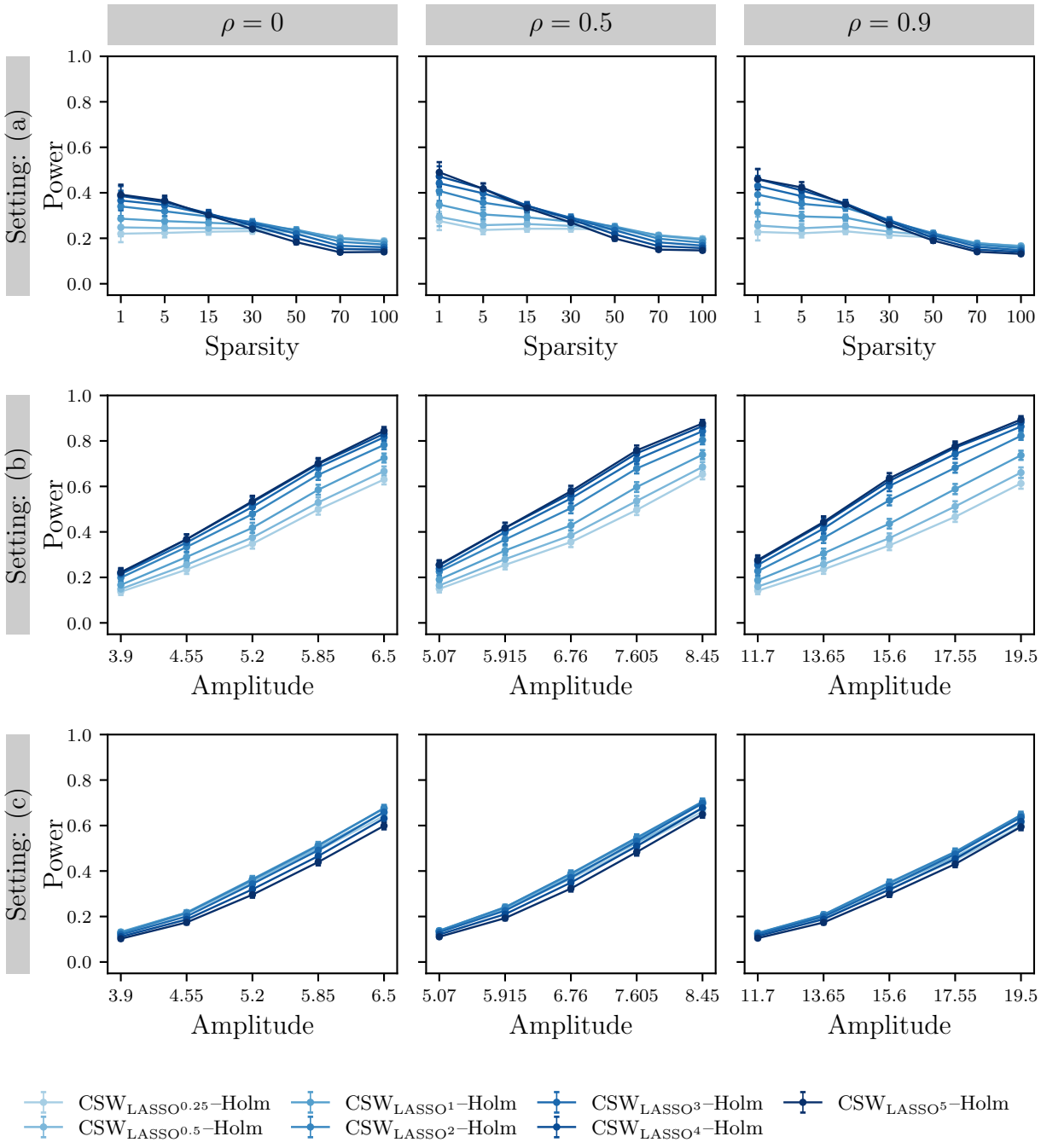


Figure 43: The empirical power of exponentiated versions of CSW$_{\text{LASSO}}$–Holm under discrete signal generation, $n/d = 1.5$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

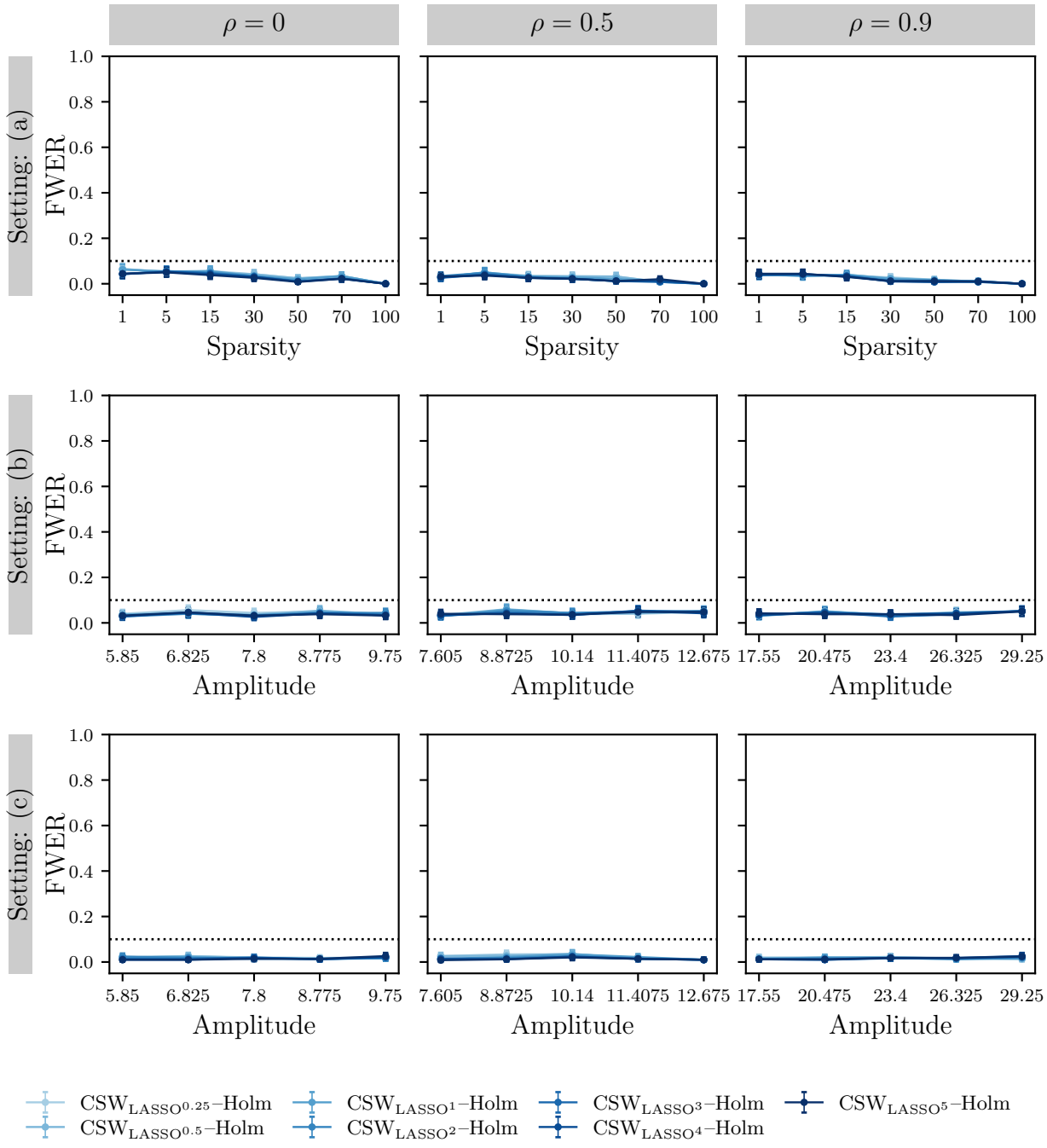


Figure 44: The empirical FWER of exponentiated versions of $\text{CSW}_{\text{LASSO}}$–Holm under continuous signal generation, $n/d = 1.5$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.

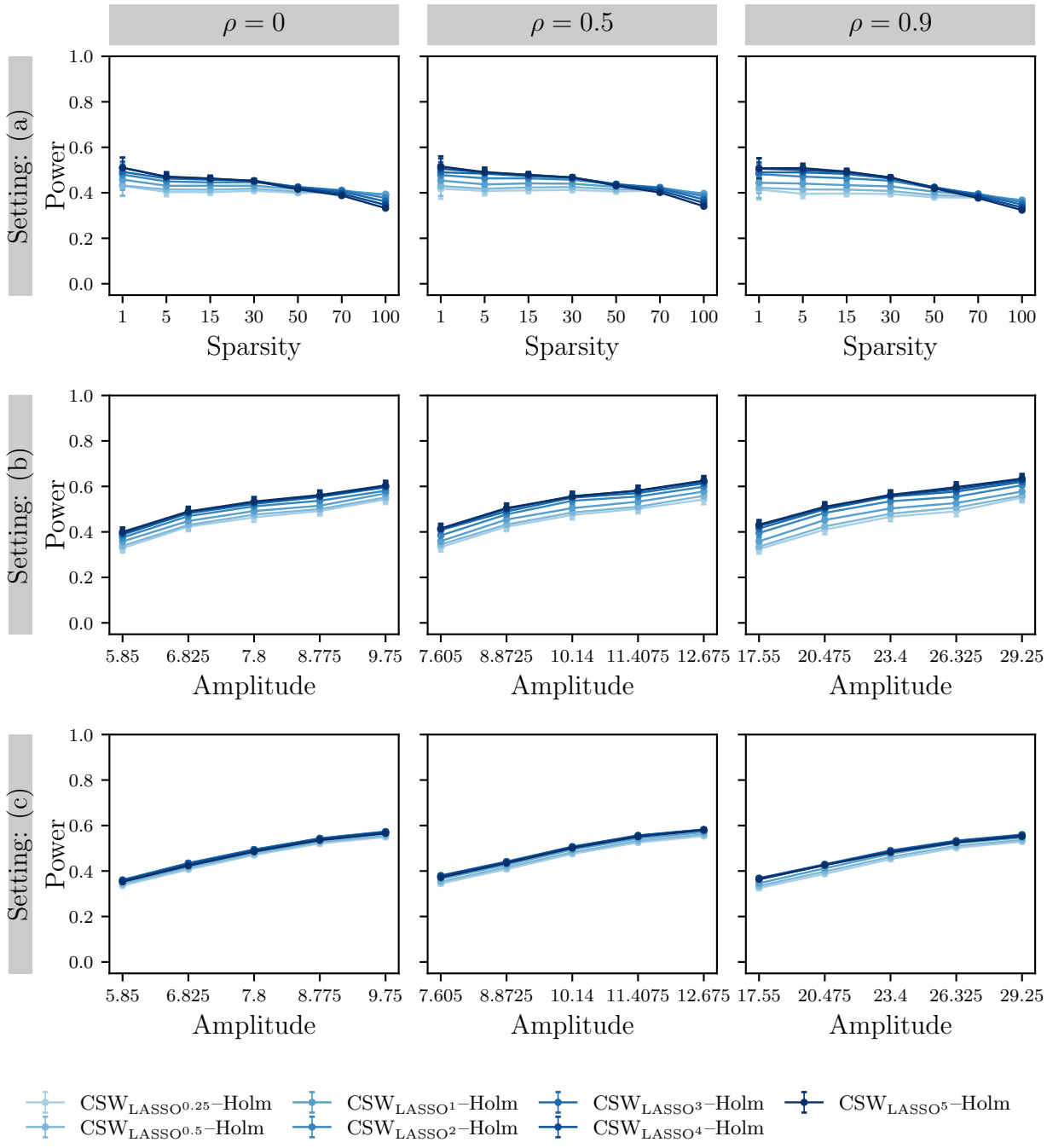


Figure 45: The empirical power of exponentiated versions of $\text{CSW}_{\text{LASSO}}$–Holm under continuous signal generation, $n/d = 1.5$, $d = 100$ and simulation regimes (a), (b) and (c). The error bars represent plus or minus two standard errors.